

Markups and Markdowns in Indian Organised Manufacturing: Plant-Level Evidence from the Annual Survey of Industries, 2012–2022

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Abstract

Using 423,127 plant-years from India’s Annual Survey of Industries, 2012–2022, we jointly estimate price markups and wage markdowns under a translog technology identified by the moment conditions of [Akerberg et al. \(2015\)](#). Product-market power is modest: the median markup is 1.22, and the sales-weighted aggregate rose 4.6 per cent, two-thirds of it through reallocation towards high-markup plants. Labour-market power is concentrated: 48 per cent of plants pay labour below its marginal revenue product (44 per cent unweighted), although the median plant does not. The two margins rank industries in opposite directions and have different geographies. Monopsony incidence ranges from 31 to 67 per cent across states, and three-quarters of the gap between high- and low-incidence states remains within industries. Contract labour, 46 per cent of the workforce by 2022, is barely related to the markdown. Because the ASI records no geography finer than the state, which corresponds to no commuting zone, the thin-market mechanism remains untested.

JEL: D22, D24, D43, J42, L11, L13, L60, O14. **Keywords:** markups; markdowns; monopsony; production function; ACF; translog; ASI; India.

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1 Introduction

The earnings of Indian factory workers relative to the value they produce have changed substantially over the past four decades, and the causes of that change remain unclear. In 1980–81, labour received 38.9 per cent of the value added in India’s registered manufacturing sector; by 2007–08 it received 25.7 per cent, a fall of thirteen percentage points, most of it after the reforms of 1991. The decline then stopped. Over the decade examined in this paper, 2012–13 to 2022–23, the share drifted back up to 32.6 per cent. Figure 1 plots the series, computed from the 2024 vintage of the India KLEMS database (RBI, 2024). The long-run performance of the organised sector over these decades, by sub-period and by industry, is documented by Basole & Narayan (2020). What caused the decline, and why it stopped, remain open questions.

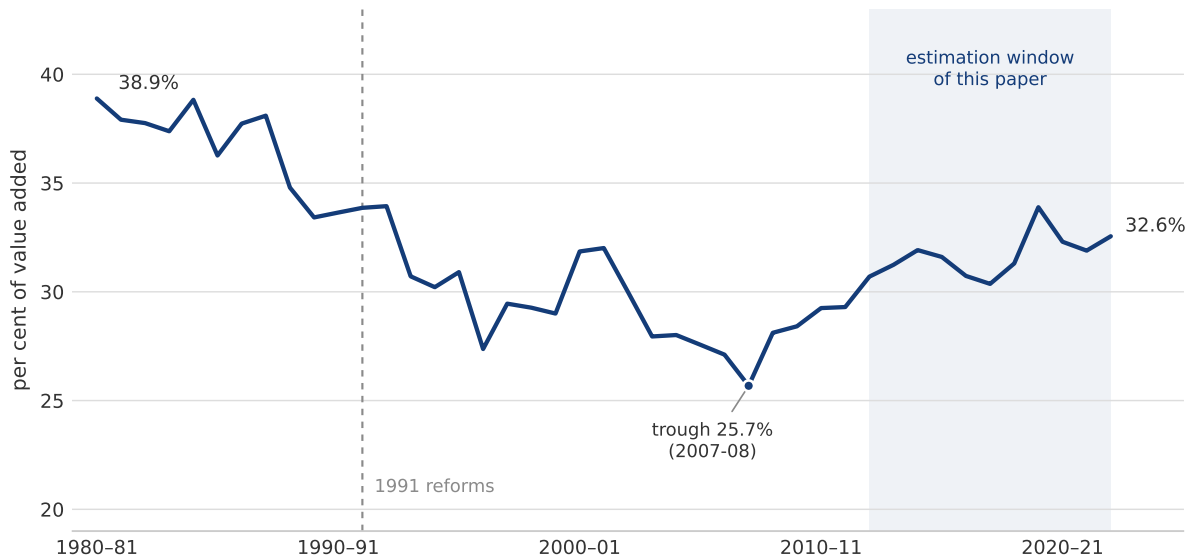


Figure 1: Labour’s share of value added in Indian registered manufacturing, 1980–81 to 2022–23. Computed from the India KLEMS 2024 vintage (RBI, 2024) as total labour income over total value added across the thirteen registered-manufacturing industries. The shaded band is the estimation window of this paper.

One caution about this series should be registered at the outset, because “the labour share” refers to several different objects in this literature, and they differ by more than a factor of two. Figure 1 uses KLEMS labour income, which includes an imputation for the labour earnings of the self-employed and of proprietors, divided by value added. A narrower and equally common construction, the ASI wage bill divided by value added, is far lower: Shukla (2026) reports it at 13.7 per cent in 2023–24, against 22.2 per cent in 2000–01, alongside a doubling of output per worker and real wages that have been flat since about 2014. The two series measure different things, and we do not treat them as interchangeable. What matters for this paper is that both describe the same qualitative history; that the wedge between what a worker produces and what a worker is paid

is the object that both approach indirectly; and that this paper measures that wedge directly, at the plant, rather than inferring it from an aggregate ratio. Our own labour share, α^L , defined as emoluments over gross output, is a third construction. It enters the markdown directly, which is one reason why this paper emphasises incidence, dispersion and geography rather than the level of the markdown.

The international literature has converged on a leading explanation. Labour’s share has fallen in most large economies since the 1980s (Karabarbounis & Neiman, 2014), and the most influential accounts place firms’ market power at the centre: industries have become more concentrated in a few highly productive “superstar” firms with low labour shares (Autor et al., 2020), pure profits have risen at the expense of both labour and capital (Barkai, 2020), and measured markups have risen steeply (De Loecker et al., 2020). A parallel literature argues that the change occurred on the other side of the firm: that what fell was not labour’s productivity but its bargaining power (Stansbury & Summers, 2020), and that employers hold more wage-setting power than competitive models allow (Manning, 2021).

Both explanations concern *market power*, and both were developed using data from rich countries. Neither has been assessed for Indian manufacturing in the recent period, because the object it requires—market power on both sides of the firm, measured on the same plants—has not been estimated for that period (Section 2). To ask whether market power explains what happened to Indian factory workers, one needs to know how much market power Indian factories actually hold, both in the market in which they sell and in the market in which they hire. Providing that measurement is the task of this paper, which is the first objective of the wider research programme to which it belongs.

A related Indian debate has been asking a closely connected question without a firm-level measure of the wedge. The literature on India’s industrial transformation, including a recent series in the *Economic and Political Weekly*, has documented that India’s growth has been unusually capital- and skill-intensive for its stage of development: industry absorbs about 26 per cent of employment, against roughly a third in China and Vietnam, and the labour-intensive segments that would ordinarily absorb surplus agricultural labour have grown slowly (Bishnoi et al., 2024). Closest to our question, Babu (2026) examines the size distribution of establishments together with price–cost margins, and finds increasing concentration of output and value added among larger, more capital-intensive establishments, with margins that vary widely across industries in a way that suggests uneven capture of value.

That finding and ours fit together, and the way in which they fit is instructive. Babu (2026) measures concentration in *output and value added*; the concentration measure we report is in *employment*, and it falls, with the mean industry employment Herfindahl index declining from 0.086 to 0.066 over our decade. Both can hold at once, and together they describe a familiar pattern: output concentrating in large, capital-intensive plants

while employment disperses across many smaller ones is what capital deepening looks like from two different angles. It is also exactly the configuration in which one should not expect product-market power to be the binding distortion, and should look instead at the market in which those dispersed workers are hired. Our estimates point to that market.

The stakes are also formal. Target 9.2 of the Sustainable Development Goals commits signatories to “promote inclusive and sustainable industrialization and, by 2030, significantly raise industry’s share of employment and gross domestic product” (United Nations, 2015), and its second indicator, 9.2.2, is manufacturing employment as a proportion of total employment. Monopsony bears on that indicator directly. An employer with wage-setting power restricts hiring below the competitive level, and at the markdown and labour elasticity implied for our typical plant, the restriction is more than a quarter (Section 4.4). A wedge that suppresses manufacturing employment is not a distributional footnote to the industrialisation target; it is an obstacle to the target itself, and one that output-side policy does not remove.

The question also has a sharp policy dimension, because the two kinds of market power call for very different remedies. If Indian factories are marking up prices, the remedies lie in competition policy, entry and trade. If instead they are marking *down* wages—paying workers less than the revenue their labour generates—then competition policy is beside the point, and the relevant levers are labour-market ones: mobility, information, outside options and the thickness of local labour markets. For a decade, India’s industrial policy has pulled the first set of levers. Make in India (2014) and the Production Linked Incentive schemes, with an outlay of Rs. 1.97 lakh crore across fourteen sectors, are instruments aimed at output, investment and export competitiveness (PIB, 2025). Whether that is the binding margin for workers is an empirical question, and it is the question this paper answers.

Market power has two faces. A firm with power in its *product* market sets its price above marginal cost (the markup $\mu \equiv p/c > 1$), transferring surplus from consumers and, when markups differ across producers, distorting the allocation of demand. A firm with power in its *labour* market pays a wage below the marginal revenue product of labour (the markdown $\psi \equiv \text{MRPL}/w > 1$), transferring surplus from workers and distorting the allocation of labour. The two faces are formally symmetric (Robinson, 1933): monopoly restricts output to raise the price, and monopsony restricts hiring to lower the wage. The static distortion implied by each wedge is textbook material but worth stating. A monopolist facing a demand elasticity η restricts output until $p = \frac{\eta}{\eta-1}c$; a monopsonist facing a labour-supply elasticity ε_L pays $w = \frac{\varepsilon_L}{1+\varepsilon_L}\text{MRPL}$ and hires fewer workers than it otherwise would. How many fewer depends on how quickly the marginal revenue product falls as the plant expands, which is governed by exactly the labour elasticity θ^L that this paper estimates. Section 4.4 derives this restriction and evaluates it at our estimates.

The two wedges are also difficult to separate in production data. As Section 4 shows,

the object that the production approach recovers from labour alone is the *product* of the two wedges. Credible measurement therefore requires both to be estimated at once. Providing that joint measurement for India’s organised manufacturing plants over 2012–2022 is the task of this paper.

The question matters for India with particular force, for three reasons grounded in theory and in the country’s policy debate. First, the welfare economics of the two wedges differ. A common markup is a transfer with modest Harberger losses, but *dispersed* markups misallocate inputs across plants and reduce aggregate productivity (Baqaee & Farhi, 2020; Edmond et al., 2023), and India is the canonical high-dispersion economy (Hsieh & Klenow, 2009). Whether Indian markups are large, and above all whether they are dispersed, is therefore an input into the country’s productivity accounting, which our companion paper takes up. Second, the labour-market face bears on India’s central development challenge: manufacturing wages and the labour share. If plants systematically pay less than the marginal revenue product of labour, the gains from industrial growth accrue disproportionately to profits, and policies that raise labour demand or worker mobility yield first-order efficiency gains, not merely distributional ones (Manning, 2003; Berger et al., 2022). Third, the international context has raised the stakes: the documented rise of markups in the United States (De Loecker et al., 2020), together with evidence that markdowns are substantial (Yeh et al., 2022) and may account for part of what is measured as rising markups (Ren & Zhang, 2025), raises the question of where a large developing manufacturing sector stands on both margins.

Stated as questions, the paper asks four things. *Q1 (product market)*: how large are markups in Indian organised manufacturing, how dispersed are they across plants, and how did they move over 2012–2022—and through which margin, within-plant pricing or the reallocation of sales across plants? *Q2 (labour market)*: what fraction of plants pay labour below its marginal revenue product, how large is the implied wedge, and what labour-supply elasticity does it imply that plants face? *Q3 (geography)*: do the two wedges share a geography—across states, and across the rural–urban divide within states—or do product-market and labour-market power arise in different places? *Q4 (composition or place)*: where the wedges differ across states, is that because different states host different industries, or because the same industry behaves differently in different states? The identifying lever for the first three questions is the same. Cost minimisation ties output elasticities to revenue shares, input by input, so a flexible input priced without frictions (materials) reveals the product-market wedge, and the *gap* between the labour-based and materials-based versions of the same ratio reveals the labour-market wedge. Everything else in the paper is the careful implementation of that lever. Q4 needs no further identification, because it is a decomposition of the answer to Q3.

Answering these questions raises five well-known challenges, which shape our design. *(i) Identifying the elasticities.* The markup formula $\mu = \theta^X / \alpha^X$ (the output elasticity of

a flexible input over its revenue share) is only as good as the output elasticity, and input choices respond to unobserved productivity—the classic simultaneity problem (Marschak & Andrews, 1944). We estimate gross-output production functions separately for each two-digit industry using the control-function approach (Olley & Pakes, 1996; Levinsohn & Petrin, 2003), implementing the correction of Akerberg et al. (2015), whose two-stage moment structure is set out in equations (11)–(13).

(ii) *Restrictive technology.* Cobb–Douglas imposes a single elasticity per industry, assuming away the plant-level heterogeneity in market power that is the object of interest. Our headline technology is therefore the translog, with plant-specific elasticities (10), estimated by a two-step GMM procedure that we implement directly.

(iii) *Revenue data.* Like nearly all plant surveys, the ASI records revenues, not quantities. The estimated objects are therefore revenue elasticities, and markup *levels* are subject to the caveats of Bond et al. (2021) and Raval (2023). We respond in three ways: our questions emphasise dispersion, trends, incidence and geography, which are robust to level scalings; we cross-validate two estimators and report where they disagree; and we state the Hicks-neutrality assumption explicitly, together with the direction in which it would bias each wedge if it were violated (Section 8), following Demirer (2025).

(iv) *Inference on constructed objects.* Both wedges are *constructed* from estimated elasticities, so a regression with $\ln \mu$ or $\ln \psi$ on the left-hand side that reports conventional standard errors treats estimates as data. We therefore bootstrap the whole pipeline—resampling plants and re-running the first-stage polynomial, the instruments, the GMM, the share correction and the winsorisation within every replication—and report the resulting standard errors in place of, not alongside, the conventional ones.

(v) *Identifying the markdown.* The markdown is the *ratio* of two elasticity-to-share ratios (14). It requires both labour and materials to be flexible, and it is sensitive to anomalies in reported factor shares. We document two industries, printing and other transport equipment, in which the value added implied by reported shares is negative for 36.4 and 14.9 per cent of plants respectively. Printing, where the problem is severe and specific to the industry, is excluded from the markdown aggregates (Section 4.6); other transport equipment, whose rate is close to the all-division average, is retained.

We apply these methods to a newly constructed panel. From the raw block-level ASI files with common factory identifiers, we build a factory-year panel of 935,975 observations (2008–2022), deflate it to 2011–12 prices using industry-matched WPI series, and retain for estimation the 423,127 plant-years (on 84,298 factories in divisions NIC 10–33) that are operating, have positive real output, materials, labour and capital, and appear in at least two consecutive rounds, as the ACF moment conditions require. Appendix A documents the construction. Every number in the paper can be traced to a saved result file and the logged script that produced it; where a component of the pipeline did not run cleanly, or where a check failed, this is reported at the point of use rather than folded

into a global claim of verification.

Five sets of findings emerge. *First, product-market power is modest and rising only slightly.* The plant-level median markup is about 1.22, with industry medians ranging from 1.00 to 1.72 and the middle half of divisions lying within 1.17–1.33, and it is close to flat over the decade. The sales-weighted aggregate rises from 1.120 in 2012 to 1.171 in 2022, an increase of 4.6 per cent, and a Melitz & Polanec (2015) decomposition assigns two-thirds of it to the reallocation of sales towards plants with high markups and only a third to the plants’ own pricing. An earlier draft of this paper reported a falling markup; Section 5.2 and Appendix L show that the decline came from a trend in the first-stage residual and disappears once the first stage includes year effects. Concentration measures point the same way: the mean industry employment HHI fell from 0.086 to 0.066, and the four-firm concentration ratio from 0.39 to 0.32.

Second, monopsony is concentrated. The markdown exceeds one for 44.0 per cent of plants unweighted, and for 48.4 per cent once the ASI’s sampling multipliers are applied (40.2–49.6 per cent year by year, unweighted, excluding NIC18 throughout). The median plant’s ψ of 0.82, however, lies below one, so this is a large minority of plants, not the typical plant. Among the plants that do mark wages down, the median markdown of 2.03 implies a labour-supply elasticity, $\varepsilon_L = 1/(\psi - 1)$, close to one, at the low end of the range of international estimates. Across the aggregation concepts that lie in the region where this conversion is numerically stable, the implied elasticity ranges from 0.35 to 1.64, a sensitivity we document rather than conceal.

Third, the two margins have opposite geographies. Median markups differ by about 0.05 across India’s ten largest manufacturing states, but monopsony incidence ranges from 0.31 to 0.67 across the 21 states large enough to be ranked, peaks in the central-eastern mineral belt, and is higher among rural than among urban plants in *every* year (46.4 against 41.9 per cent, pooled), with rural–urban gaps approaching twenty percentage points in West Bengal (+19.5 points) and Uttar Pradesh (+17.3 points). The sign of that gap is stable—positive in all eleven years—but its magnitude is not: in the bootstrap of the earlier specification the paired interval for the pooled gap ran from -1.8 to $+7.2$ points and included zero, so we report the rural gradient as a consistent pattern rather than as a precisely measured difference. The gap reverses in eight of the twenty-seven states with adequate data, materially in six—Bihar (-31.4 points), Nagaland (-22.4), Tripura (-18.1), Assam (-12.7), Goa (-8.9) and Punjab (-7.0), the Nagaland and Tripura figures resting on small urban samples—and no single explanation covers them all. Figures 8 and 2 map both facts.

Fourth, the geography of monopsony reflects the state, not the industry mix. The gap in incidence between the five highest- and five lowest-monopsony states is 22.4 percentage points, and decomposing it into a composition term and a within-industry term assigns 73.9 per cent to the latter: twenty-one of the twenty-two divisions common to both

groups have higher incidence in the high-incidence states. The gap therefore reflects the same industry behaving differently across state boundaries, not a different portfolio of industries, which moves the policy question from which sectors a state hosts to how its labour markets work. Section 7 develops this result, together with the related finding that monopsony is systematically higher where organised manufacturing is smaller. The industry dimension tells the same story from the other side. Only 31.5 per cent of the plant-level variance of the log markdown lies *between* the twenty-two divisions, and 68.5 per cent within them, so knowing a plant’s industry reveals under a third of what one would want to know about its labour-market power. Across divisions, the two margins are also negatively related: the correlation between a division’s median markup and its monopsony incidence is -0.53 , and the rank correlation -0.52 . Under the baseline measure of labour, then, the industries with the most pricing power are not those in which workers are furthest below the marginal revenue product of their labour, although, as Section 8.8 shows, division-level orderings are sensitive to how labour is measured. Section 6 describes the divisions and develops both results.

Section 8 sets out the identifying assumptions on which these findings rest—revenue rather than quantity data, and Hicks-neutral productivity—together with the direction in which each would bias the wedges if violated, and the evidence that would be required to relax them.

The paper contributes to three literatures. To the literature on *markup measurement*, it adds a large, recent panel for a developing economy, estimated under the field’s preferred technology with full cross-estimator diagnostics, together with an explicit statement of the direction in which the Hicks-neutrality assumption biases each wedge on the same plants. To the *monopsony* literature, it adds the first economy-wide geography of markdowns for India, by state and by rural–urban location, and evidence that the choice of aggregation concept, rather than the estimation itself, can account for much of the wide range of published labour-supply elasticities. To the literature on *Indian industrial development*, it offers a revised picture: after two decades in which evidence on the liberalisation of the 1990s dominated the field (De Loecker et al., 2016), our evidence indicates that in the 2010s and early 2020s the binding margin of market power in Indian factories is not the price of goods but the price of labour.

A fourth contribution is methodological and, we believe, extends beyond this paper. The production approach recovers the markup from a materials elasticity close to one and the markdown from a labour elasticity of about 0.1, jointly, from a just-identified system whose objective is nearly flat around its optimum. We show that this makes the two wedges very unequally determined. Across four perturbations that change only the numerical path to the solution—leaving the data, the sample, the moment conditions and the starting values provably identical—the materials elasticity is reproduced to within a few hundredths of a per cent, while division-level labour elasticities move by up to 84 per

cent and division-level markdowns by up to 82 per cent (diagnostics computed under the earlier first stage; Section 8.2). Aggregates survive, because the displacements are roughly symmetric and cancel; the markdowns of individual industries do not. Section 8.2 documents this, and it is the reason why we pair the headline statistics with a plant-clustered bootstrap that re-runs the entire pipeline, first stage included, rather than reporting the analytic standard errors of a regression that treats the estimated wedges as data. Those analytic standard errors are not merely optimistic; they answer a different question. We know of no other application of the production approach that quantifies this, and the point applies to any study that computes a markdown as a ratio of elasticity-to-share ratios.

Table 1 collects the headline numbers for reference throughout.

Table 1: Findings at a glance (pooled 2012–2022 unless noted).

Object	Estimate
Estimation sample	423,127 plant-years; 84,298 factories; 23 industries
Manufacturing frame (before the panel-length rule)	481,322 plant-years; 142,493 factories
Translog estimation sample (NIC33 excluded)	422,820 plant-years; 23 divisions
Median markup μ (translog, aggregate)	≈ 1.22 ; industry medians 1.00–1.72
Sales-weighted aggregate markup Ω	1.120 (2012) $\rightarrow$ 1.171 (2022); two-thirds reallocation
Monopsony incidence $\Pr(\psi > 1)$	44.0% pooled unweighted, 48.4% survey-weighted; 40.2–49.6% by year (excl. NIC18)
Pooled median ψ	0.82 — below one
Typical-plant implied ε_L	≈ 1.6 ; concept range [0.35, 1.64]
Employment HHI (NIC-4 mean)	0.086 (2012) $\rightarrow$ 0.066 (2022)
Rural vs urban monopsony incidence	46.4% vs 41.9%, every year
Across-state spread of incidence	0.31–0.67 (21 ranked states)
Contract share of the workforce	37.7% (2012) $\rightarrow$ 45.7% (2022), aggregate
Contract wage penalty	0.751 of the direct wage per head
Markdown on contract share	+0.018 (conventional s.e. 0.004) — under 2% across the range
Largest state rural–urban gaps	West Bengal +19.5pp; Uttar Pradesh +17.3pp
Monopsony incidence, highest vs lowest state	Chhattisgarh 66.9% vs Tamil Nadu 31.0%
Within-industry share of the inter-state gap	73.9% (composition 26.1%)
Between-division share of plant variance	$\ln \mu$ 42.4%; $\ln \psi$ 31.5%
Markup vs monopsony across divisions	$r = -0.53$; Spearman -0.52
Product DWL / labour DWL (sample aggregates)	3.7–13.6% of sales / 2.6–3.4% of wage bill

Section 2 situates the paper in the literature. Section 3 describes the institutional setting, the panel and every variable. Section 4 develops the empirical framework, equation by equation. Section 5 presents the results, Section 6 the inter-industry analysis and

Section 7 the inter-state analysis. Section 8 covers robustness and measurement, Section 9 extends the analysis to contract labour, Section 10 presents the welfare accounting, and Section 11 concludes. Twelve appendices document the panel construction (A), the year-by-year supporting series (B), the industry-level estimates (C), the Cobb–Douglas benchmark (D), the estimation procedure (E) and its convergence diagnostics (F), the state and state $\times$ industry detail (G), the data-quality checks (H), the division bias in the size gradient (I), the algebra of the aggregation concepts (J), a division-by-division comparison of the estimates obtained with the finite-difference and analytic-gradient optimisers (K), and the comparison of the first stage with and without year effects (L).

2 Related literature

A large and growing literature measures market power in both product and factor markets, and it falls into two strands. The first measures the power that firms hold over the prices they charge in product markets; the second measures the power they hold over the wages they pay in labour markets. The two share a common algebra, with only minor differences in the equations, rely on the same kind of data and confront the same identification problem, yet they are rarely estimated on the same plants. As a result, a good deal is known about each wedge in isolation, but very little about how the two compare within the same firms. This section reviews what each literature has established, shows where the two meet, and identifies the question that remains open for India. It moves from the common theory (Section 2.1), to the measurement of markups (Sections 2.2 and 2.3) and of markdowns (Sections 2.4 and 2.5), and closes with the Indian evidence and the contribution of this paper (Section 2.6).

2.1 Basic theory of market power

We begin with the theory, because it explains why a single estimator can recover both forms of market power. In the product market, market power is conventionally summarised by the Lerner index, in its textbook definition: the share of the price that is not accounted for by marginal cost. It is a monotonic transformation of the markup:

$$\mathcal{L} \equiv \frac{P - c}{P} = 1 - \frac{1}{\mu}, \quad \mu \equiv \frac{P}{c} \geq 1. \quad (1)$$

Here P is the price the firm charges, c the marginal cost the firm incurs, μ the markup and $\mathcal{L}$ the Lerner index. A price-taking firm sets $P = c$, so that $\mu = 1$ and $\mathcal{L} = 0$. For a firm that sets its price optimally, the first-order condition links the Lerner index to the price elasticity of demand, $\mathcal{L} = 1/\varepsilon_D$. The markup is therefore, at root, a statement about how insensitive the firm’s customers are to its price: the less elastic the demand it faces, the larger the gap between price and marginal cost.

The labour market has the same structure, except that the friction lies on the supply side rather than the demand side. Suppose the firm faces an upward-sloping labour-supply curve with elasticity ε_L . To hire an additional worker it must raise the wage, and the higher wage is paid to every worker it already employs. The marginal cost of labour therefore exceeds the wage itself. Equating this marginal cost to the marginal revenue product of labour gives

$$\psi \equiv \frac{\text{MRPL}}{w} = 1 + \frac{1}{\varepsilon_L}, \quad (2)$$

where MRPL is the marginal revenue product of labour, w the wage and ψ the markdown. In a competitive labour market $\varepsilon_L \rightarrow \infty$ and $\psi = 1$, so that workers are paid exactly what they add to revenue. Any finite elasticity opens a wedge between what a worker produces and what the worker is paid, and the less elastic labour supply is, the wider that wedge becomes. The parallel with (1) is close: each wedge is governed by a single elasticity—the markup by the elasticity of product demand, since $\mu = \varepsilon_D/(\varepsilon_D - 1)$, and the markdown by the elasticity of labour supply. [Robinson \(1933\)](#) was the first to recognise this symmetry. It matters for everything that follows, because it is the reason the same empirical framework can recover both wedges.

2.2 Measuring the markup with production data

The symmetry is useful only if the wedges can be measured, and for the markup the decisive step was to stop trying to estimate demand at all. [Hall \(1988\)](#) observed that cost minimisation, for any input N bought in a competitive market, implies an identity between that input’s share in revenue and its output elasticity:

$$\alpha_N \equiv \frac{wN}{PQ} = \frac{\theta_N}{\mu}, \quad \theta_N \equiv \frac{\partial \ln Q}{\partial \ln N}, \quad (3)$$

where α_N is the observed expenditure on input N as a share of revenue, w here denotes the price of that input, θ_N is its output elasticity and Q is output. The economic content of the identity is simple. A firm with market power spends a *smaller* share of its revenue on each input than its technology alone would warrant, and the ratio between the two is exactly μ . Observing the revenue share and estimating the output elasticity is therefore enough to recover the markup. Hall identified μ from time-series covariation in industry data, under the assumption of constant returns to scale.

[De Loecker & Warzynski \(2012\)](#) removed both of these restrictions—constant returns and aggregation to the industry level—by applying the same identity plant by plant to any input that is free of adjustment frictions:

$$\mu_{it} = \frac{\theta_{it}^V}{\alpha_{it}^V}, \quad (4)$$

where V indexes the flexible input, θ_{it}^V is plant i 's output elasticity with respect to that input in year t , and α_{it}^V is the input's share in revenue. Two features have made this approach the workhorse of the recent revival of research on market power (De Loecker et al., 2020; Syverson, 2019). First, it requires no demand-side data and no assumption about how many firms compete or how they compete. Second, it yields a markup for every plant-year rather than a single industry average, so that the whole distribution of market power, and not only its mean, can be studied. De Loecker & Syverson (2021) provide an authoritative account of what the approach can and cannot deliver.

2.3 Determining the output elasticity

Equation (4) is an identity, so in empirical work its entire content lies in the output elasticity: the revenue share α^V is observed directly, and every contestable choice concerns how θ^V is estimated. Estimating it runs into two identification problems first set out by Marschak & Andrews (1944). Firms choose their inputs in response to productivity shocks that the researcher cannot directly observe, which biases the estimates; and unproductive plants exit, which leaves a selected sample of survivors. The subsequent literature is best read as a sequence of solutions to these two problems, each of which exposed a new difficulty.

Olley & Pakes (1996) invert the investment demand function to obtain a proxy for unobserved productivity, and model exit explicitly. Because investment is lumpy and frequently zero in plant data, Levinsohn & Petrin (2003) use intermediate inputs as the proxy instead. Akerberg et al. (2015) then show that both approaches suffer from a functional-dependence problem: if labour is chosen at the same time, and on the same information, as the proxy variable, its coefficient cannot be identified in the first stage. Their solution is to identify no coefficients in the first stage and to recover all of them in the second, from moment conditions on the innovation to productivity. This is the estimator used in this paper. Wooldridge (2009) shows that the same moments can be imposed in a single GMM step, which is more efficient but less transparent about which moment identifies which parameter. Gandhi et al. (2020) take a different route altogether, showing that the share equation of the flexible input identifies its output elasticity non-parametrically, without any timing assumption.

The critiques of this toolkit matter as much as the methods themselves, and several of them bear directly on the estimates reported here. Bond et al. (2021) show that when only revenue is observed—as in the ASI, and in almost every plant survey—the estimated elasticity is a revenue elasticity, and the markup recovered from it is not, in general, identified. Raval (2023) proposes a test of internal consistency: markups estimated from different flexible inputs should coincide, but in practice they frequently do not. Hashemi et al. (2022) argue that what the production approach often measures is

an input distortion rather than a markup. [De Ridder et al. \(2026\)](#) assess how poorly the approach performs when it is applied to financial rather than production data. Finally, [Demirer \(2025\)](#) and [Rubens et al. \(2026\)](#) show that imposing Hicks-neutral technology when productivity is in fact factor-augmenting inflates the estimated elasticities of flexible inputs and depresses that of labour; as [Section 8.7](#) explains, this biases markups upward and markdowns downward. This paper does not claim to resolve these critiques. Instead, it states which of them affect which of its claims and, for that reason, places more weight on comparisons than on levels.

2.4 From the markup to the markdown

The discussion so far has assumed that the firm has power only in the product market. If it also has power in the labour market, applying (4) to labour returns a contaminated object, because labour’s first-order condition now contains two wedges rather than one. [Brooks et al. \(2021\)](#) resolve this problem by applying the identity to two flexible inputs and taking the ratio. The product-market markup, which is common to both inputs, cancels, and only the labour-market wedge remains:

$$\psi_{it} = \frac{\theta_{it}^L / \alpha_{it}^L}{\theta_{it}^M / \alpha_{it}^M}, \quad (5)$$

where L denotes labour and M materials, and θ and α are, as before, the output elasticity and the revenue share. The strategy rests on one requirement: the firm must have no buying power over materials, so that materials are purchased at a competitive price. This is the maintained assumption of this literature, and of this paper.

[Ren & Zhang \(2025\)](#) sharpen the same warning from the opposite direction. If the input used to estimate the markup is itself subject to wage-setting power—as with a cost-of-goods-sold aggregate that bundles labour together with other inputs—then

$$\ln \mu_{it}^{\text{measured}} = \ln \mu_{it} + \ln \psi_{it}, \quad (6)$$

where $\mu_{it}^{\text{measured}}$ is what the estimator returns, μ_{it} the true product-market markup and ψ_{it} the labour markdown of (2). The estimator then recovers the *product* of the two wedges rather than the markup alone, so a literature that reports rising markups may, in part, be reporting rising markdowns. This is a direct argument for estimating the two wedges jointly, from inputs that are kept separate, which is the approach taken in this paper. [Mertens \(2022\)](#) makes the same case empirically for Germany, finding that markdowns rather than markups account for much of the decline in the labour share there.

2.5 What is known about labour-market power

Having set out how a markdown can be measured, we now turn to what the monopsony literature has learned about its sources and its size, since this provides the benchmark against which the estimates in this paper should be judged. The modern view of monopsony does not rest on employer concentration. Search frictions and workers’ idiosyncratic preferences over employers give each firm an upward-sloping labour-supply curve even when there are many employers (Manning, 2003; Card et al., 2018). Manning (2021) shows that the firm’s labour-supply elasticity can be decomposed into a recruitment margin and a separation margin, and that under the search structure of Burdett & Mortensen (1998) the two are linked:

$$\varepsilon_L = \varepsilon_R(w) - \varepsilon_s(w) \approx -2\varepsilon_s(w), \quad (7)$$

where ε_R is the elasticity of recruitment and ε_s the elasticity of separations with respect to the wage. This relationship is useful here because it turns the separation elasticity, which can be estimated from worker flows, into an external check on any ε_L implied by a production-side markdown.

Three families of measures coexist in this literature. *Direct* estimates exploit quasi-experimental variation in wages (Card et al., 2018; Lamadon et al., 2022) and typically find firm-level labour-supply elasticities between 2 and 5. Sokolova & Sørensen (2021) meta-analyse 53 studies and report a mean close to 2 once publication bias is taken into account; this is the benchmark against which the elasticities implied by this paper’s markdowns should be read. *Concentration-based* measures regress wages on the employer Herfindahl–Hirschman index (HHI) (Azar et al., 2022; Berger et al., 2022), but they require a definition of the local labour market, which the ASI does not permit. *Production-approach* measures, based on (5), require only production data. This is why they can be applied in settings that the first two families cannot reach, and why they are the appropriate tool here. Yeh et al. (2022) apply the approach to US manufacturing, Morlacco (2020) to French importers and Amodio & de Roux (2023) to Colombian plants.

2.6 The Indian evidence, and what it leaves open

Against this international background, the Indian evidence is strikingly uneven: there is a substantial literature on markups but almost none on markdowns. On the product-market side, Balakrishnan et al. (2006) document the links between market power and productivity growth after liberalisation; Sivadasan (2009) finds that entry deregulation raised productivity; Krishna & Mitra (1998) and Goldberg et al. (2010) study price–cost margins and the effects of input tariffs; Topalova & Khandelwal (2011) identify the productivity gains from trade reform; and Mukherjee & Chanda (2021) estimate firm-level

markups under tariff liberalisation. The labour side is far thinner. Aggregate studies of the labour share (RBI, 2024; Basole & Narayan, 2020) document its decline without attributing it to a cause, and Dasgupta (2006) is an early structural attempt to measure market imperfection in Indian manufacturing directly, rather than to infer it as a residual.

The gap, then, is not a shortage of evidence on either wedge in isolation, but the absence of evidence on both together. Apart from Brooks et al. (2021), who include Indian plants in a cross-country comparison, no study has measured both wedges on the same Indian plants with the same estimator over the past decade. This is the gap that this paper fills, and it is what makes the comparison in Table 2 possible. Until both objects are estimated together, it is impossible to say which margin is larger, whether the two move together, or even whether they arise in the same places.

The paper makes four contributions, each of which follows from this gap. First, and most directly, it provides the measurement itself: markups and markdowns for the same 84,298 Indian factories, estimated from a single translog production function over a decade. Second, it makes possible a comparison that no pair of separate studies could have delivered: under the baseline measure of labour, the two margins rank industries in *opposite* order, and they share no common geography. Third, it extends the analysis to the category of labour on which the Indian policy debate actually turns—contract workers—recovered from survey blocks that the standard panel discards. Fourth, and methodologically, it shows that the two wedges are *not* equally well identified by this estimator. The markup rests on an output elasticity close to unity, whereas the markdown rests on one close to 0.10; the same estimation error is therefore negligible for the first but can overwhelm the second. Section 8.3 makes this argument precise using the geometry of the moment Jacobian. We believe that this asymmetry extends beyond the present paper, and that reporting it should be treated as part of reporting the estimates themselves.

3 Institutions, data, and variables

This section describes the data on which every estimate in the paper rests. We first introduce the Annual Survey of Industries (ASI) and the factory identifier that allows plants to be followed over time, and then explain how the panel is built and how each variable is defined. We next place the sample period in its macroeconomic context, and close by describing the two cost shares that enter every object estimated later, together with the coverage of the final sample.

Table 2: Where this paper sits. Columns are the choices that determine whether two estimates are comparable at all: the data source, whether output is observed as revenue or as physical quantity, and which input is treated as free of adjustment frictions — the markup in (4) is only as good as that choice. “Joint” means the markup and the markdown are estimated on the same units from the same production function, so that the two can be compared rather than juxtaposed. Cells marked n.r. are not reported in a form we can state confidently. PF = production function; CF = control function; COGS = cost of goods sold.

Study	Data source	Period	Output measure	Flexible input(s)	Elasticity method	Mark-up	Mark-down	Key finding
<i>Panel A. The methodological lineage</i>								
Hall (1988)	US industry accounts	1953–84	Revenue	Labour (aggregate)	Time series, CRS imposed	✓	—	Price exceeds marginal cost; the Solow residual co-moves with input growth
De Loecker & Warzynski (2012)	Slovenian firm census	1994–2000	Revenue	Materials	CF (LP, then ACF)	✓	—	Firm-level markups; exporters charge more than non-exporters
De Loecker et al. (2020)	US Compustat	1955–2016	Revenue	COGS	CF on a single aggregate input	✓	—	Aggregate markup rises 1.21 to 1.61; concentrated in the upper tail
Gandhi et al. (2020)	Colombia, Chile manufacturing	1980s–90s	Revenue	Intermediates	Share equation, non-parametric	✓	—	The flexible input’s share equation identifies its elasticity without timing assumptions
Bond et al. (2021)	Analytical	—	Revenue	—	Critique of the above	not identified	—	With revenue data the markup is not generally identified
Ren & Zhang (2025)	Analytical, US	—	Revenue	Contaminated	Analytical	$\mu\psi$	—	If the flexible input carries wage-setting power the estimate is the product of both wedges
<i>Panel B. Studies that estimate both wedges</i>								
Brooks et al. (2021)	China ASIF; India ASI	1998–2007	Revenue	Materials and labour	CF, two-input ratio	✓	✓	Markdowns substantial in both countries; the two-input ratio identifies them
Yeh et al. (2022)	US Census of Manufactures	1976–2014	Revenue	Materials and labour	CF, two-input ratio	✓	✓	Markdown ≈ 1.53 ; monopsony widespread in US manufacturing
Morlacco (2020)	French customs and firm data	1996–2007	Revenue	Imported inputs	CF, input-market wedge	—	✓	Buyer power over foreign input suppliers
Mertens (2022)	German manufacturing (AFiD)	1995–2014	Revenue	Materials and labour	CF, two-input ratio	✓	✓	Markdowns, not markups, account for much of the falling labour share
Amodio & de Roux (2023)	Colombian plant survey	1982–2012	Revenue	Materials and labour	CF, two-input ratio	✓	✓	Monopsony in a developing-country setting
<i>Panel C. The Indian evidence</i>								
Dasgupta (2006)	India, ASI	1980s–90s	Revenue	n.r.	Structural, not residual	✓	—	Market imperfection measured directly rather than inferred
Sivadasan (2009)	India, ASI	1980s–90s	Revenue	—	Productivity, not markup	—	—	Entry deregulation and FDI reform raised plant productivity
Balakrishnan et al. (2006)	India, ASI industry data	1988–98	Revenue	n.r.	Price–cost margin	✓	—	Liberalisation, market power and productivity growth
Mukherjee & Chanda (2021)	India, firm panel	1998–2012	Revenue	Materials	CF	✓	—	Input-tariff liberalisation lowers firm markups
This paper	India, ASI plant panel (common factory id)	2012–22	Revenue (ex-factory)	Materials and labour	Translog, ACF two-step	✓	✓	Joint measurement on 84,298 plants: markups modest and rising slightly through reallocation, markdowns concentrated, the two with opposite geographies and opposite industry rankings

3.1 The Annual Survey of Industries

The ASI is the census-cum-sample survey of India’s registered factories, that is, factories registered under the Factories Act, 1948: those employing ten or more workers with the use of power, or twenty or more without it. It is conducted under the Collection of Statistics Act. Larger units, including all units with 100 or more employees, are enumerated every year under the census scheme; the remaining units are sampled with known probabilities, and the survey supplies the corresponding inverse-probability multipliers w_{it} (‘wt’). The unit of observation is the factory (establishment).

We use the unit-level data distributed with *common factory identifiers*. Each identifier is an eight-character code that remains constant for a plant across years, which is what allows the same factory to be followed through time. We verified the structure of this code against the official record layouts: characters 1–5 are a serial number, characters 6–7 give the 2011-census *state* code, and character 8 is a scheme character. The state coding is also confirmed empirically, since the most frequent codes correspond, in order, to Tamil Nadu, Maharashtra, Gujarat and Uttar Pradesh. In addition, Block A records whether each factory is located in a rural or an urban area (item A9); this is the information on which the geographical analysis of the paper draws.

3.2 Panel construction in brief

Each survey round is distributed as ten fixed-width block files (A–J), and the positions of fields within these files vary across six format eras. We parsed every file against a column dictionary built from the official layout records. Blocks that contain several items per factory—fixed assets, employment, indigenous and imported materials, and products—are collapsed to factory totals; the blocks are then merged one-to-one on the factory identifier, and the rounds are appended. The result is a file of 935,975 factory-years covering 2008–2022, in which each factory×year combination has been verified to be unique.

Three further steps lead from this file to the estimation sample. First, we retain operating units, defined as those reporting positive output, materials, labour, capital and survey multiplier. Second, because the 2011-12-base deflators are available only from April 2012, the real panel begins in 2012. Together, these two steps yield a 2012–2022 real panel of 495,148 plant-years. Third, within manufacturing divisions NIC 10–33, we require each factory to be observed in at least two consecutive rounds, because the ACF moment conditions need a lagged observation. This panel-length restriction reduces the manufacturing frame of 481,322 plant-years on 142,493 factories—the figures reported in Table 1—to an estimation panel of 423,127 plant-years on 84,298 factories. The translog estimates rest on somewhat smaller subsets of this panel, which are defined precisely at the end of this section. Appendix A documents the full construction chain and its validation.

3.3 Variable definitions

Table 3 defines every variable used in the analysis. Two conventions deserve emphasis, because they determine how the theory is taken to the data. First, revenue shares are measured in *nominal* terms, $\alpha^M = p^M M / (pY)$ and $\alpha^L = wL / (pY)$, with both numerator and denominator at current prices. The shares in the theory are expenditure shares—what the plant pays for an input relative to what it earns from its sales—so they are measured at the prices the plant actually faces, and no deflator enters them. Second, the inputs and output of the production function are measured in *real* terms, at 2011-12 base prices. Output and materials are deflated with industry-matched wholesale price indices (WPI), with documented fallbacks for petroleum and repair; capital is deflated with the machinery WPI; and labour enters as a headcount.

Table 3: Variable definitions

Symbol	Definition	ASI source (blocks)
Y_{it}	Real gross output	Ex-factory value of products (J), deflated by NIC-2 WPI
M_{it}	Real materials	Indigenous (H) + imported (I) inputs, deflated by NIC-2 WPI
L_{it}	Labour	Total persons engaged (E)
K_{it}	Real capital	Net fixed assets (C), deflated by machinery WPI
wL_{it}	Wage bill (nominal)	Wages + bonus + PF + welfare (E; A for 2016+ formats)
$p^M M_{it}$	Materials bill (nominal)	Purchase value totals (H+I)
$\alpha_{it}^M, \alpha_{it}^L$	Nominal revenue shares	Materials (wage) bill over nominal output
w_{it} (‘wt’)	Survey multiplier	Frame (A); stored always, applied in aggregation only
$state_{it}$	2011-census state code	Characters 6–7 of the factory identifier (A)
$rural_{it}$	Rural (1) / urban (2)	Item A9 (A), all rounds
NIC2/3/4/5	Industry (NIC-2008)	Activity code (A); no cross-classification concordance needed in-window

3.4 The macro backdrop, 2012–2022

The decade covered by the panel was one of unusual macroeconomic variation, and any trend in market power has to be read against it. Trade openness fell from 55.8 per cent of GDP in 2012 to 37.8 per cent in 2020, before recovering to 50.7 per cent. Applied tariffs on manufactured goods, after two decades of decline, turned upward after 2018. Two major reforms—the Insolvency and Bankruptcy Code (2016) and the Goods and Services Tax (2017)—reshaped formalisation and the exit of firms. Finally, the pandemic produced a

year in which output fell by 5.8 per cent, followed by a rebound of 9.7 per cent. Whether these forces *cause* movements in market power is the subject of a companion paper on its determinants. Here, they serve as a caution against reading too much into any single year. This is why the statements in this paper refer to the decade as a whole, and why the welfare series is reported annually.

3.5 Cost shares at a glance

Two revenue shares enter every object estimated in this paper, so their distribution deserves attention before any estimation. Both are widely dispersed and heavy-tailed (Table 4; statistics refer to the manufacturing estimation sample). The materials share α^M has a median of 0.756, with a 10th–90th percentile range of 0.462 to 0.959; the labour share α^L has a median of 0.098, with a range of 0.023 to 0.347. Two features of these distributions matter for what follows. The first is the dominance of materials in costs, which, as Section 4.1 shows, stabilises the measurement of markups. The second is a pathological upper tail: before cleaning, 69,031 observations report $\alpha^L + \alpha^M > 1$, that is, spending on labour and materials alone that exceeds revenue. The problem is most severe in printing (NIC 18), which is excluded from the markdown aggregates for this reason (Section 4.6).

Table 4: Nominal revenue shares, estimation sample.

	p10	median	p90
Materials share α^M	0.462	0.756	0.959
Labour share α^L	0.023	0.098	0.347

3.6 Sample coverage

Coverage is stable over time, at between 33,000 and 41,000 estimation plants per round: 33,199 in 2012, rising to 40,905 in 2021, and 39,078 in 2022 (Table 16 in Appendix B). Rural plants account for 41.8 per cent of the panel in 2012 and 45.6 per cent in 2022, and for 44.0 per cent over the pooled period. Measured by plant-years, the largest divisions are food products (62,535), non-metallic minerals (39,980), textiles (29,560), chemicals (28,962), machinery (27,019), fabricated metals (22,969) and rubber and plastics (22,780); Appendix Table 18 reports all 23 divisions.

Four sample counts must be kept distinct, and we set them out here because every subsequent table rests on one of them. First, the Cobb–Douglas/ACF sample consists of 423,127 plant-years. Second, the translog is estimated for 23 divisions, NIC33 being excluded for insufficient support in the second estimation stage; output elasticities, markups and markdowns are computed for the 422,820 plant-years that these 23 divisions contain.

Third, all 23 divisions pass the admissibility screen, so these 422,820 plant-years form the basis of every translog markup statistic reported below, while the markdown aggregates exclude NIC18 (Section 4.6). Fourth, the nine translog parameters themselves are identified only from the subset with a valid $t - 1$ observation, which comprises 290,023 plant-years, or 69 per cent of the sample on which the elasticities are subsequently evaluated. In food products, for example, the parameters are estimated from 43,332 plant-years and then applied to all 62,535. This is standard practice—the estimated technology is used to construct plant-specific elasticities wherever the inputs are observed—but it means that the effective estimation sample is materially smaller than the reported N , a distinction that is not visible in the N column of any table.

4 Empirical framework

This section explains how the two wedges are measured. We begin with the markup, which follows from cost minimisation alone (Section 4.1), and then describe the production technology (Section 4.2) and the estimator used to recover its parameters (Section 4.3). Section 4.4 applies the same logic to labour to obtain the markdown, and shows what the markdown implies for employment. A worked example using actual numbers from the data follows (Section 4.5), and the section closes with the conventions used throughout the estimation (Section 4.6).

4.1 The markup from cost minimisation

Let plant i produce $Y_{it} = F(K_{it}, L_{it}, M_{it}) e^{\omega_{it}}$, and suppose it chooses the flexible input M to minimise the cost of producing Y_{it} , taking the materials price p_{it}^M as given. The Lagrangian $\mathcal{L} = p_{it}^M M_{it} + \dots + \lambda_{it}(Y_{it} - F(\cdot)e^{\omega_{it}})$ yields the first-order condition $p_{it}^M = \lambda_{it} \partial Y_{it} / \partial M_{it}$, where the multiplier λ_{it} is marginal cost. Multiplying both sides by M_{it}/Y_{it} and dividing by the output price p_{it} gives

$$\underbrace{\frac{\partial Y_{it}}{\partial M_{it}} \frac{M_{it}}{Y_{it}}}_{\theta_{it}^M} = \underbrace{\frac{p_{it}}{\lambda_{it}}}_{\mu_{it}} \underbrace{\frac{p_{it}^M M_{it}}{p_{it} Y_{it}}}_{\alpha_{it}^M} \implies \boxed{\mu_{it} = \frac{\theta_{it}^M}{\alpha_{it}^M}} \quad (8)$$

(De Loecker & Warzynski, 2012). Here Y_{it} and M_{it} are plant i 's output and materials in year t , p_{it} and p_{it}^M are their prices, and λ_{it} is the Lagrange multiplier on the output constraint, which is marginal cost. Thus $\mu_{it} = p_{it}/\lambda_{it}$ is the ratio of price to marginal cost, θ_{it}^M is the output elasticity of materials, and α_{it}^M is the share of revenue spent on materials.

Everything in the paper runs through equation (8), so it is worth being clear about what it says. The output elasticity of materials tells us what share of its revenue a cost-

minimising plant *would* spend on materials if it priced at marginal cost; the observed share tells us what it *actually* spends. The ratio of the two is exactly the factor by which price exceeds marginal cost.

Nothing beyond cost minimisation is needed for this result. No assumption about demand, market structure or pricing conduct enters the derivation: the plant may be a monopolist, an oligopolist or a Bertrand competitor, and as long as it does not waste money on inputs, (8) holds. It is this independence from conduct that allows one formula to be applied across 23 very different industries.

The derivation does, however, rely on two conditions: materials are priced at their marginal cost of acquisition, and the plant can adjust them freely within the year. Adjustment costs, long-term contracts or buying power over suppliers would each drive a wedge between p^M and the true shadow price of materials, and so contaminate the level of μ . This is one reason why our emphasis falls on dispersion, trends and incidence rather than on levels. It is also precisely the kind of failure that, when it occurs in the labour market, becomes our opportunity to measure the markdown (Section 4.4).

Of the two ingredients in (8), the share α^M is read directly from the accounts. The entire econometric burden therefore falls on the elasticity θ^M , to which we now turn.

4.2 Technology: translog, and why

Estimating θ^M requires a model of the technology. With output and inputs in logs (y, k, l, m), we specify the translog gross-output technology

$$y_{it} = \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \beta_{kk} k_{it}^2 + \beta_{ll} l_{it}^2 + \beta_{mm} m_{it}^2 + \beta_{kl} k_{it} l_{it} + \beta_{km} k_{it} m_{it} + \beta_{lm} l_{it} m_{it} + \omega_{it} + \varepsilon_{it}, \quad (9)$$

where y_{it} , k_{it} , l_{it} and m_{it} are the natural logs of real gross output, capital, labour and materials, and the β coefficients describe the technology. The two unobserved terms play very different roles. ω_{it} is plant productivity, which the plant observes when it chooses its inputs but the econometrician never does. ε_{it} collects measurement error and shocks realised after inputs are set, which nobody observes in time to act on. This distinction lies at the heart of the identification problem discussed below: ω is what makes input choices endogenous, and ε is what the first stage is designed to remove. The equation is estimated separately for each NIC-2 industry, and yields the plant-specific elasticities

$$\theta_{it}^M = \beta_m + 2\beta_{mm}m_{it} + \beta_{km}k_{it} + \beta_{lm}l_{it}, \quad \theta_{it}^L = \beta_l + 2\beta_{ll}l_{it} + \beta_{kl}k_{it} + \beta_{lm}m_{it}, \quad (10)$$

with θ_{it}^K defined symmetrically.

The second-order coefficients are what give the translog its economic content. The coefficient β_{mm} allows the materials elasticity to vary with the plant's materials intensity, and the cross-terms β_{km} and β_{lm} allow it to vary with the plant's mix of capital and

labour. Two plants in the same industry but with different input mixes can therefore operate at genuinely different points of the technology, which is exactly the plant-level heterogeneity that (8) needs. Cobb–Douglas, by contrast, sets all second-order terms to zero and so forces every plant in an industry to share a single θ^M . Differences in markups across plants then simply mirror, inversely, differences in their materials shares. This assumes away the very variation that (8) is meant to measure, and passes every anomaly in reported shares straight into the markup.

We therefore treat the translog as the headline specification and Cobb–Douglas (Appendix D) as a diagnostic benchmark. Where the two agree, as they do on incidence, rankings and trends, we regard the finding as robust to the choice of technology. Where they disagree, as they do on dispersion and levels, the disagreement itself tells us how much of the measured heterogeneity is technological. The sample means of the translog elasticities are $\bar{\theta}^M = 0.905$, $\bar{\theta}^L = 0.102$ and $\bar{\theta}^K = 0.016$, implying returns to scale of about 1.02. This is the materials-heavy gross-output technology typical of plant-level data.

4.3 Identification: the ACF control function

Estimating (9) by OLS is inconsistent, because input choices respond to productivity that the econometrician does not observe ($\mathbb{E}[\omega_{it} | \text{inputs}] \neq 0$). The control-function approach deals with this by inverting an observable decision of the plant to obtain a proxy for ω . We use materials as the proxy (Levinsohn & Petrin, 2003), together with the timing assumptions of Akerberg et al. (2015): labour is chosen at the same time as, or before, materials, and capital is predetermined. Under these assumptions the first stage removes only measurement error:

$$y_{it} = \Phi_t(k_{it}, l_{it}, m_{it}) + \varepsilon_{it}, \quad \hat{\Phi}_{it} \text{ estimated by a high-order polynomial}, \quad (11)$$

so that, for any candidate β , implied productivity is $\omega_{it}(\beta) = \hat{\Phi}_{it} - f(k_{it}, l_{it}, m_{it}; \beta)$, where f is the translog aggregate in (9). Productivity is assumed to follow a first-order Markov process,

$$\omega_{it} = g(\omega_{it-1}) + \xi_{it}, \quad \mathbb{E}[\xi_{it} | \mathcal{I}_{it-1}] = 0, \quad (12)$$

with g approximated by a polynomial. The innovation $\xi_{it}(\beta)$ is then orthogonal to anything in the plant's $t - 1$ information set and to predetermined capital, which gives the second-stage GMM moment conditions

$$\mathbb{E}[\xi_{it}(\beta) \mathbf{z}_{it}] = 0, \quad \mathbf{z}_{it} = (k_{it}, k_{it}^2, l_{it-1}, l_{it-1}^2, m_{it-1}, m_{it-1}^2, k_{it}l_{it-1}, k_{it}m_{it-1}, l_{it-1}m_{it-1}), \quad (13)$$

in which current capital and *lagged* flexible inputs serve as instruments for the translog terms.

The markup estimates can be no more credible than this machinery, so we set out the economics behind each step of (11)–(13). Consider first why OLS fails, and in which direction. A plant that draws a high ω buys more materials, so OLS attributes part of productivity to materials and biases θ^M upward. Since $\mu = \theta^M/\alpha^M$, uncorrected simultaneity would manufacture markups out of productivity.

The first stage has a deliberately limited role. Under the scalar-unobservable and monotonicity conditions of Levinsohn & Petrin (2003), materials demand $m_{it} = m_t(k_{it}, l_{it}, \omega_{it})$ can be inverted for ω , so output conditional on (k, l, m) is purged of everything except the shock ε . For the same reason, however, no coefficient of (9) is identified in the first stage: conditioning on inputs that are themselves functions of ω absorbs the very variation that would identify them. This is the functional-dependence point of Akerberg et al. (2015), and it is why *all* parameters are recovered in the second stage.

The Markov assumption (12) is what gives the second stage its content. It turns unobserved productivity from an unrestricted object into one with a law of motion, so that its innovation ξ_{it} is a genuine surprise relative to what the plant knew at $t - 1$.

The instruments in (13) are both valid and relevant. Capital is predetermined because new investment takes time to become productive, so k_{it} cannot respond to ξ_{it} . Lagged materials and labour were chosen before ξ_{it} was realised, so they are uncorrelated with it; yet, because both productivity and input demand are persistent, they remain strongly correlated with current input choices. Validity comes from timing, and relevance from persistence.

Three things could undermine this argument: mismeasurement of materials that varies with ω , which would violate scalar unobservability; productivity that is not Markov, for example anticipated demand shifts that enter input choices but not g ; and labour that is fully flexible *and* chosen strictly after materials. The exclusion of two industries and the diagnostic tests in Appendix H address the first of these; the robustness of trends and incidence across estimators (Section 8) speaks to the other two.

We minimise the GMM objective separately for each industry, using Nelder–Mead followed by BFGS (as two separate optimisation calls). The full run proceeds only after the procedure has passed a reproduction check on a single industry. Identification requires at least two consecutive observations per plant, and this requirement defines the estimation sample (Section 3). The Cobb–Douglas benchmark uses the same moments, implemented through `prodest`, and reports analytic (Wooldridge-form) standard errors. The accompanying 50-replication block bootstrap is degenerate for several industries, so we report it as a diagnostic rather than as corroboration (Appendix E).

4.4 The markdown and monopsony

We now apply the logic of (8) to labour. If labour were free of frictions and competitively priced, the ratio $\theta_{it}^L/\alpha_{it}^L$ would also equal μ_{it} . If instead the plant faces an upward-sloping labour supply curve or other wage-setting frictions, the labour condition picks up an additional wedge, ψ_{it} , between the marginal revenue product of labour and the wage:

$$\frac{\theta_{it}^L}{\alpha_{it}^L} = \mu_{it} \psi_{it} \quad \Longrightarrow \quad \boxed{\psi_{it} = \frac{\theta_{it}^L/\alpha_{it}^L}{\theta_{it}^M/\alpha_{it}^M}} \quad (14)$$

(Brooks et al., 2021; Yeh et al., 2022). Under static monopsony, with a firm-level labour-supply elasticity ε_L ,

$$\psi = 1 + \frac{1}{\varepsilon_L} \quad \Longleftrightarrow \quad \varepsilon_L = \frac{1}{\psi - 1}, \quad (15)$$

so a value of $\psi > 1$ signals monopsony, and its size maps into the elasticity of the labour supply that the plant faces.

The intuition behind the boxed ratio in (14) is that the materials condition acts as the plant’s own control group. Both input conditions contain the same markup μ , because both inputs produce the same marked-up output. Dividing the labour ratio by the materials ratio therefore removes the markup, and whatever wedge remains is specific to the labour transaction. This is why (14) needs no data beyond what (8) already uses. It is also why the markdown inherits the estimation error of *both* elasticities, which is why we winsorise ψ more aggressively and emphasise incidence, $\Pr(\psi > 1)$, over levels. Equation (14) also carries a lesson about measurement: the labour-based “markup” θ^L/α^L is in fact the *product* $\mu\psi$. Markups and markdowns cannot be told apart using a single input, which is why both flexible inputs, estimated jointly, are essential.

Two caveats on interpretation are in order. First, (15) reads ψ through the lens of static wage-posting monopsony (Manning, 2003). The measured wedge itself does not depend on any model—it is simply MRPL/w —but its translation into an elasticity does: under bargaining, efficiency wages or dynamic contracts, the same ψ implies different underlying primitives. Second, labour quasi-fixity offers an alternative reading. If hiring involves adjustment costs, the static labour condition fails mechanically, and $\psi > 1$ may reflect the shadow cost of adjustment rather than wage-setting power. Three features of our results suggest that frictional wage-setting does most of the work. The first is geography: ψ is highest precisely where outside options are thin—in rural plants and in the mineral-belt states—rather than where turnover or skill-specificity is highest. The second is the gradient of incidence with employer concentration. The third is the persistence of plant-level ψ over horizons at which adjustment costs should have washed out. We flag this caveat wherever the argument rests on levels rather than on cross-sectional contrasts.

How much employment the wedge costs. The markdown has a quantity counterpart as well as a price counterpart. A plant with wage-setting power hires fewer workers than a wage-taker would, and how many fewer depends on how quickly its marginal revenue product falls as it expands. That rate is governed by the labour elasticity θ^L estimated here. With isoelastic labour supply and revenue $R \propto L^{\theta^L}$, comparing the monopsonist's optimum with that of a wage-taker gives

$$\frac{L}{L^*} = \psi^{-1/(\psi-\theta^L)} < 1, \quad \psi = 1 + \frac{1}{\varepsilon_L}, \quad (16)$$

Here L is employment at a plant with wage-setting power and L^* is what the same plant would hire if it faced a competitive wage, so the ratio is the share of competitive employment that the monopsonist actually offers. As before, ψ is the markdown, the marginal revenue product of labour over the wage, and θ^L is the output elasticity of labour, which governs how fast that marginal revenue product falls as the plant expands. ε_L is the elasticity of the labour supply curve the plant faces, and it is the primitive behind ψ : a plant that can hire more workers only by raising wages steeply faces a small ε_L and marks the wage down heavily.

Two properties can be read directly from the expression. First, the exponent is negative whenever $\psi > \theta^L$, which holds throughout our estimates, so $L/L^* < 1$: the plant hires fewer workers than a wage-taker would. Second, the shortfall grows as θ^L rises: when the marginal revenue product falls slowly, a larger cut in employment is needed to move the wage by any given amount. When $\theta^L = 1$, so that the marginal revenue product does not fall at all, the expression reduces to the familiar $(\varepsilon_L/(1 + \varepsilon_L))^{\varepsilon_L}$. That special case is a poor guide here. Our estimated labour elasticities are an order of magnitude smaller than one, and using it would overstate the employment distortion roughly twofold. We therefore evaluate (16) at the values we actually estimate: a typical-plant markdown of $\psi = 1.61$ and a pooled median $\theta^L = 0.102$. At these values, employment at affected plants is about 27 per cent below its competitive level. The figure is not sensitive to θ^L across the range we observe: it is 26.3 per cent when $\theta^L = 0.05$ and 30.1 per cent when $\theta^L = 0.28$. None of these figures includes the misallocation across plants that the same wedges induce.

4.5 A worked example

The algebra is short enough to walk through once with actual numbers. This is useful because the magnitudes are not obvious, and because the arithmetic itself shows why one of the two wedges is measured precisely and the other is not. We take food products (NIC 10), the largest division in the sample, with 62,535 plant-years.

The median plant in this division has a materials elasticity of $\theta^M = 0.925$ and spends

a share $\alpha^M = 0.839$ of its revenue on materials, so (8) gives

$$\mu = \frac{\theta^M}{\alpha^M} = \frac{0.925}{0.839} = 1.10,$$

that is, a price about ten per cent above marginal cost. The number is small, and the reason is visible in its ingredients: a plant that spends 84 paise of every rupee of revenue on materials, and whose output responds almost one-for-one to them, has very little room between price and marginal cost. Food processing is exactly the kind of homogeneous, thin-margin activity in which one would expect to find this.

Consider next the same plant's labour. The median labour elasticity is $\theta^L = 0.0792$ and the median wage bill is a share $\alpha^L = 0.0521$ of revenue, so

$$\frac{\theta^L}{\alpha^L} = \frac{0.0792}{0.0521} = 1.52, \quad \psi = \frac{1.52}{1.10} = 1.38.$$

In words, labour's output elasticity (about 7.9 per cent) is well above its share of revenue (about 5.2 per cent). Dividing out the markup that both inputs share leaves $\psi = 1.38$: the marginal revenue product of labour is about 38 per cent above the wage. Read through (15), this corresponds to a labour-supply elasticity of $\varepsilon_L = 1/(\psi - 1) \approx 2.6$, meaning that the plant could expand employment only by bidding wages up appreciably—which is precisely what wage-setting power means. Across all food-products plant-years, 62.4 per cent have $\psi > 1$.

One arithmetic point is easy to get wrong and is worth stating. The medians above are taken variable by variable, and the median of a ratio is not the ratio of the medians. The ratios of the median ingredients are $\mu = 1.1022$ and $\psi = 1.3810$, whereas the medians of the plant-level ratios—the figures this paper reports throughout—are $\mu = 1.1028$ and $\psi = 1.3585$. The two agree to within about a tenth of a per cent for the markup and under two per cent for the markdown, so nothing substantive depends on the distinction; the reported figures, however, are always the medians of the plant-level ratios.

The example also shows where the fragility documented in Section 8.2 comes from. The markup divides a number near 0.93 by one near 0.84: both are large and well determined, and a one per cent error in either moves μ by roughly one per cent. The markdown divides 0.0792 by 0.0521. Both numbers are small, the numerator is the least precisely determined parameter the estimator produces, and the result is then divided by the markup a second time. There is nothing wrong with the algebra. Dividing small, imprecisely estimated quantities simply propagates error in a way that dividing large, precisely estimated ones does not. This asymmetry is why we report the markup as a point estimate and the markdown with a bootstrap interval.

4.6 Estimation conventions

Markups are winsorised at the 1st and 99th percentiles, and markdowns at the 2nd and 98th, within NIC-2×year cells before any aggregation; monetary variables are winsorised at 1/99 by year during cleaning. Survey multipliers are never used in estimation, which targets the model, but are always used when aggregating to the population. Monopsony incidence is defined as $\Pr(\psi_{it} > 1)$.

One industry is excluded from all markdown aggregates: printing (NIC 18). The problem lies in its reported shares: 36.4 per cent of its plant-years report $\alpha^L + \alpha^M > 1$, which implies negative value added. This is a reporting artefact associated with job work, and it undermines the logic of the denominator of (14). It is, by a wide margin, the highest rate among the estimated divisions. Printing remains in every markup statistic.

Earlier drafts also excluded other transport equipment (NIC 30) from every aggregate, and the change deserves a sentence. Its rate of negative implied value added, 14.9 per cent, is barely above the all-division average of 14.1 per cent and *below* that of four divisions that were always retained (NIC 26 at 19.7, NIC 13 at 19.5, NIC 24 at 17.0 and NIC 10 at 16.7 per cent), so the share criterion never singled it out. The exclusion rested instead on the admissibility screen $\bar{\theta}^L > 0$, $\bar{\theta}^M > 0$: under a first stage without year effects its mean labour elasticity was negative (−0.020). With year effects in the first stage it is +0.073, the division passes the screen, and it now enters every aggregate on the same footing as the others. The screen itself is unchanged and is still applied within every bootstrap replication.

Concentration is measured by the employment Herfindahl index $HHI_{jt} = \sum_i s_{ijt}^2$ within NIC-4 markets, where s_{ijt} is plant i 's share of employment in market j in year t . Aggregate markup dynamics are decomposed following Melitz & Polanec (2015):

$$\Delta \bar{\mu}_t = \Delta \tilde{\mu}_t^{\text{within}} + \Delta \text{cov}_t(s, \mu) + \text{entry}_t + \text{exit}_t, \quad (17)$$

where $\bar{\mu}_t$ is the sales-weighted aggregate markup, $\tilde{\mu}_t$ the unweighted average among plants present in both years, $\text{cov}_t(s, \mu)$ the covariance across plants between sales share and markup, and the final two terms the contributions of plants entering and leaving between $t-1$ and t . The decomposition thus separates change within continuing plants, reallocation of sales among them, and the net contribution of entry and exit. It answers a question that the aggregate series alone cannot. A rising sales-weighted markup is consistent with plants raising their own prices over cost, but it is equally consistent with sales shifting towards plants that already price high, while the typical plant's margin barely moves. These two readings carry different policy implications, and only (17) can distinguish between them. Section 5.2 shows that in India the second accounts for two-thirds of the rise.

5 Results

This section presents the estimates in the order in which they build on one another. We begin with the output elasticities, since every markup and markdown is constructed from them, and then report the markup (its level, trend, links with concentration and decomposition) and the markdown. We next show how strongly the aggregate markdown depends on the way it is aggregated, and close with two dimensions along which the two wedges behave very differently: plant size and geography.

5.1 Output elasticities

Appendix Table 18 reports, for each of the 23 divisions, the sample size, the mean elasticities θ^L , θ^K and θ^M , returns to scale, monopsony incidence, and the median values of μ and ψ . Three regularities matter for what follows, and the third is the reason this paper reports intervals rather than precise digits.

First, materials elasticities cluster tightly, between 0.79 and 0.99. The gross-output technology is dominated by materials in every industry, which stabilises the numerator of the markup. Second, labour elasticities are both small and widely spread (0.02–0.28). Every division now has a positive mean labour elasticity, so all 23 pass the admissibility screen; the one remaining anomaly, near-zero monopsony in NIC 18 (printing), motivates the single exclusion described in Section 4.6. Third, returns to scale lie close to constant (0.94–1.12), which is consistent with the identification assumptions. The contrast between the first two facts is more than a matter of presentation. A numerator close to one and a numerator of about 0.1, estimated jointly from the same nine moments, are not determined with the same precision, and Section 8.2 shows what follows from this.

5.2 Markups: levels, trends, concentration, decomposition

Table 5 contains the headline series. Consider the markup columns first. The median markup is close to flat over the decade: 1.199 in 2012, a peak of 1.258 in the pandemic year 2020, and 1.213 in 2022. The multiplier-weighted (population) series moves in parallel, at a small premium. This series should not be confused with the sales-weighted aggregate discussed below: the two answer different questions and differ in level, and we keep their labels separate throughout.

An earlier draft of this paper reported a markup that *fell* after 2015, to 1.154 in 2022. That decline did not survive a correction to the first stage of the estimator. The first-stage regression, which purges output of measurement error before the moment conditions are formed, did not include year effects, so a common trend in its residual $\hat{\varepsilon}_{it}$ passed straight into the share correction $\alpha^M e^{\hat{\varepsilon}}$ and from there into every markup. The mean residual rose from -0.030 in 2012 to $+0.054$ in 2022, which is enough on its own to generate an eight

Table 5: Translog estimates year by year. Columns 2–5 cover all 23 estimated divisions, every one of which passes the admissibility screen (N sums to 422,820); NIC33 is not estimated under the translog (fewer than 500 usable plant-years). “wtd μ ” applies the ASI survey multiplier wt and so estimates the organised-manufacturing *population* mean — it is *not* sales-weighted, and differs from the sales-weighted aggregate of Table 6. The final two columns give unweighted monopsony incidence on all 23 divisions and on the sample that excludes NIC18, the exclusion rule used for the markdown aggregates in Section 5.3; the latter is the series quoted in the text. Standard errors for every entry are in Table 9.

Year	N	med μ	wtd μ	med ψ	Pr($\psi > 1$) all 23	Pr($\psi > 1$) excl. 18
2012	33,130	1.199	1.225	0.968	0.488	0.496
2013	36,332	1.191	1.221	0.932	0.476	0.483
2014	38,370	1.199	1.226	0.891	0.458	0.466
2015	37,322	1.218	1.247	0.803	0.426	0.433
2016	39,087	1.221	1.257	0.768	0.413	0.420
2017	39,212	1.221	1.257	0.770	0.413	0.420
2018	39,402	1.221	1.254	0.792	0.419	0.426
2019	39,738	1.239	1.269	0.730	0.395	0.402
2020	40,256	1.258	1.278	0.741	0.399	0.405
2021	40,899	1.221	1.240	0.837	0.439	0.445
2022	39,072	1.213	1.230	0.858	0.449	0.455
Pooled	422,820	1.218	1.247	0.821	0.433	0.440

per cent decline in the measured markup with no change in pricing. With year effects in the first stage the residual is mean-zero in every year by construction, and the decline disappears; Appendix L reports the comparison in full. The markdown is not exposed to this channel in the same way, because the share correction cancels in the ratio that defines it, but it moves through the re-estimated elasticities, and we report those changes too.

The markdown columns tell a different story. The median ψ falls from 0.968 in 2012 to a trough of 0.730 in 2019, and recovers to 0.858 by 2022. Monopsony incidence traces the same U-shape. On the sample consistent with the exclusion rule, it is 49.6 per cent in 2012, 40.2 per cent in 2019 and 45.5 per cent in 2022; retaining NIC18 lowers it by between six- and eight-tenths of a point in each year. From year to year, the two medians tend to move in opposite directions, a joint dynamic that the companion paper on determinants links to trade exposure and to the thickening of local labour markets.

The industry pattern. The ordering of industries (Appendix Table 18) makes economic sense. Median markups are highest where products are differentiated, branded or regulated—wearing apparel (1.72), tobacco (1.70), computer and electronic products (1.43), pharmaceuticals (1.40) and non-metallic minerals (1.37)—and lowest in homogeneous intermediate goods: wood (1.06), basic metals (1.10), food products (1.10) and coke and refined petroleum (1.15). Monopsony incidence orders industries quite differ-

ently. It exceeds 60 per cent in wood products (0.87), basic metals (0.68), coke and refined petroleum (0.68), chemicals (0.64), electrical equipment (0.64), food products (0.62) and pharmaceuticals (0.62). Most of these are process industries, whose plants are large, anchored by heavy capital and often the dominant employer in their local labour market; wood products, at the other end of the size distribution, is dominated by small plants in rural areas, where the alternative employer is frequently none. Incidence is lowest in wearing apparel (0.05), textiles (0.07), leather (0.17) and non-metallic minerals (0.25). We offer these interpretations as conjectures consistent with the data, not as mechanisms we test. What the data do establish is the divergence itself. Wearing apparel illustrates it in a single row: the highest median markup in manufacturing coincides with the second-lowest monopsony incidence. A specification that confused the two margins could not produce such a pattern, so the divergence in rankings is itself evidence that the estimation separates them. Division-level incidence is, however, the least robust number in this paper: several divisions move by tens of percentage points between the two first-stage specifications reported in Appendix L, the reason for which Section 8.2 documents.

Across industries, the median translog markup ranges from 1.00 (printing) to 1.72 (wearing apparel), and the pooled plant-level median is 1.22 (Table 5). The economy-wide winsorised mean drifts up from 1.24 in 2012 to 1.31 in 2020 and stands at 1.26 in 2022. Markups of this size are *modest*: they lie at the bottom of the international range of production-approach estimates, and far below the sales-weighted US aggregates of De Loecker et al. (2020). Two pieces of evidence suggest that this modesty is real rather than an artefact of the estimator, and a third shows where the modest increase over the decade comes from.

The first is that the Cobb–Douglas benchmark (Appendix D) produces the same ordering of industries and a similar aggregate level (year-by-year means of 1.31–1.38, with no trend). The level of the markup therefore does not depend on the translog specification.

The second is that *concentration fell*. The mean employment HHI across NIC-4 industries declined from 0.086 in 2012 to 0.066 in 2022, and the four-firm concentration ratio from 0.39 to 0.32. An economy that is becoming less concentrated is not one in which product-market power is building up. Two qualifications apply to these figures. They are computed over NIC-4 cells with an adequate number of plants; including thinly populated cells raises the HHI to 0.109 and 0.097 respectively, but leaves the direction of change unchanged. The decline is also a net movement between the two end-points rather than a steady one: the series rises in 2017, 2019 and 2021 before resuming its fall.

The third is the Melitz & Polanec (2015) decomposition (17) of the sales-weighted aggregate markup Ω , which rises from 1.1195 in 2012 to 1.1711 in 2022, an increase of 0.052, or 4.6 per cent. Two-thirds of that increase is reallocation: the covariance between market share and markup among surviving plants contributes +0.0396, the change within those plants only +0.0193, and entry and exit together slightly less than nothing (−0.0017

and -0.0056). The four components sum exactly to the net change. Economic activity, in other words, shifted towards plants with high markups, while the typical plant’s own pricing power barely moved. This is the pattern documented for the United States (De Loecker et al., 2020), in which the rise of the aggregate markup is driven by the reallocation of activity towards high-markup firms, but here it operates at about one-tenth of the US level and over a much smaller change.

Two cautions apply to the size of that change. The decomposition is a point estimate; the bootstrap of Section 8.4 does not re-form Ω within each replication, so we cannot attach an interval to it. And under the earlier first-stage specification the net change was -0.0009 , with alternative optimisers placing it between that value and $+0.0020$; the present figure is some eighteen times the width of that numerical range, so its sign is no longer in doubt on numerical grounds, but its magnitude should be read as indicative. The entry term is negative in nine of the ten year-on-year transitions, because entrants generally price below incumbents and so discipline margins from below; the only exception is 2013, when it is essentially zero ($+0.0004$). Table 6 summarises the cumulative decomposition for 2012–2022.

Table 6: Melitz–Polanec anatomy of the sales-weighted aggregate markup, decomposing the change from 2012 to 2022. The aggregate rises by 0.052, two-thirds of it through reallocation toward high-markup plants. Ω is the sales-weighted aggregate over the pooled sample, weighting by nominal ex-factory sales without survey multipliers. The survivor-based year-on-year series is a different object — it weights by multiplier-scaled real sales and re-forms the survivor, entrant and exiter groups for each adjacent pair of years — runs 1.119–1.216, and is comparable neither to Ω nor to the multiplier-weighted series of Table 5. Components sum to $\Delta\Omega$ exactly. Ω is not re-formed inside the bootstrap, so no interval is attached; see the text.

Component (cumulative, 2012–2022)	Contribution
Sales-weighted aggregate Ω_{2012}	1.1195
Sales-weighted aggregate Ω_{2022}	1.1711
Net change $\Delta\Omega$	+0.0516
Within-plant markup change (survivors)	+0.0193
Reallocation covariance $\Delta\text{cov}(s, \mu)$ (survivors)	+0.0396
Entry	−0.0017
Exit	−0.0056
Sum of components (identity check)	+0.0516

5.3 Markdowns and monopsony

The markdown tells the opposite story. Its median lies below one in every year (0.73–0.97), so the *typical* plant pays its workers at or above the static competitive benchmark, which is consistent with efficiency wages and regulatory wage floors. The upper half of the distribution, however, is heavy. The share of plants with $\psi > 1$ ranges from 40.2 to 49.6 per cent across years, and is 44.0 per cent in the pooled sample (48.4 per cent

once the sampling multipliers are applied). The unweighted mean markdown across all plants, 1.61, implies through (15) a labour-supply elasticity of about 1.6—within the range that modern quasi-experimental studies find for much richer labour markets (Card et al., 2018). Among the monopsonistic plants alone the wedge is wider: their median markdown is 2.03, which implies an elasticity close to one.

Monopsony incidence falls by 9.4 percentage points between 2012 and its trough in 2019, and by 9.0 points between 2012 and 2020, before recovering to 45.5 per cent in 2022. Over the decade as a whole, therefore, incidence ends 4.0 points below its starting level. This erosion of wage-setting power coincides with the thickening of local labour markets documented in the companion paper on determinants.

5.4 The aggregation-concept range

Published estimates of the “aggregate markdown” differ enormously, and the algebra in Appendix J shows why: the markdown is a ratio, and the choice of weights passes through it non-linearly. On a common trimmed sample of 403,196 plant-years we compute several concepts. The *typical-plant* (unweighted) markdown is 1.610 ($\varepsilon_L \approx 1.64$), and the *employment-weighted* markdown is 1.194. The *sales-weighted* markdown is 3.862 ($\varepsilon_L \approx 0.35$): plants that account for more of manufacturing sales carry higher measured markdowns. This is an accounting statement about where the wedge is concentrated, not evidence that size causes wage-setting power. Finally, the harmonic aggregate of Yeh et al. (2022) falls below one, to 0.408, an artefact of harmonic weighting under extreme dispersion, which we document rather than adopt. The spread across concepts on identical plants—a factor of more than three between the employment-weighted and sales-weighted figures, and of more than nine between the harmonic and sales-weighted figures—is the point of the exercise.

We convert only two of these concepts into an implied labour-supply elasticity, and the reason is arithmetic rather than convenience. The mapping $\varepsilon_L = 1/(\psi - 1)$ is a hyperbola with a pole at $\psi = 1$. It is well behaved only when ψ is bounded away from one, and it has no economic meaning when $\psi < 1$. The employment-weighted aggregate, at $\psi = 1.194$, lies close enough to the pole that its implied elasticity is 5.14, and a change in ψ of one percentage point moves ε_L by roughly a quarter of a unit. The median plant ($\psi = 0.879$) and the harmonic aggregate ($\psi = 0.408$) lie below one, where the formula returns negative values (−8.27 and −1.69) that cannot be read as labour-supply elasticities at all.

We therefore report every concept as a markdown (Table 7), but headline the typical-plant concept and convert only the typical-plant and sales-weighted aggregates, which lie in the stable region of the mapping. This gives the range $\varepsilon_L \in [0.35, 1.64]$ quoted in the introduction. The employment-weighted row of Table 7, which implies an elasticity of 5.2, is shown because the markdown itself is informative, but for the reason just given it

Table 7: The aggregate markdown under alternative concepts, common trimmed sample ($N = 403,196$ — plant-years on which both estimators deliver a positive markup and markdown, identically winsorised). Implied elasticity from $\varepsilon_L = 1/(\psi - 1)$ where $\psi > 1$. The Cobb–Douglas estimates were not re-estimated for this draft (Appendix D), but these rows are recomputed here, because the common sample is not the same one.

Concept	Estimator	$\bar{\psi}$	implied ε_L
Typical plant (unweighted mean)	translog	1.610	1.64
Employment-weighted mean	translog	1.194	5.14
Sales-weighted mean	translog	3.862	0.35
Harmonic (Yeh et al.)	translog	0.408	—
Typical plant (unweighted mean)	Cobb–Douglas	1.768	1.30
Harmonic (Yeh et al.)	Cobb–Douglas	1.366	2.73

is not part of the quoted range. Reporting a single tidy elasticity band across all concepts would hide the fact that most of the spread is generated by the conversion itself, not by the data; we treat this sensitivity as a finding about the literature in its own right. The concepts do agree on everything except the scalar level: on incidence (86.9 per cent classification agreement across estimators), on the ranking of industries, and on the size gradient.

5.5 Size gradients

Within industry-years, the markup rises mildly with plant size (a coefficient on log employment of +0.017, and +0.046 within plants), which is consistent with pricing power that grows with market share. The raw relationship between the markdown and size is contaminated by division bias, because the denominator of α^L contains output; we therefore estimate it against *lagged* size. All four lagged specifications give the same, positive answer. The coefficient on lagged output is +0.096 ($t = 24.4$) within plants and +0.036 ($t = 16.3$) within industry-years; on lagged market share it is +0.052 ($t = 20.5$) and +0.041 ($t = 18.8$) respectively. These are conventional t -statistics; Section 8.4 shows that conventional standard errors understate the uncertainty of such regressions substantially, so we rely on these estimates for their sign rather than their precision. Larger plants open a wider gap between the marginal revenue product of labour and the wage, both as they grow and relative to their peers in the same industry. This is the pattern predicted by the granular-employer model of Berger et al. (2022).

This result has been sensitive to the numerical implementation, and we report that sensitivity openly because it once changed an inference, not merely a digit. Under an alternative optimiser and the earlier first-stage specification, the two industry $\times$ year specifications returned +0.0003 ($t = 0.15$) and +0.007 ($t = 3.1$). Read at face value, the first of these is a precisely estimated zero, which would imply that the size gradient is purely a *within-plant* phenomenon—describing how a given plant’s markdown changes as

it grows—rather than a ranking of large against small plants in the same industry. On the present estimates that distinction does not survive: the cross-sectional gradient is of the same order as the within-plant gradient, with a conventional t -statistic above 16. The economic reading is therefore simpler and stronger, and it no longer rests on a contrast between two specifications that in fact agree. The episode is also a caution in its own right: a coefficient of 0.0003 with $t = 0.15$ invited a null reading that a different numerical path to the same moment conditions did not support. We therefore report the gradient as a conditional association. The sign of the raw markdown–size gradient depends on the specification, for the mechanical reason given above; lagging size mitigates division bias but does not provide an exogenous instrument; and nothing here identifies a causal effect of size on wage-setting power.

5.6 Geography: states and the rural–urban margin

The two margins part ways in space, and the contrast is informative about mechanism. Median markups across the major states span only 1.15–1.25, with the single exception of Bihar (1.48; see Section 7). Goods travel, so a plant’s pricing power is determined in product markets that are national or international in scope, and state boundaries barely register. Monopsony incidence, by contrast, ranges from 0.31 to 0.67 across the 21 states with at least 5,000 plant-years. Workers do not move nearly as freely as goods, so the labour wedge is plausibly set in *local* labour markets and inherits their geography. Section 7 examines the state dimension in full. Here we report the margin that state-level tables cannot show, because it lies within states rather than between them.

Within states, the sharper margin is between rural and urban areas. Rural plants are monopsonistic in 46.4 per cent of plant-years, against 41.9 per cent for urban plants, and the gap is present in *every* year of the sample (51.4 against 48.1 per cent in 2012; 48.5 against 42.9 per cent in 2022). Median markups, by contrast, are almost identical in the two sectors (1.216 rural and 1.220 urban). Table 17 reports the sectoral series year by year.

Figure 2 maps the gap by state. It approaches twenty percentage points in West Bengal (+19.5) and exceeds seventeen in Uttar Pradesh (+17.3); it is twelve to sixteen points in Dadra and Nagar Haveli, Puducherry and Telangana, and nine to ten in Odisha and Karnataka. Among the 27 states with at least 300 rural and 300 urban plant-years, it *reverses* in eight, but six of these reversals are material: Bihar (−31.4), Nagaland (−22.4), Tripura (−18.1), Assam (−12.7), Goa (−8.9) and Punjab (−7.0); the other two, Andhra Pradesh and Rajasthan, are below 1.5 points. Bihar and Assam are consistent with the thin-market reading: rural manufacturing there is dominated by large agro-processing employers, but the urban sector is thinner still. The Nagaland and Tripura figures rest on small urban samples (342 and 463 plant-years) and should not be pressed. We report the

reversals because they are real, and because no single mechanism explains all of them.

Taken together, labour-market power in India looks like a rural, thin-market phenomenon, whereas product-market power has almost no geography at all. The word “looks” matters here, in two respects. Section 8.5 shows, for the earlier specification, that the spread across states survives resampling intact, but that the pooled rural–urban gap does not reach conventional significance, and that the *ranking* of states is far less certain than a map suggests. Section 9.4 then asks what lies behind the rural gradient, using labour categories recovered from the raw returns, and rules out the explanation most readers will reach for first. Rural plants do use more contract labour, but the gradient survives within both contract strata, and it widens slightly, rather than narrowing, when contract intensity is held fixed.

6 Inter-industry analysis

So far the results have been reported either for manufacturing as a whole or by state. Yet “Indian organised manufacturing” consists of twenty-three divisions that differ from one another almost as much as countries do, and up to this point the paper has summarised them in a single appendix table. This section treats the division as a unit of analysis in its own right. We first describe what these industries look like, then ask how market power varies across them, and finally ask how much of the plant-level variation in the two wedges is a matter of industry at all.

6.1 What the divisions look like

Table 8 sets out the structural profile of each division alongside its two wedges. The first thing to note is how large the structural differences are, because they frame everything that follows.

Median plant size ranges from 13 persons in wood products and 20 in furniture to 225 in motor vehicles, 184 in pharmaceuticals and 169 in wearing apparel—a seventeen-fold range within a single sector of a single economy. The median labour share ranges from 0.036 in coke and refined petroleum to 0.185 in printing, a five-fold range, and the materials share from 0.51 in tobacco to 0.87 in printing. Employment concentration varies even more: the employment Herfindahl index is 0.00007 in food products, an industry of many thousands of small plants, compared with 0.0059 in tobacco, a difference of a factor of about eighty-five.

These differences are not incidental. A division of 13-person plants with a labour share of 0.09 and a division of 225-person plants with a labour share of 0.11 are different economic objects, and any statement about “the” Indian manufacturing markup is an average across such objects. This is the case for reporting results by division, and for the

Rural minus urban monopsony incidence (pp gap), 2012-2022

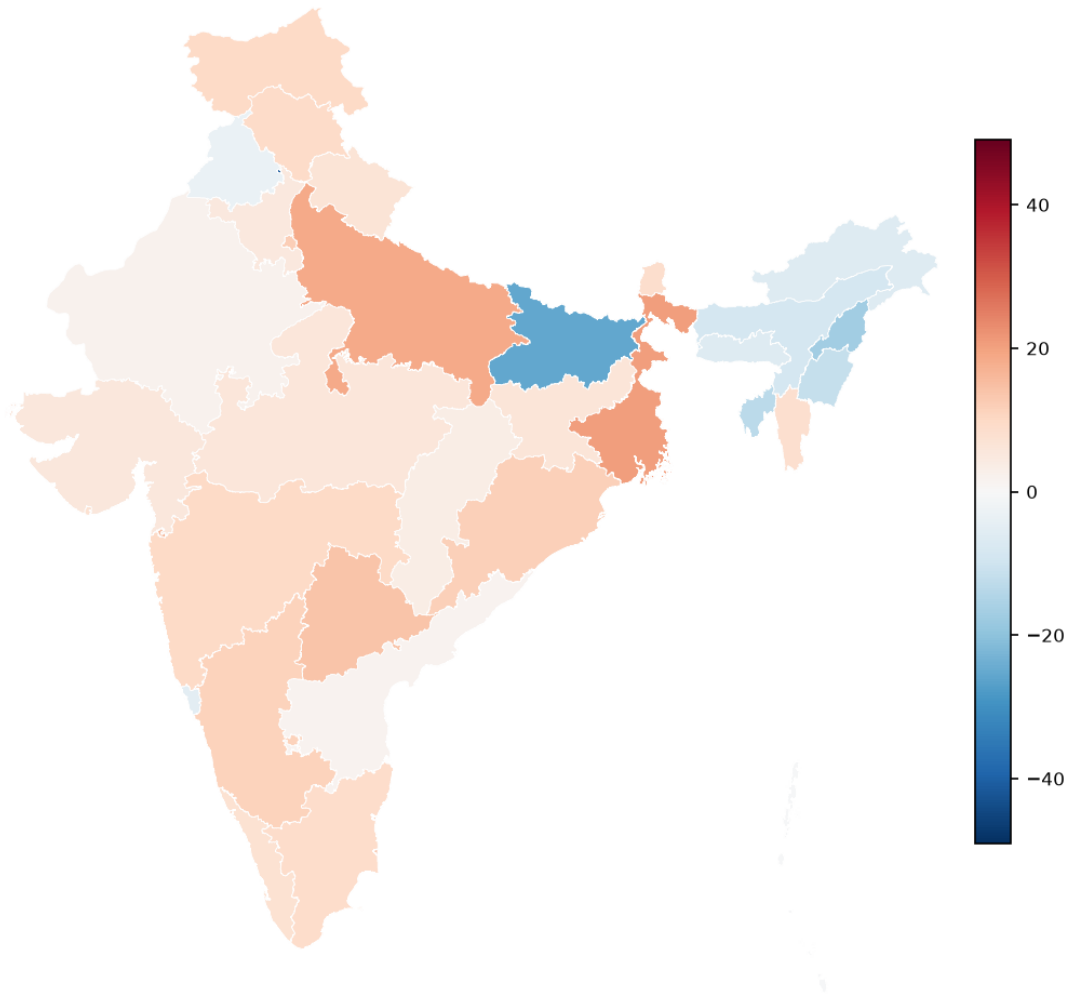


Figure 2: The rural–urban monopsony gap by state (rural minus urban incidence, percentage points), pooled 2012–2022. Red: rural excess; blue: reversal.

variance decomposition below.

6.2 The two margins rank industries in opposite directions

Section 5.2 noted that wearing apparel has the highest markup but the second-lowest monopsony incidence. This is not an isolated case. Across the 22 divisions for which both statistics are available, the correlation between the median markup and monopsony incidence is -0.53 , and the rank correlation is -0.52 . Industries with the most product-market power tend to have the *least* labour-market power, and vice versa. Figures 3 and 4 show this in two ways.

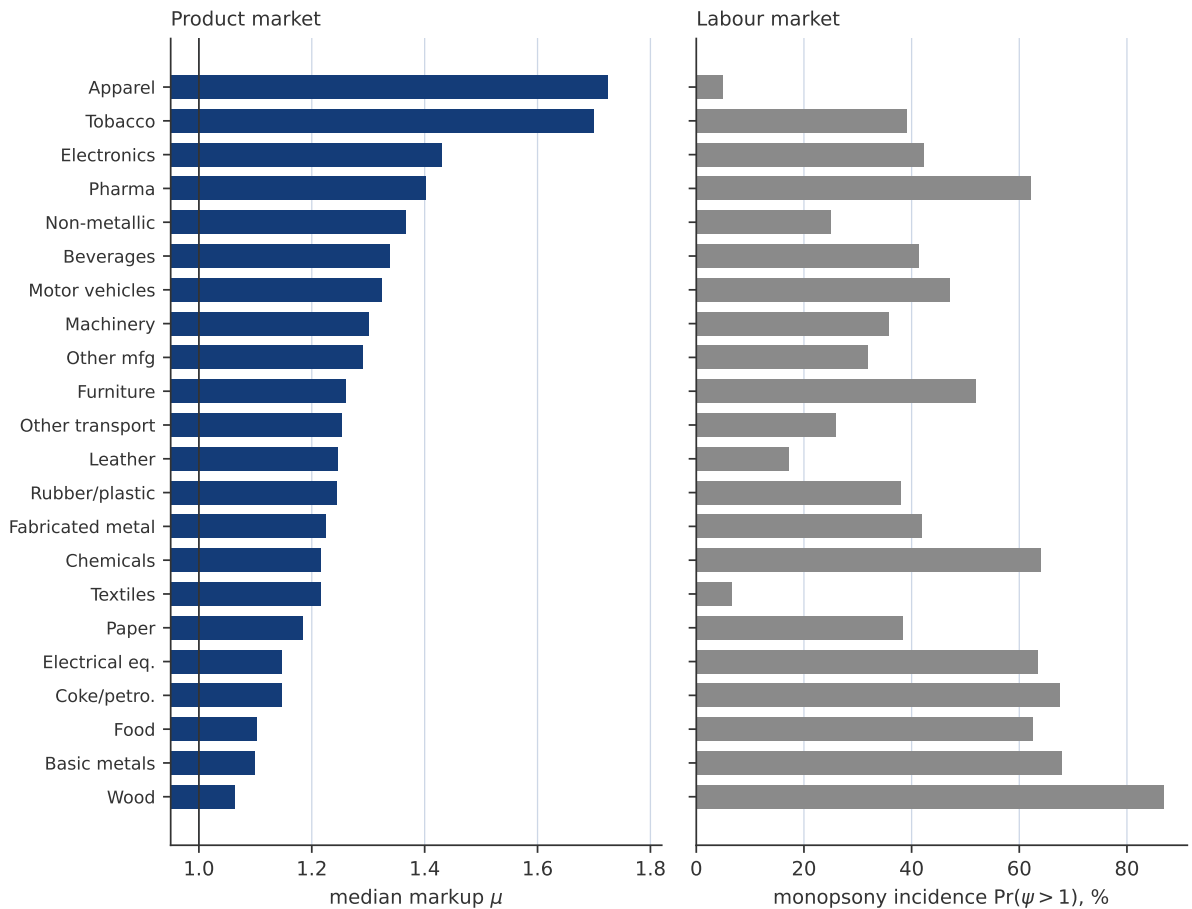


Figure 3: The two wedges by division, sorted by median markup. If the two margins ordered industries alike, the right-hand panel would descend with the left. It does not.

The two lists make the point directly. The highest markups are found in wearing apparel (1.72), tobacco (1.70), computer and electronic products (1.43) and pharmaceuticals (1.40), all industries with differentiated, branded or regulated products. The highest monopsony incidence is found in wood products (0.87), basic metals (0.68), coke and refined petroleum (0.68), chemicals (0.64) and electrical equipment (0.64): mostly process industries, anchored by heavy capital, whose plants are often the dominant employer

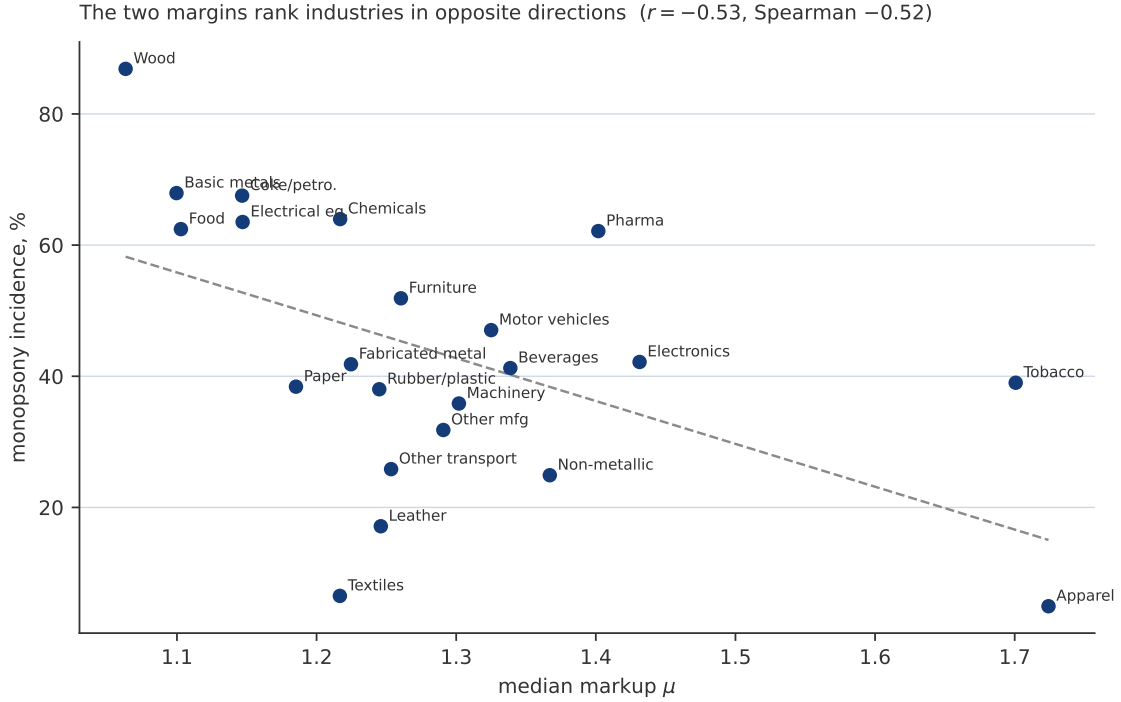


Figure 4: Median markup against monopsony incidence, 22 divisions. Dashed line: ordinary least squares. The two margins are negatively related across industries.

wherever they are located, together with a small-plant rural industry in which the alternative employer is frequently absent. The lowest incidence is in wearing apparel (0.049), textiles (0.065), leather (0.171) and non-metallic minerals (0.249). The first three are labour-intensive industries concentrated in dense clusters such as Tiruppur, Ludhiana and Kanpur, where a worker has many potential employers nearby.

As in Section 5.6, we offer this reading as a conjecture consistent with the data rather than as a mechanism we test. What the data do establish is the divergence itself, and this divergence says something about the estimation as well as about the economics: a specification that confused the two margins could not produce a negative correlation between them across industries.

6.3 How much of market power is an industry fact?

Given how much industries differ, a natural question is how much of the plant-level spread in each wedge lies *between* divisions rather than within them. Decomposing the total variance of each log wedge into a between-division and a within-division component gives a clear answer (Figure 5).

For the markup, 42.4 per cent of the variance lies between divisions and 57.6 per cent within them. For the markdown, industry matters less: 31.5 per cent of the variance lies between divisions and 68.5 per cent within. Knowing an Indian plant's two-digit division therefore tells us less than half of what we would want to know about its product-market

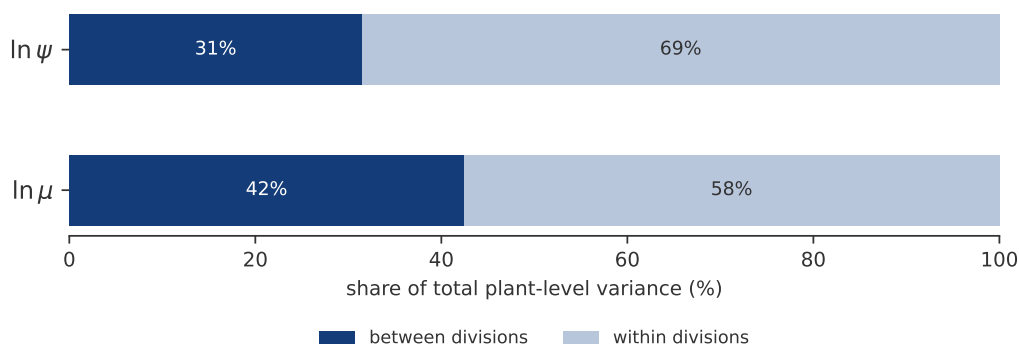


Figure 5: Between- and within-division shares of the total plant-level variance of the log markup ($N = 422,820$) and the log markdown ($N = 403,196$, the plant-years with a positive markdown in the 22 divisions carrying one).

power, and under a third of what we would want to know about its labour-market power.

This is the industry counterpart of the inter-state result in Section 7, and the two point in the same direction. There, 73.9 per cent of the gap in monopsony between high- and low-incidence states survived once the industry mix was held fixed. Here, 68.5 per cent of the plant-level variance in the markdown remains once the division is held fixed. Neither the industry a plant belongs to nor the state in which it is located comes close to determining how much it pays its workers relative to what they produce. Whatever generates labour-market power in India operates largely below the level of both.

6.4 The decade, division by division

The aggregate series conceal which industries moved over the decade. Figure 6 shows the change in each division.

Division by division, the decade shows no general compression of markups. The median markup fell in only 7 of 23 divisions, and in six of those by less than 0.05: wood (-0.049), motor vehicles (-0.041), and electronics, basic metals, food products and textiles (each between -0.010 and -0.013). The one large fall is in tobacco (-0.40), a small division whose 2012 level was unusually high. The rises are more numerous and larger: wearing apparel ($+0.10$), printing ($+0.09$), other manufacturing ($+0.09$), pharmaceuticals ($+0.09$), leather ($+0.07$) and fabricated metal products ($+0.07$). The modest rise in the aggregate markup is therefore not the work of one or two industries. It reflects small increases in most of them, set against a sharp fall in one.

Monopsony incidence fell in most divisions (16 of 22), but unevenly. The largest falls are in other manufacturing (-16.4 percentage points), rubber and plastics (-11.3), leather (-10.2) and non-metallic minerals (-9.8), whereas incidence rose by 13.4 points in tobacco, 8.4 in coke and refined petroleum and 3.5 in basic metals. The dispersion of these movements is a further reason why the paper reports division-level markdowns as

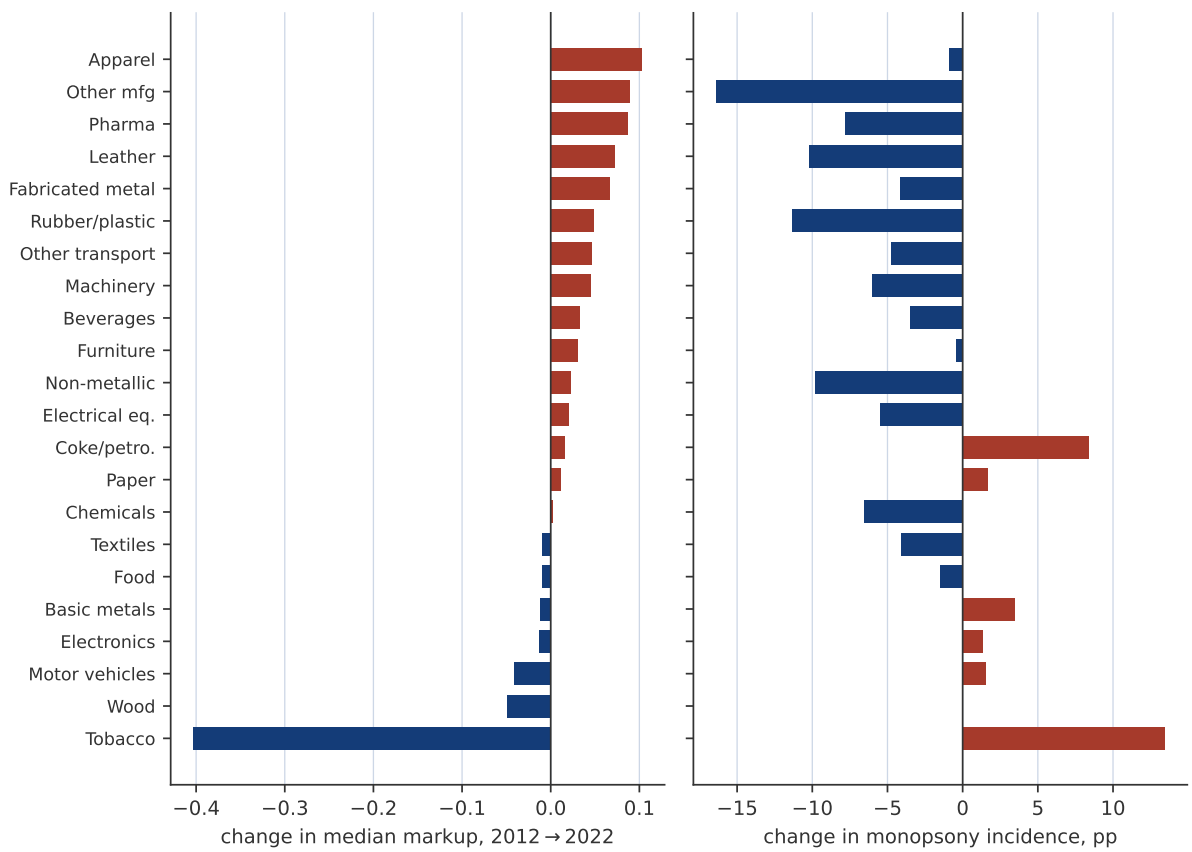


Figure 6: Change from 2012 to 2022 in the median markup (left) and in monopsony incidence (right), by division, sorted by the markup change. Navy denotes a fall, rust a rise.

an ordering rather than as levels (Section 8.2).

6.5 Structure and the two wedges

Finally, the descriptive columns of Table 8 can be compared with the two wedges. Here a caution has to come first rather than last.

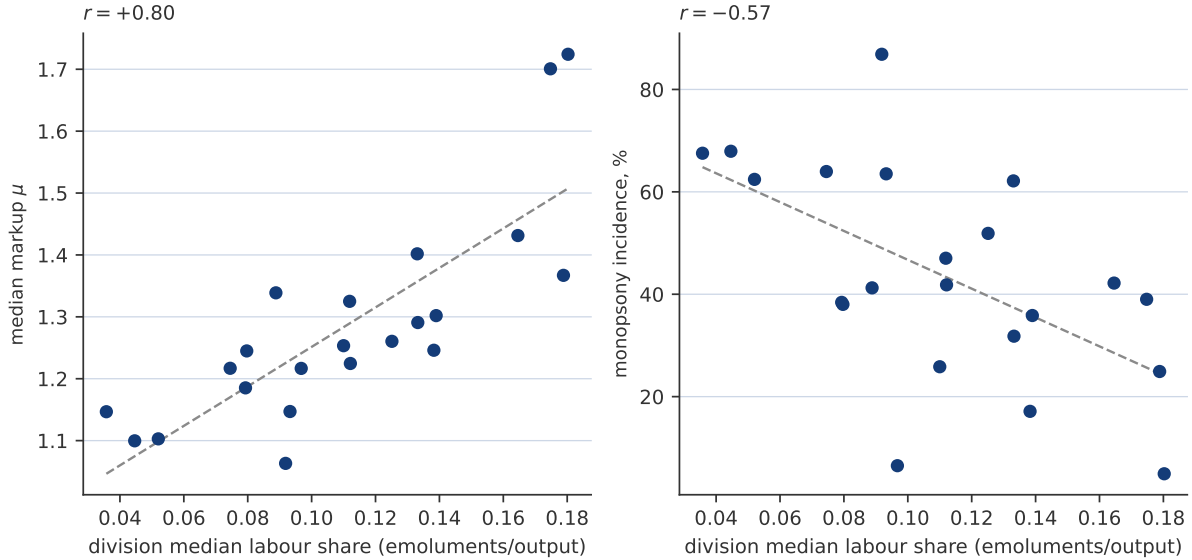


Figure 7: Division median labour share against the markup (left) and against monopsony incidence (right), 22 divisions. Both associations carry a mechanical component, discussed in the text, and are reported for description rather than as evidence of a mechanism.

Divisions with higher labour shares have markedly higher markups ($r = +0.80$) and markedly lower monopsony incidence ($r = -0.57$). Both correlations are present in the data, but both are partly definitional, and we prefer to say so rather than present them as findings. The markdown is $\psi = (\theta^L/\alpha^L)/\mu$, so α^L appears in its denominator: a division with a high labour share will tend to show a low markdown, whatever is happening to wage-setting power. The markup is θ^M/α^M , and because the materials and labour shares together are bounded by revenue, a division with a high labour share tends to have a low materials share, and therefore a high measured markup. Neither correlation should be read as a behavioural relationship. We report them because a reader comparing the columns of Table 8 will notice them, and because the mechanical channel deserves to be named.

Two associations are free of this objection and are worth recording. First, across divisions, monopsony incidence is weakly *negatively* related to median plant size ($r = -0.43$). This has the same sign as the thin-market association found across states in Section 7; we note it without pressing it. Second, the relationship between markups and employment concentration depends on how it is measured: the Pearson correlation is $+0.41$, but the rank correlation is only $+0.13$. The difference is driven by tobacco, which is both the

most concentrated division and the one with the second-highest markup. Judged by the ranks, concentration and markups are at best loosely related across Indian manufacturing divisions. This is consistent with the finding in Section 5.2 that concentration fell over a decade in which the sales-weighted aggregate markup rose.

Table 8: Structure and market power by NIC-2008 division, pooled 2012–2022. Plant-years are those over which elasticities are evaluated. “Size” is median persons engaged; “labour share” is median emoluments over gross output; “materials share” is the median nominal materials share α^M ; “HHI” is the employment Herfindahl computed over all plant-years in the division, so it measures how concentrated that industry’s employment is. Medians of winsorised μ and ψ ; incidence is $\Pr(\psi > 1)$. [†] NIC18 is excluded from every markdown and incidence statistic (Section 4.6), so its last two cells are empty by construction rather than missing. NIC33 does not appear: it does not enter the paper’s aggregates.

Division	Plant-years	Size	Lab. sh.	Mat. sh.	HHI	med μ	med ψ	$\Pr(\psi > 1)$
Food products	62,535	73	0.052	0.839	0.0001	1.103	1.358	0.624
Beverages	9,375	56	0.089	0.686	0.0005	1.339	0.672	0.412
Tobacco	5,542	39	0.175	0.509	0.0059	1.701	0.662	0.390
Textiles	29,560	162	0.097	0.826	0.0002	1.217	0.191	0.065
Wearing apparel	18,947	169	0.180	0.556	0.0010	1.724	0.051	0.049
Leather	8,448	129	0.138	0.708	0.0009	1.246	0.225	0.171
Wood	9,049	13	0.092	0.753	0.0016	1.063	2.879	0.869
Paper	12,113	33	0.079	0.780	0.0006	1.185	0.780	0.384
Printing [†]	6,690	62	0.185	0.868	0.0006	0.995	—	—
Coke & refined petroleum	5,182	30	0.036	0.816	0.0031	1.147	2.113	0.675
Chemicals	28,962	67	0.075	0.728	0.0002	1.217	1.343	0.640
Pharmaceuticals	15,237	184	0.133	0.603	0.0003	1.402	1.437	0.621
Rubber & plastic	22,780	78	0.080	0.753	0.0003	1.245	0.788	0.380
Non-metallic minerals	39,980	66	0.179	0.618	0.0002	1.367	0.586	0.249
Basic metals	24,838	99	0.045	0.846	0.0007	1.100	1.774	0.679
Fabricated metal	22,969	52	0.112	0.734	0.0003	1.225	0.845	0.418
Electronics	8,497	96	0.165	0.649	0.0009	1.431	0.786	0.422
Electrical equipment	22,357	66	0.093	0.763	0.0003	1.147	1.348	0.635
Machinery	27,019	85	0.139	0.668	0.0002	1.302	0.675	0.358
Motor vehicles	18,629	225	0.112	0.714	0.0004	1.325	0.935	0.470
Other transport	7,775	133	0.110	0.730	0.0007	1.253	0.334	0.258
Furniture	5,593	20	0.125	0.692	0.0020	1.260	1.050	0.519
Other manufacturing	10,743	107	0.133	0.649	0.0005	1.291	0.579	0.318

7 Inter-state analysis

Section 5.6 established that the two wedges have opposite geographies: markups barely vary across states, whereas monopsony incidence ranges from 0.31 to 0.67 across the 21 states large enough to be ranked. A spread of this size, in a country as internally diverse as India, can be explained in one of two ways. Either it reflects the *industries* that happen to be located in different places, or it reflects something about the *states* themselves.

The two explanations carry different policy implications, and they can be distinguished empirically. This section does so. We first describe how incidence varies across states, then show that it is higher where organised manufacturing is smaller, and then decompose the gap between high- and low-incidence states into the part due to industry mix and the part due to the states themselves. We close with the policy implications.

7.1 What varies across states

Incidence is highest in the central-eastern mineral belt and lowest, for the most part, in the diversified south, where dense industrial clusters plausibly give workers many alternative employers. Figure 8 maps incidence across the country, and Table 22 in Appendix G reports the markup, markdown, incidence, labour share and median plant size for every state with at least 1,000 plant-years. Figure 9 plots incidence for the 21 states with at least 5,000 plant-years. This is the threshold a state must meet before it enters any ranking, so that no comparison rests on a handful of factories.

The ordering is not what a reader familiar with India’s industrial policy might expect. The five highest-incidence states are Chhattisgarh (66.9 per cent), Odisha (57.0), Jharkhand (56.9), Madhya Pradesh (55.4) and Uttarakhand (55.1). The five lowest are Tamil Nadu (31.0), Bihar (35.1), Kerala (36.0), Karnataka (37.5) and Maharashtra (38.9). The gap between the top and the bottom of this list is thirty-six percentage points. That is several times the movement in the all-India series over the decade, and far larger than any variation in the markup: median markups across the same states range from 1.15 to 1.48, and from 1.15 to 1.25 if Bihar is set aside. As Section 8.5 shows, the precise ranking of individual states is less certain than these point estimates suggest, although the overall spread survives resampling.

We report this ordering as heterogeneity, not as evidence of a particular mechanism. States are coarse proxies for labour markets. The data contain no commuting-zone boundaries, no district identifiers and no measure of employer concentration at the level at which workers actually search for jobs. The “one-employer town” reading of the mineral belt is therefore a plausible interpretation of the pattern, not a tested one. Establishing it would require definitions of local labour markets of the kind used by Berger et al. (2022), which the state-level geography of the ASI cannot support.

7.2 Monopsony is higher where manufacturing is smaller

Ranking states by the size of their organised manufacturing sector, rather than by incidence, produces almost the opposite order. This is the first substantive finding of the section, and Figure 10 plots the two measures against each other.

Tamil Nadu, the largest organised-manufacturing state in the sample, has the *lowest* monopsony incidence, and Maharashtra, the second largest, is fifth lowest (Table 22).

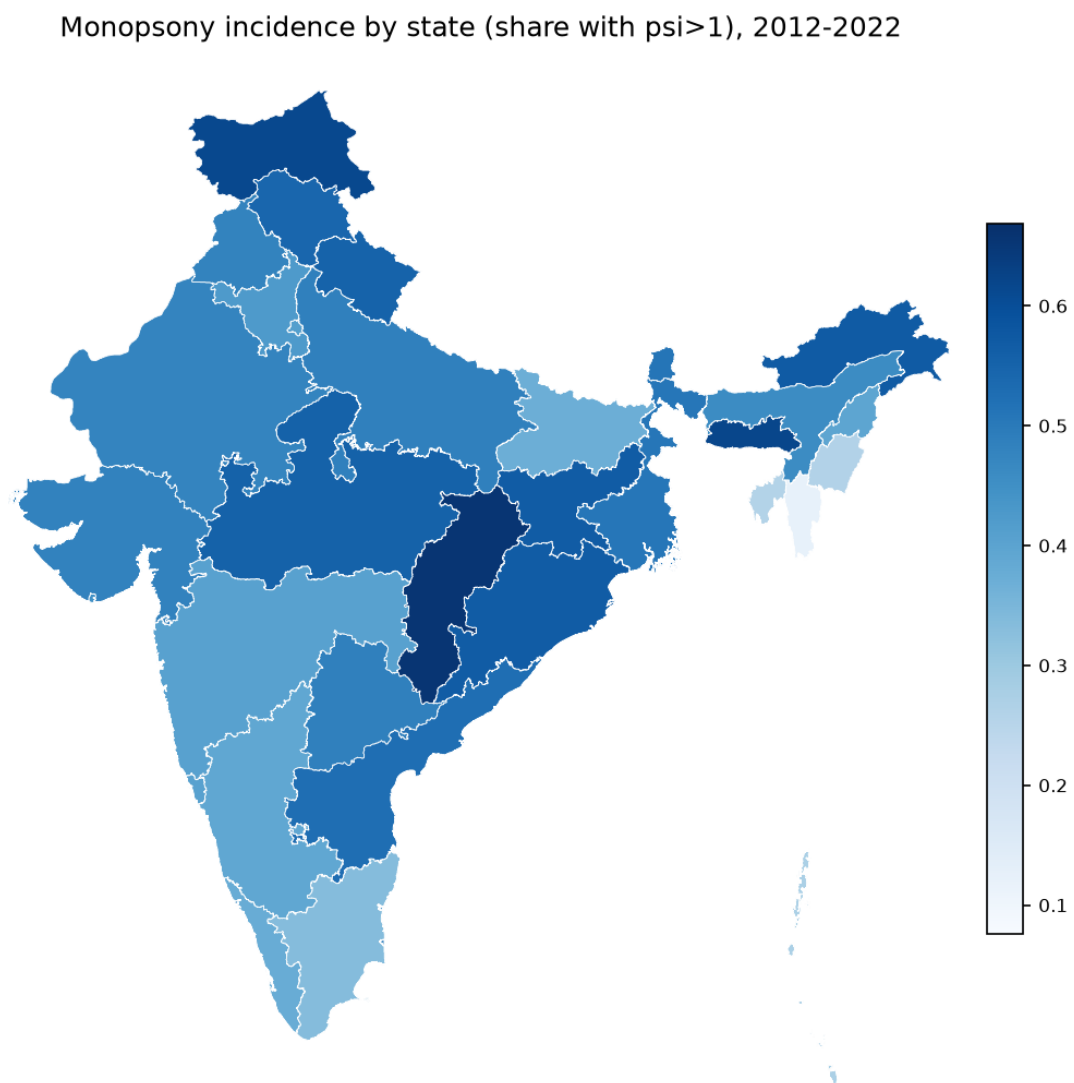


Figure 8: Monopsony incidence by state: share of plants with $\psi > 1$, pooled 2012–2022. Grey: no/thin data. Source: authors' estimates from the ASI panel.

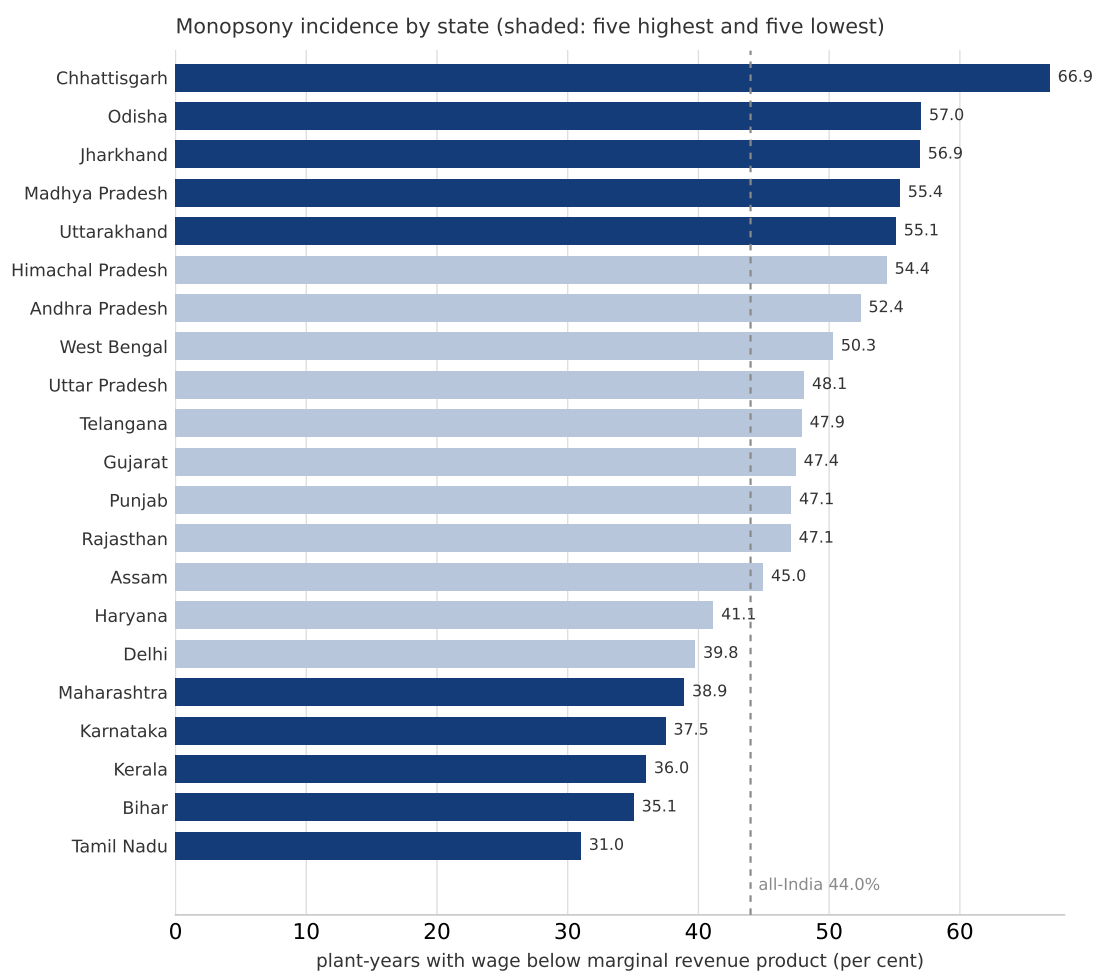


Figure 9: Monopsony incidence by state, 2012–2022 pooled: the share of plant-years in which the wage lies below the marginal revenue product of labour. States with at least 5,000 plant-years. The five highest and five lowest are shaded. Incidence excludes NIC18 throughout.

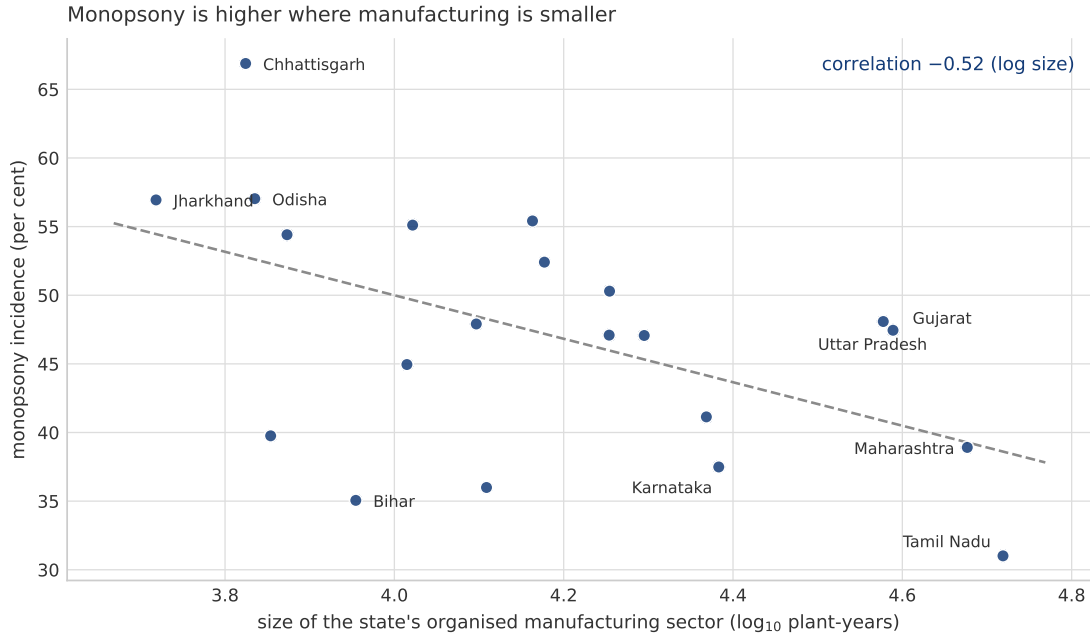


Figure 10: Monopsony incidence against the size of the state’s organised manufacturing sector, 21 states with at least 5,000 plant-years. The dashed line is an ordinary least-squares fit. The correlation in levels is -0.50 and in logs -0.52 (Spearman -0.51).

Chhattisgarh, Odisha and Jharkhand, which rank twentieth, nineteenth and twenty-first by size, rank first, second and third by incidence. The correlation between size and incidence is -0.50 in levels and -0.52 in logs. This is a cross-sectional association, and we do not claim a causal direction, since states are not randomly assigned an industrial base. The sign, however, is the one that the monopsony mechanism predicts. Where organised manufacturing is thin, a worker’s realistic set of alternative employers is small, and an employer’s wage-setting power is correspondingly large. This is the outside-options logic of wage-posting models (Manning, 2003; Berger et al., 2022), in which an employer’s power is the inverse of the labour-supply elasticity it faces, and that elasticity rises with the density of competing employers.

Two states complicate any simple reading, and we report them rather than smooth them over. Gujarat and Uttar Pradesh are third and fourth by size, yet their incidence, at 47.4 and 48.1 per cent, lies above the national level: size is not destiny. Bihar, for its part, combines incidence in the lowest quintile (35.1 per cent) with by far the highest median markup in the country (1.48, against a national figure of 1.22). This combination fits neither the thin-market story nor a story of dominant “superstar” firms, and we flag it as unexplained. One feature of the data is relevant, however: almost half of Bihar’s plant-years (4,316 of 9,004) are in non-metallic minerals, where the state’s median markup is 1.81 and its incidence only 0.03 (Table 23), so Bihar’s aggregate figures largely reflect a single division.

7.3 Composition, or the state itself?

The obvious alternative explanation is composition. Perhaps the high-incidence states simply contain more of the industries that are monopsonistic everywhere; mining, metals and chemicals are, after all, concentrated in exactly the states at the top of the list. This can be tested. Writing each group’s incidence as the plant-year-weighted average of its industry cells, the gap between the five highest- and five lowest-incidence states decomposes exactly as

$$\underbrace{\overline{\text{Pr}}_H - \overline{\text{Pr}}_L}_{22.4 \text{ pp}} = \underbrace{\sum_j (w_{Hj} - w_{Lj}) \text{Pr}_{Lj}}_{\text{composition}} + \underbrace{\sum_j w_{Hj} (\text{Pr}_{Hj} - \text{Pr}_{Lj})}_{\text{within-industry}}, \quad (18)$$

where Pr_{Hj} and Pr_{Lj} are monopsony incidence in division j among the five highest- and five lowest-incidence states, and w_{Hj} and w_{Lj} are that division’s shares of each group’s plant-years. The first term asks what the gap would be if every division had the low group’s incidence and only the industry mix differed between the groups. The second asks what the gap would be if both groups had the high group’s industry mix and only incidence within divisions differed. Because the two terms sum to the observed gap by construction, they partition it exactly. Both are computed over the 22 divisions that are present in both groups and have an incidence statistic; NIC18 is excluded from incidence throughout, and so cannot enter. The result is clear:

	contribution	share of gap
Composition (different industry mix)	+0.059	26.1%
Within-industry (same industry, different state)	+0.166	73.9%
Total gap	+0.224	100%

The identity holds exactly. Roughly seventy per cent of the gap between states reflects the same industry behaving differently in different states, and about a quarter reflects a different mix of industries. Twenty-one of the 22 divisions have higher incidence in the high-incidence states, tobacco being the exception. Figure 11 shows the pattern industry by industry: incidence in chemicals is 73 per cent in the high group against 53 per cent in the low group, in machinery 49 against 25 per cent, and in fabricated metal products 52 against 31 per cent.

This is the central result of the section, and it matters for policy. If the geography of monopsony were a matter of composition, the policy question would be an industrial one: which sectors a state hosts. It is not. It is largely a question about the state itself—plausibly about how thick its labour markets are, how mobile its workers are and how well its labour standards are enforced, although, as noted above, our data cannot distinguish among these. Two chemicals plants in Chhattisgarh and Tamil Nadu belong

The same industries carry different monopsony in different states

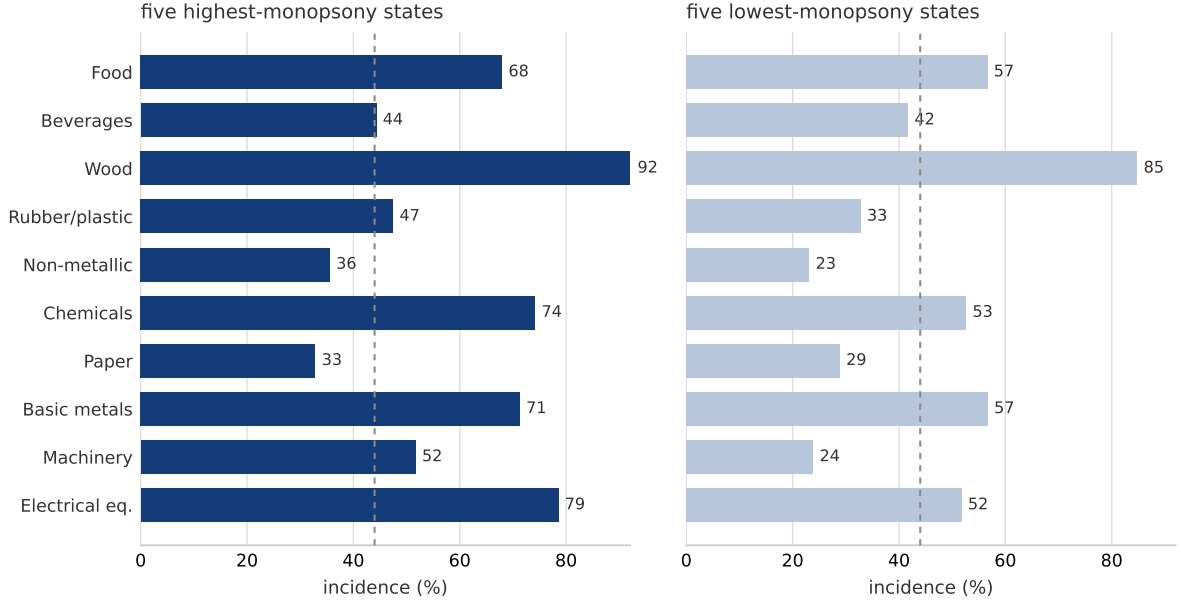


Figure 11: Monopsony incidence by division, in the five highest-monopsony states and the five lowest, for the ten divisions best represented in both groups. Dashed line: the all-India incidence of 44.0 per cent. The same industry carries systematically more monopsony in the high-incidence states, which is what (18) quantifies.

to the same division and are assessed against the same estimated technology, yet they face very different labour markets, and it is this difference that our estimates pick up.

7.4 Three-digit detail

Repeating the exercise at the three-digit level retains 799 state×NIC-3 cells with at least 100 plant-years each, covering 92.5 per cent of the estimation sample; the full set is tabulated in Appendix G. The three-digit results add detail rather than a new conclusion, and one caution governs their use. As Section 8.2 establishes, markdown *levels* at the division level are not determined to three significant figures, and state×three-digit cells are finer still, so their levels are correspondingly noisier. We therefore use the three-digit layer only for incidence and for orderings, never for the second decimal place of ψ . We do not compute standard errors at this level, because the bootstrap does not resample at it.

7.5 Policy: the state is the operative unit

Three implications follow. They are more specific than the national implications discussed in Section 11, because the variation across states is sharper.

First, the binding constraint differs by state, whereas much of industrial policy is set nationally. A sectoral incentive applied uniformly across India meets a labour market in

Chhattisgarh in which 67 per cent of plant-years already pay wages below the marginal revenue product of labour, and one in Tamil Nadu in which 31 per cent do. The same subsidy to the same industry will therefore not have the same effect on workers in the two places. [NITI Aayog \(2026\)](#) assesses sectors for their employment potential without a state dimension of this kind; our estimates supply one.

Second, the states that most need labour-market remedies are those least equipped to pay for them. With the partial exception of Madhya Pradesh, the high-incidence states are smaller industrial economies with correspondingly smaller fiscal capacity. Improving connectivity, enforcement and the provision of labour-market information all cost money, and the correlation in [Figure 10](#) suggests that the need is largest where the base to pay for it is smallest. This is an argument for a central role in precisely the instruments identified in [Section 11](#).

Third, variation across states is an opportunity for identification, not only a policy problem. Indian states differ in their labour regulation, and this variation has been used before to identify effects on manufacturing outcomes ([Besley & Burgess, 2004](#)), as have size-dependent thresholds in central legislation ([Bertrand et al., 2021](#)). Our within-industry decomposition shows that whatever drives the gap between states operates on the same industry across state boundaries, which is exactly the setting in which such variation can be exploited. We do not attempt this here, since the purpose of this paper is measurement, but the measurement makes such a design possible, and it is the natural next step for the companion paper on determinants.

8 Robustness and measurement

This section asks how far the estimates can be trusted. We first compare the translog and Cobb–Douglas estimators, and then show that the two wedges are not equally well determined, first by perturbing the numerical path to the solution and then by inspecting the geometry of the estimator directly. We next replace conventional standard errors with a bootstrap over the entire estimation pipeline, and apply the same bootstrap to the spatial results. The section closes with three measurement issues: the flagged industries, the assumption of Hicks-neutral technology, and the choice between measuring labour in heads or in efficiency units.

8.1 Cross-estimator diagnostics

The translog and Cobb–Douglas estimators agree where they should, and disagree where theory says they must. The markdown has a Spearman correlation of 0.88 across the two estimators, and the two classify 86.9 per cent of plants in the same way as monopsonistic or not. The rank correlation of the markup is lower, at 0.55, precisely because the translog’s

plant-specific θ^M absorbs within-industry heterogeneity that Cobb–Douglas forces into the markup. The analytic (Wooldridge-form) standard errors of the Cobb–Douglas elasticities lie between 0.0015 and 0.019. The 50-replication bootstrap run alongside them is degenerate for a third of industries, and we do not use it as corroboration (Appendix E). A reproduction suite re-derives the NIC-31 estimates to machine precision.

8.2 What is well determined, and what is not

The two wedges are not pinned down equally well by this estimator, and the asymmetry is structural rather than incidental. We state it explicitly because it determines how much weight the numbers in this paper can bear, and because it applies to every study that computes a markdown as a ratio of two elasticity-to-share ratios.

The diagnostics in this subsection and the next were computed under the previous specification of the first stage, without year effects (Appendix L), and have not been re-run. They concern a property of the estimator’s design, the small size of θ^L relative to θ^M , which the year effects leave intact, and we report them for that reason; the division counts and percentages below belong to that earlier specification.

We subjected the estimates to four perturbations. Each changes only the numerical path to the solution; none touches the data, the sample, the moment conditions or the starting values. The first replaces the optimiser’s finite-difference gradient with the analytic gradient. The second restarts the optimiser from the converged full-sample solution rather than from the OLS starting values. The third substitutes an algebraically identical, but differently ordered, implementation of the same analytic gradient, which agrees with the first element by element to within 10^{-14} . The fourth compares two independent programs that build *provably identical* moment conditions—with identical second-stage samples, first-stage fitted values and starting values, all verified to the last bit—and that differ only in the sequence of floating-point operations that follows.

All four perturbations produce the same pattern. The materials elasticity is reproduced to within a few hundredths of a per cent; the labour elasticity is not. Across the 22 admissible divisions, the change of gradient moves the mean θ^M by a median of 0.00 per cent, and the mean θ^L by a median of +0.33 per cent, but with a range of -84 to $+34$ per cent. Median markups move by a median of -0.004 per cent (range -3.0 to $+8.6$), and median markdowns by $+0.23$ per cent (range -82 to $+76$). Monopsony incidence is unchanged in the median division, but moves by up to 24 percentage points in the most extreme one. Even the fourth perturbation, in which the inputs are provably identical, moves the mean θ^L by about 0.2 per cent, while the mean θ^M moves by 0.01 per cent.

The reason is visible in the magnitudes. The markup is $\mu = \theta^M / \alpha^M$, with θ^M between 0.79 and 0.99 across divisions; the markdown is $\psi = (\theta^L / \alpha^L) / \mu$, with θ^L between 0.02 and 0.28. The same absolute displacement of the nine-coefficient vector is a rounding error

relative to a numerator close to one, but a large proportional shock to a numerator of about 0.1. Behind this arithmetic lies the estimator itself. The system is just-identified, with nine parameters and nine moments, and its objective function is nearly flat in several directions around the optimum, so the coefficient vector is only loosely pinned down; because the labour elasticity is small, this looseness matters far more for it, in proportional terms, than for the materials elasticity. Section 8.3 confirms this and refines the explanation. This is a property of the design, not of the Indian data, and it will arise wherever the production approach is used to recover a labour wedge from a gross-output technology dominated by materials.

Three consequences guide the rest of the paper. First, markup levels, trends, dispersion and rankings are robust, and we report them as findings. Second, aggregate markdown statistics are robust enough to carry the paper’s claims, and we report them, with standard errors, as findings. Pooled monopsony incidence moved by about two percentage points under the change of gradient, and by 0.09 of a percentage point (39.20 against 39.11 per cent, on the identical sample and under the earlier first stage) between the two programs of the fourth perturbation. Aggregation is what rescues the markdown: the division-level displacements are large, but roughly symmetric around zero (the median division moves by +0.23 per cent), so they largely cancel rather than accumulate. Third, *division-level* markdowns are not robust to three significant figures; we do not report them in that way, nor do we build arguments on the markdown level of any individual industry. Appendix K gives the full division-by-division comparison behind these figures, and the bootstrap of Section 8.4 quantifies all three statements rather than leaving them as assertions. This is also why a paper that could have reported a markdown to four decimal places reports an interval instead.

8.3 The geometry behind the fragility

Section 8.2 establishes the asymmetry by perturbation: when the numerical path changes, θ^M stays put while θ^L moves. That is evidence about how the estimator *behaves*. Its *geometry* can also be inspected directly, and doing so both confirms the conclusion and corrects the explanation that one would naturally give for it.

The system is just-identified—nine translog parameters and nine moments—so at the converged estimate the Jacobian $G = \partial \bar{g} / \partial \theta'$ is square, and $\hat{\theta} - \theta \approx -G^{-1} \bar{g}$. Everything about how noise in the moments is transmitted into error in the parameters is contained in that inverse. G is already formed analytically inside the verified gradient evaluator of Appendix E, and then discarded, since only $-2 \bar{g}' W G$ is returned. We recompute it at $\hat{\theta}$ for each division, take its singular value decomposition, and form delta-method standard errors for the mean elasticities that the paper actually reports. For a translog, these are linear combinations of four parameters each, rather than single coefficients.

The first finding is that the system is severely ill-conditioned in every division, without exception. With the columns of G normalised, so that the condition number measures the near-collinearity of the nine identification directions rather than the units of nine regressors, the condition number has a median of 2.9×10^6 and ranges from 1.7×10^5 to 1.8×10^7 . All twenty-three divisions exceed 10^5 . A condition number of this order means that the moment conditions place almost no restriction on some directions in parameter space. The “nearly flat objective” of Section 8.2 is therefore not a figure of speech but a measurable property, and this is the quantity that measures it.

The second finding is that the flat direction is not the labour direction, and this matters. The natural conjecture—that θ^L is poorly determined because it loads on the near-null direction while θ^M does not—is false here. The squared loading of the labour coefficient on the smallest right singular vector has a median of 0.06, against 0.13 for the materials coefficient. Nor are the two elasticities unequally determined in absolute terms: the delta-method standard error of the mean θ^L has a median of 0.416 across divisions, and that of the mean θ^M a median of 0.334; the median of the division-level *ratios* is 1.07. (The two are not the same statistic: the ratio of the medians is 1.24. We report the median ratio because it is the division-level comparison the argument needs.)

In absolute terms, the estimator determines the two about equally poorly.

The third finding is that the asymmetry is one of level, exactly as Section 8.2 argued, and that it can now be measured. The median mean elasticities are $\theta^L = 0.103$ and $\theta^M = 0.888$, a factor of 8.7 apart. The same absolute uncertainty is therefore a moderate relative error in the markup, but a disabling one in the markdown. Measured by coefficients of variation, θ^L has a median of 3.75 across divisions against 0.363 for θ^M , and the median of the division-level ratios is 12.3 (the ratio of the two medians is 10.3), and in eighteen of the twenty-three divisions the standard error of θ^L exceeds θ^L itself. At the division level, the labour elasticity cannot be distinguished from zero. This justifies the decision not to report division-level markdowns as precise quantities, and it explains why aggregation rescues the markdown: errors of this kind are roughly symmetric around zero, and so cancel across divisions rather than accumulating.

Two further points keep this diagnostic in proportion. First, the conditioning does *not* predict which divisions actually moved under the change of gradient: the correlation between the normalised condition number and the realised displacement of θ^L is -0.00 . This is what one should expect when ill-conditioning is universal. With every division at 10^5 or worse, there is no cross-sectional variation left for it to explain, and which division moves is then a matter of where the optimiser happens to stop. Second, these are division-level asymptotic standard errors for a mean elasticity. They are not the bootstrap standard errors of Section 8.4, which are clustered by plant, computed on the pooled sample, and attached to the reported statistics. The two answer different questions. This subsection asks whether the θ^L of any individual division can be trusted, and the

answer is no; Section 8.4 asks how precise the pooled markdown is, and the answer, under the earlier specification, is that its standard error is about fifteen per cent of its value. The two are consistent, and together they repeat the message of the paper as a whole: aggregate markdown statistics can carry the argument, and division-level ones cannot.

8.4 Inference

The bootstrap reported in this subsection and in Section 8.5 was run under the earlier first stage, without year effects and with NIC30 excluded, and it has not been repeated for the present specification. Its point estimates therefore differ from those reported elsewhere in the paper (the pooled median markup is 1.227 rather than 1.218, the median markdown 0.72 rather than 0.82, and incidence 39.1 rather than 44.0 per cent). We report it for what it says about *precision*: how much sampling uncertainty the whole pipeline carries, and by how much conventional standard errors understate it. Those are properties of the estimator and the sample, which the year effects do not alter in kind, rather than of a particular point estimate.

Every wedge in this paper is *constructed* from estimated quantities. The markup μ divides an estimated elasticity by an observed share; the markdown ψ divides one such ratio by another; and the elasticities themselves come from a two-stage procedure in which the first stage supplies a generated regressor to the second. A standard error that treats μ_{it} or ψ_{it} as data therefore answers the wrong question: it reports how precisely a regression fits numbers that are themselves estimates, on the assumption that those estimates are exactly right.

We replace such standard errors rather than supplement them. The procedure is a plant-clustered bootstrap over the *entire* estimation pipeline. Within each two-digit division we resample plants with replacement up to the original number of plants, stratifying by division so that small divisions cannot disappear from a replication. Every drawn copy of a plant receives a fresh identifier; without this, the specifications with plant fixed effects would pool duplicate copies into a single effect and understate the very uncertainty we are trying to measure. Within each replication we then re-run every step: the first-stage polynomial, the lags and instruments, the OLS starting values, the second-stage GMM for all divisions, the admissibility screen, the share correction, the winsorisation by year, and the aggregation. Re-running the first stage inside the loop is the point of the exercise, because it is what carries the uncertainty of the generated regressor through to the final numbers.

Three conclusions follow from Table 9, and the third should change how earlier work in this literature is read.

First, the two wedges are not equally well measured, and we can now say by how much. The pooled median markup has a standard error of about one per cent of its

Table 9: Bootstrap inference, earlier first stage (without year effects; NIC30 excluded), so the point estimates differ from those reported elsewhere. Point estimates are the estimator applied to the original sample; standard errors are the standard deviation across replications; intervals are percentile (2.5th, 97.5th). B is the number of replications in which the statistic was computable, and travels with the number rather than being averaged away. The four regression rows have a smaller B for a reason unconnected to the data, given in the text; for those rows the interval should be read as indicative and the standard error as the reliable quantity.

Statistic	Point	Bootstrap SE	95% interval	B
Median markup μ	1.2272	0.0137	[1.1986, 1.2507]	142
Median markdown ψ	0.7227	0.1107	[0.4996, 0.9090]	142
Monopsony incidence $\Pr(\psi > 1)$	0.3911	0.0463	[0.2994, 0.4729]	142
$\ln \mu$ on $\ln L$, NIC4 $\times$ year	0.0138	0.0043	[0.0080, 0.0243]	101
$\ln \mu$ on $\ln L$, plant + year	0.0436	0.0042	[0.0373, 0.0548]	101
$\ln \psi$ on $\ln L$, NIC4 $\times$ year	-0.1249	0.0520	[-0.2308, -0.0094]	101
$\ln \psi$ on $\ln L$, plant + year	-0.1701	0.0709	[-0.3422, -0.0689]	101

value; the pooled median markdown, of about fifteen per cent. This is not a difference of degree between two similar objects. It is the arithmetic of Section 8.2 appearing in the only place that matters for inference: a markup built on a materials elasticity close to one is sharply determined, and a markdown built on a labour elasticity of about 0.1 is not. Monopsony incidence inherits the imprecision of the markdown. Its standard error is about five percentage points, and its percentile interval runs from roughly thirty to forty-seven per cent. That is a wide band, and it is why this paper argues that monopsony is *concentrated in a large minority* of plants rather than defending the second decimal place of 39.1. The interval comfortably supports that claim, since even its lower end implies that almost a third of plants pay wages below the marginal revenue product of labour; it does not support anything stronger. The rural claim needs more care, which Section 8.5 provides. The pooled median ψ is 0.72, with a bootstrap interval of [0.50, 0.91], so the median plant is significantly *not* a markdown payer, and from 2014 onwards this holds in every year. Incidence is the share in the tail of a right-skewed distribution, and we report it as such.

Second, conventional regression standard errors are too small by roughly an order of magnitude. The conventional standard error on the markup-size gradient with industry $\times$ year fixed effects is 0.00031; the bootstrap standard error is fourteen times larger. Across the four specifications the inflation factor ranges from seven to twenty-one. A reported t -statistic of 43.4 becomes 3.2 under proper inference.

All four intervals nonetheless exclude zero, so every gradient in Table 9 survives as a finding with a definite sign. What does not survive is the precision that the conventional t -statistics advertised.

One of the four deserves a further comment, because it shows how much of that precision was illusory, and how many draws proper inference actually needs. The *con-*

temporaneous markdown–size gradient with industry×year fixed effects, -0.125 , has a percentile interval of $[-0.231, -0.009]$ on the 101 draws available for it. On the first 59 draws, the same interval was $[-0.193, +0.013]$ and included zero, while the standard error was almost unchanged (0.053 against 0.052). The tail moved; the spread did not. This is the normal behaviour of order statistics: a 2.5th percentile is estimated from the few most extreme draws, and needs many more draws than a standard deviation does. It is a concrete reason to rely on the standard error rather than on the endpoints of the interval whenever B is modest. We report both, and say which one carries the weight.

This is, in any case, a coefficient that the paper declines to interpret. Section 5.5 and Appendix I explain why the *contemporaneous* markdown–size gradient is mechanically contaminated: current employment enters α^L , and therefore ψ . The paper’s conclusions about size therefore rest on the *lagged* specifications instead. Those lagged specifications are *not* bootstrapped here, so their standard errors remain the conventional ones and are, by exactly the argument of this section, too small. We report their point estimates and t -statistics in the conventional way, note that their true precision is lower than stated, and do not rely on them for anything finer than a sign.

The general point extends well beyond this paper. Wherever the production approach places a constructed wedge on the left-hand side of a regression and reports conventional standard errors, those standard errors answer the wrong question—here, by a factor of seven to twenty-one.

Third, failures are counted, not dropped. A division-replication is recorded as failed when it yields no usable estimate: when there are fewer than 500 identifying observations, when the optimiser returns an error, or when the solution fails the admissibility screen. NIC30 fails the screen in almost every replication, as it did on the original sample under that specification, and accounts for most of the failures; the remaining failure rate is about one per cent of division-replications. The standard errors are therefore conditional on the estimator succeeding. Since the dominant failure concerns a division that is excluded from the reported aggregates in any case, this conditioning is close to innocuous, but we report the rate rather than simply asserting that it is.

One limitation arises from our implementation rather than from the data, and it explains why the table contains two values of B . In the first part of the first batch of replications, the four regression coefficients were not recorded: `reghdfe` did not run because a package it depends on was not installed, and the error trapping in the driver program—designed so that a failed regression could not abort a nine-hour run—suppressed any warning. Once the dependency was installed, the remaining replications recorded the coefficients, and a second batch with an independent seed was run specifically to increase their number of draws. Their B is therefore 101, against 142 for every other statistic. This does not create a selection problem: the failure was deterministic and unrelated to which plants were drawn, so the replications that carry the coefficients are a random subsample

from the same distribution. The two batches differ only in their seeds, and are pooled on that basis.

8.5 Inference on the spatial claims

Because the bootstrap of Section 8.4 resamples plants and re-runs the whole pipeline, it can be used to assess the spatial results as well as the aggregate ones. Since rural and urban incidence are computed on the same resampled plants and move together, the appropriate statistic is the *paired* difference, formed within each draw. Comparing two separate intervals, which here overlap almost entirely, would seriously understate the precision of the gap. Table 10 reports the paired results.

Table 10: Inference on the spatial claims, earlier first stage (without year effects; NIC30 excluded). Each statistic is formed inside every bootstrap replication and the percentile interval taken across replications, so correlations between the pieces are preserved. 143 replications. Source: `spatial_inference.csv`, `state_rank_stability.csv`.

Statistic	Point	95% interval	Draws > 0
Incidence, rural – urban	+0.032	[−0.018, +0.072]	85%
Across states, max – min	+0.415	[+0.322, +0.526]	100%
Across states, top 5 – bottom 5	+0.273	[+0.221, +0.330]	100%

The three results support different readings.

The spread across states is as solid as anything in this paper. Not one of the 143 replications closes it, and the interval for the gap between the top five and bottom five states, [+0.221, +0.330], lies well clear of zero. The fact that monopsony varies across Indian states by tens of percentage points is not an artefact of which plants happened to be sampled.

The ranking of states, by contrast, is much weaker than a map suggests. When the states are re-ranked within every draw, the 95 per cent rank interval of the median state spans nine positions out of twenty-seven. Only the extremes are pinned down: the highest-incidence state stays in the top five and the lowest stays in the bottom three, whereas a state ranked eleventh on the point estimate has a rank interval running from second to twenty-first. Readers should therefore treat the map as evidence that geography matters, not as a league table, and we have avoided building any argument on the position of an individual state.

Finally, the pooled rural–urban gap does not reach conventional significance, and two numbers have to be kept apart to see why. Under that specification the full-sample point estimate was $42.1 - 36.6 = 5.5$ percentage points (it is $46.4 - 41.9 = 4.5$ under the present one), and the gap ran between 4.0 and 6.8 points in every individual year. The *paired* bootstrap distribution — the rural-minus-urban difference recomputed inside each replication, which is the object a significance statement needs — is centred lower,

at a mean of +3.2 points and a median of +3.8. Its interval, $[-0.018, +0.072]$, includes zero, and 85 per cent of draws place it above zero: suggestive, but not decisive. The distance between 5.5 and 3.2 is bootstrap bias rather than a second estimate of the same thing, and it sits almost entirely in the urban arm, whose bootstrap mean exceeds its own point estimate by 2.1 points. We report the interval rather than the point estimate for the significance claim, and flag that the sampling distribution is centred below the point estimate rather than on it. The sign is stable in ways that matter—it is positive in all eleven years and in the large majority of draws—and the state-level gaps in West Bengal and Uttar Pradesh are large. The honest summary, however, is that this paper establishes a rural–urban *pattern* whose magnitude its estimator cannot pin down. The reason is the one identified in Section 8.2: incidence inherits the imprecision of the markdown, and a three-point difference is small relative to a five-point standard error on the level. Separating the two would require either a more sharply estimated labour elasticity or a design that differences out the first stage.

8.6 The flagged industries

In NIC 18 (printing) and NIC 30 (other transport equipment), 36.4 and 14.9 per cent of plants respectively report $\alpha^L + \alpha^M > 1$: their reported factor bills exceed their reported output, implying negative value added. This is a known reporting pattern associated with job work and contract manufacturing. The two industries are treated differently (Section 4.6). In NIC 18 the problem is severe and specific to the industry; because (14) divides by share-based objects, these reports produce pathological markdowns, and NIC 18 is excluded from every markdown aggregate. In NIC 30 the rate is close to the all-division average, and the industry is retained everywhere. Under the earlier first stage NIC 30 had a negative mean θ^L and failed the admissibility screen; with year effects its mean θ^L is 0.073, it passes, and there is no longer a ground for treating it differently from any other division.

8.7 Bounding the Hicks-neutrality bias

Both estimators impose Hicks-neutral productivity. If differences in technology are instead labour-augmenting, the measured elasticities of flexible inputs are inflated and labour elasticities are depressed (Demirer, 2025). The sign of this bias is known from theory. If labour-augmenting productivity varies across plants but the estimator imposes Hicks neutrality, the elasticity of the flexible input absorbs variation that belongs to technology. Markups are then overstated and, because ψ scales with θ^L , markdowns are understated. Demirer (2025) quantifies this bias for several countries and finds it to be material. We do not quantify it here, and no result in this paper depends on a re-estimation with factor-augmenting technology. The markup levels reported in this paper should therefore be

read as revenue-based estimates under maintained Hicks neutrality, with the direction of any bias as stated and its magnitude outside the scope of this paper.

Separating factor-augmenting technology from a labour markdown is not a matter of a better optimiser. [Rubens et al. \(2026\)](#) show that the two are not separately identified from the first-order conditions of production alone, because the same variation in cost shares is consistent with either. Identification requires information from outside the production function: a model of labour supply with a credible definition of the local market, employer shares, an outside option and an excluded instrument for the wage. These ingredients are not available in validated form for the ASI at the level of geography this paper can observe. Assembling them is the subject of separate work, not a robustness check that could simply be added to the present estimates.

8.8 Labour in heads or in efficiency units

One measurement choice deserves separate treatment, because it is the only point on which the two papers in this series disagree. The translog is estimated with labour measured as a *headcount*, $\ell = \ln(\text{persons})$, whereas the labour cost share is the *wage bill* over revenue, $\alpha^L = \text{emoluments}/(Py)$. Arithmetically the pairing is correct, since emoluments are the expenditure on persons. The difficulty is heterogeneity. θ^L is estimated as the elasticity with respect to one more worker, as though workers were interchangeable. A plant that pays above-average wages per worker therefore has a mechanically larger α^L and, since $\psi = (\theta^L/\alpha^L)/\mu$, a mechanically smaller measured markdown. Differences in skill composition would then appear as competitive wage-setting. The companion paper on misallocation measures labour as the wage bill precisely to avoid this problem, so the question cannot be left to a footnote.

The issue can be tested. Within an industry-year, regressing $\ln \psi$ on the log wage per person gives a slope of -0.374 (conventional s.e. 0.002): measured monopsony falls steeply with the wage level. The same regression for the markup gives -0.032 , which is the placebo result one would want, since μ depends on the *materials* share and has no such dependence by construction. A negative slope, however, is also what the economics predicts, since monopsony should be concentrated where wages are low, so this diagnostic alone cannot distinguish an artefact from a genuine finding.

Re-estimation settles the question. We rebuild the entire estimator with labour measured as the deflated wage bill, and change nothing else: the sample, the first-stage polynomial, the instruments, the weight matrix, the starting values, the analytic gradient, the acceptance rule and the winsorisation are all the same. The variant is generated automatically from the production script, so that the difference is provably a single line of economics. [Table 11](#) reports what changes.

Three conclusions follow, and they do not all point in the same direction.

Table 11: Labour as a headcount (baseline) against labour as the deflated wage bill. Same sample, same estimator, same numerical settings; only the labour variable differs. Source: `wagebill_vs_headcount.csv`, `wagebill_theta.csv`.

	Headcount (baseline)	Wage bill (efficiency units)
Slope of $\ln \psi$ on log wage per person	−0.374	−0.085
Monopsony incidence, unweighted	43.3%	52.9%
Monopsony incidence, survey-weighted	47.6%	57.5%
Median ψ	0.821	1.079
Interquartile range of ψ	0.35–1.79	0.51–2.12
Mean θ^L	0.115	0.136
Incidence, lowest wage decile	56.8%	53.2%
Incidence, highest wage decile	26.4%	46.4%
Correlation of division incidence across the two		0.517
Divisions with inadmissible $\theta^L < 0$	none	NIC26

First, the wage gradient was mostly mechanical. The slope falls from -0.374 to -0.085 , so more than three-quarters of the apparent concentration of monopsony among low-wage plants was due to the α^L denominator rather than to the labour market. The remaining -0.085 is consistent with the thin-market pattern this paper describes, but it is under a quarter of what the baseline implies. Across deciles of the wage per person, the corrected profile is close to flat from the first decile to the eighth (53.2 to 52.6 per cent) and falls only at the top, whereas the headcount specification produced a steady decline from 56.8 to 26.4 per cent.

Second, the correction strengthens the paper’s central claim rather than weakening it. Incidence rises from 43.3 to 52.9 per cent unweighted, and to 57.5 per cent once the sampling multipliers are applied—more than half of organised manufacturing. The median ψ rises from 0.821 to 1.079, so under this measure the median plant itself pays a markdown. A reader who considers the wage-bill measure the better one should read the headline incidence in this paper as an understatement, and the language of Section 5 as conservative.

Third, the division-level estimates do not survive the change. The correlation of division incidence across the two specifications is only 0.517, and individual divisions move by amounts that no reader should be asked to accept: fabricated metal products (NIC25) from 41.8 to 90.6 per cent, rubber and plastics (NIC22) from 38.0 to 82.6 per cent, and wearing apparel (NIC14) from 4.9 to 43.0 per cent, while electrical equipment (NIC27) moves in the opposite direction, from 63.5 to 26.7 per cent, and computer, electronic and optical products (NIC26) from 42.2 to 8.6 per cent. Under the wage-bill measure NIC26 returns an inadmissible negative labour elasticity where the baseline gave an admissible one. All of this restates the conclusion of Section 8.2 in a different form:

a labour elasticity of about 0.1, estimated from a nearly flat objective, is not robust to a defensible change in how labour is counted, and the division-level markdowns built on it are not objects that this paper reports to three significant figures.

We therefore retain the headcount specification as the baseline. It is the convention in the markdown literature to which this paper contributes, and the alternative leaves a division undefined. The wage-bill results are reported as the robustness check they are. The aggregate direction is the same under both measures, and the level is higher under the alternative. Neither measure identifies the ranking of divisions, as the paper has emphasised from the outset.

9 Extension: contract labour and the locus of monopsony

The markdown measured in this paper is a plant-level average. It tells us that, at a monopsonistic plant, the typical rupee spent on labour is paid below the revenue that labour generates, but it does not tell us *which* workers bear the wedge. Indian labour economics has a strong prior on this question. Workers hired through a contractor fall outside most of the protections of the Industrial Disputes Act, have no direct employment relationship with the plant, and face weaker outside options. The Contract Labour (Regulation and Abolition) Act of 1970 exists precisely because this category of worker was already understood to have weak bargaining power. If the markdown reflects employers' wage-setting power, it should therefore be larger where contract labour makes up a larger share of the workforce. This section tests that prediction, and finds at most a small effect in the predicted direction, one that the plant-level estimator cannot distinguish from zero. We first explain how the labour categories are recovered from the raw survey returns, then describe what they show, test whether the markdown is concentrated on contract labour, and finally ask whether the rural gradient in monopsony is itself a contract-labour gradient.

9.1 Recovering the categories

The test is possible because the ASI records employment by category, even though the standard panel discards this detail. Block E is keyed by the factory identifier crossed with the schedule serial number, and the serial number *is* the category: men, women and children employed directly, workers employed through contractors, total workers, supervisory and managerial staff, other employees, unpaid family members, and the associated subtotals. The production panel retains only the row with the largest headcount—the grand total—so the category structure was lost at the extraction stage rather than being missing from the source. We re-read Block E for 2012–2022 using the same verified column dictionary and the same fixed-width parsing, but keeping every row. This yields

5,139,634 block records across 644,815 factory-years.

The serial numbers are decoded from the arithmetic of the returns rather than from an assumed ordering, because mislabelling the contract row would invalidate everything that follows. Within a factory-year, men plus women equals directly employed workers in 100.0 per cent of cases, and directly employed plus contract workers equals total workers in 100.0 per cent of cases, on 644,815 factory-years.

Two further signatures confirm this reading: the supervisory row pays a median of Rs.412,151 per head, against Rs.113,708 for workers, and the row for unpaid family members pays exactly zero.

9.2 What the recovered data show

The recovered data yield three facts, all new to this paper; the first has the widest implications.

First, contract labour now makes up close to half of the organised manufacturing workforce, and its share is rising. Aggregating persons across all plants, the contract share rose from 37.7 per cent in 2012 to 45.7 per cent in 2022, as shown by the upper series in Figure 12(a). The mean share at the plant level rose from 23.5 to 30.4 per cent, and the proportion of plants using any contract labour at all from 37.6 to 47.4 per cent. Whatever else the decade did to employment in Indian factories, it moved a further eight percentage points of it onto contracts.

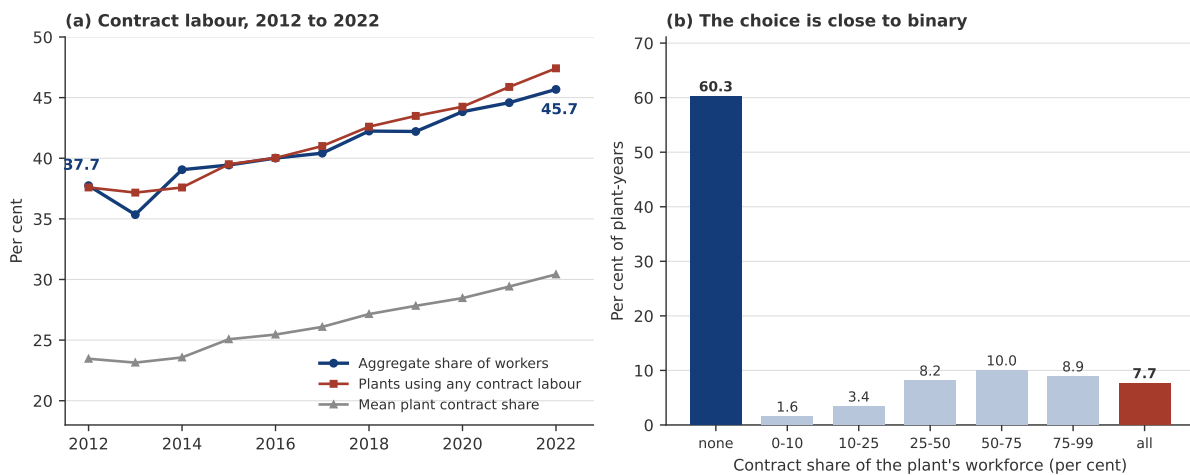


Figure 12: Contract labour in Indian organised manufacturing, 2012–2022. Panel (a): the aggregate share counts workers across all plants; the plant mean weights every plant equally; “using any” is the proportion of plants reporting a non-zero contract workforce. Panel (b): the distribution of the plant-level contract share, pooled over plant-years. The two shaded bars are the corners — plants using no contract labour and plants using nothing else — and they hold two-thirds of the sample between them, which is why the median plant share is zero in every year while the mean is close to a third. Source: `contract_descriptives.csv`, `contract_buckets.csv`.

Second, the choice to use contract labour is close to a corner solution. Sixty per cent of plant-years use no contract labour at all, and 7.7 per cent use nothing else; only 1.6 per cent lie between zero and a tenth (Figure 12(b)). This is why the median plant share is zero in every year, while the mean lies between a quarter and a third. The distribution is not a continuum of mixing but two distinct populations. It also means that the identifying variation in what follows comes mostly from comparing plants that have made opposite organisational choices, not plants that have fine-tuned a ratio.

Third, contract workers are paid about a quarter less than directly employed workers. Among plant-years with at least five workers of each kind, the median ratio of the contract wage per head to the direct wage per head is 0.751, with an interquartile range of 0.549 to 0.927. This penalty is stable over the decade and, if anything, narrows slightly, from 0.745 in 2012 to 0.767 in 2022. Contract intensity also differs sharply across industries: the aggregate contract share is 72.6 per cent in tobacco and 71.4 per cent in coke and refined petroleum, compared with 11.2 per cent in wearing apparel and 15.9 per cent in textiles. Industries with small core workforces, such as continuous-process industries, contract out; labour-intensive light manufacturing does not.

9.3 Does the markdown sit on contract labour?

Not detectably, and the way in which the test turns out is instructive.

A simple regression of the log markdown on the contract share within industry-years is not informative, for a reason established in Section 8.8. Contract workers are paid less per head, so a contract-intensive plant has a low wage per person, and a low wage per person mechanically raises ψ through the α^L denominator. Any association found without holding the wage level fixed would reflect this artefact. Table 12 therefore reports the regression in four versions: with and without a control for the wage, and under both the headcount and the wage-bill measures of labour. The wage-bill measure removes the mechanical channel by construction.

Table 12: The log markdown against the contract share of the workforce. All specifications absorb $\text{NIC-4} \times \text{year}$, so comparisons are within an industry-year. The wage-bill rows use the efficiency-units estimates of Section 8.8, in which the α^L channel is absent by construction. Source: `contract_regressions.csv`.

Labour measure	Control	Coefficient	Std. error
Headcount	none	−0.151	0.005
Headcount	log wage per person	−0.082	0.005
Wage bill	none	+0.002	0.004
Wage bill	log wage per person	+0.018	0.004
<i>Placebo: log markup</i>	log wage per person	+0.041	0.001

Under the headcount measure the coefficient is strongly negative, and it shrinks by

about half once the wage is controlled for: that is the mechanical channel. Under the wage-bill measure it is close to zero without the wage control and +0.018 with it, on 401,415 observations. This last specification is the one in which a genuine effect would have to appear, and its sign is the one the prior predicts. Two things keep us from reading it as a finding. The magnitude is small: moving from a plant with no contract labour to one with nothing else raises the log markdown by 0.018, under two per cent. And the standard error of 0.004 is conventional and treats the estimated markdown as data, which Section 8.4 shows understates uncertainty by a large factor. If the conventional standard errors are as optimistic here as they were in that bootstrap, the t -statistic of 4.0 falls below one at the smallest inflation factor found there.

The decile profile makes the same point more plainly. Sorting plants into deciles of the contract share within their industry-year, monopsony incidence under the wage-bill measure runs 53.5, 53.4, 52.9, 52.3, 52.7, 52.8, 53.6, 52.4, 52.2 and 53.2 per cent, from the decile that uses no contract labour to the decile that is 92 per cent contract. The profile is flat, as Figure 13(a) shows, while the headcount profile slopes downwards until the last decile. That slope is the mechanical channel made visible, not an economic gradient. Across states, the association runs the wrong way: the correlation between monopsony incidence and contract intensity is -0.37 .

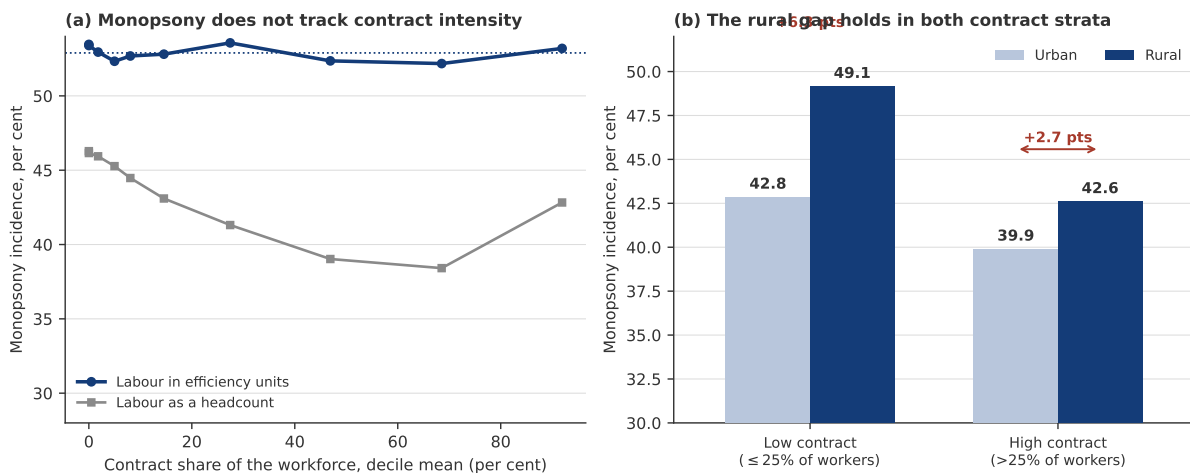


Figure 13: The markdown does not sit on contract labour. Panel (a): monopsony incidence by decile of the plant’s contract share, computed within industry-year, under both measures of labour. Under the efficiency-units measure — the one in which the mechanical α^L channel of Section 8.8 is absent by construction — the profile is flat from the plants that use no contract labour to those that use almost nothing else. The declining grey profile under the headcount measure is that mechanical channel, not an economic gradient. Panel (b): incidence by sector and contract intensity; the rural excess survives inside both contract strata. Source: `contract_deciles.csv`, `rural_contract.csv`.

What this does and does not establish. The evidence establishes that plants which contract out more of their workforce show at most slightly larger measured markdowns

than otherwise similar plants in the same industry and year. Given how strongly prior expectations point the other way, this is worth reporting. The limitation, however, is fundamental, and we state it plainly: *ψ is a plant-level average over the whole workforce, and this paper has no worker-level wage equation.* If contract workers face a steeper markdown than direct workers within the same plant, the plant-level average can be flat in the contract share even though the underlying wedge is concentrated entirely on contract workers. The observed wage penalty of 0.751 is consistent with that possibility, but does not distinguish it from an explanation based on compensating differentials. A proper test would require the markdown to be estimated separately for each category of labour, which the production function cannot deliver from these data, because output cannot be attributed to a particular category of labour.

Two readings therefore remain. Either contract status is not the margin on which Indian employers exercise wage-setting power, and the markdown is a property of the plant and its local labour market rather than of the contract; or it is that margin, and a plant-level estimator cannot see it. The first is what these estimates indicate; the second is what they cannot rule out. What can be stated without qualification is the descriptive finding: contract labour reached 45.7 per cent of the organised manufacturing workforce by 2022, it is paid about a quarter less, and neither fact has any visible counterpart in the plant-level markdown.

9.4 Is the rural gradient a contract-labour gradient?

Section 5.6 reported that monopsony incidence is higher among rural plants than among urban plants in every year of the sample, and Section 8.5 showed that the pooled size of that gap cannot be distinguished from zero once first-stage uncertainty is taken into account. The recovered categories allow us to ask a different and sharper question: whatever the rural gradient is, is it *made up of* contract labour? Rural labour markets are thinner, contract hiring is the margin through which Indian employers are usually said to exploit this, and if the two facts are really one fact, the paper’s geography and its labour-market mechanism collapse into a single story.

Answering this question requires the plant’s sector, which Block A records as item A9. We extract it for 2012–2022 using the same dictionary and parsing as for the labour categories, and validate it twice before use. Every one of the 719,496 factory-years parses to a value in $\{1, 2\}$, with none outside this set; and when matched against the sector lookup already maintained for this project, the two agree on all 719,496 plant-years. Applying the paper’s own conventions—the winsorised markdown, with NIC18 excluded—the extraction returns rural and urban incidence of 46.4 and 41.9 per cent, the figures reported in Section 5.6. It is the same variable.

Contract labour is indeed more common in rural areas. Rural plants have a mean

contract share of 31.0 per cent, against 23.1 per cent for urban plants, and 47.1 per cent of rural plants use any contract labour, against 37.5 per cent of urban plants. The difference is not due to industry composition: absorbing NIC-4×year effects, rural plants are 4.62 percentage points more contract-intensive than urban plants in the same industry and year (s.e. 0.11). Both the outcome and the candidate channel therefore point in the same direction, which is exactly the configuration in which a spurious mechanism is easiest to believe.

It is not, however, the channel. Table 13 and Figure 13(b) show monopsony incidence in the four cells defined by sector and contract intensity. Within both contract groups, and under both measures of labour, rural plants are the more monopsonistic ones. Under the headcount measure, the rural gradient is 49.1 against 42.8 per cent among plants with little contract labour, and 42.6 against 39.9 per cent among plants with a lot; under the wage-bill measure, it is 56.3 against 52.8 and 53.5 against 52.1 per cent. Within each sector, more contract-intensive plants are slightly *less* monopsonistic under both measures. Under the headcount measure this is partly the mechanical channel discussed above; under the wage-bill measure the difference is small, under three points in the rural sector and under one in the urban.

Table 13: Monopsony incidence by sector and contract intensity. “High contract” is a contract share above 25 per cent. NIC18 excluded and the markdown winsorised, as elsewhere in the paper. Source: `rural_contract.csv`.

		Headcount labour	Wage-bill labour
Urban	low contract	42.8%	52.8%
Urban	high contract	39.9%	52.1%
Rural	low contract	49.1%	56.3%
Rural	high contract	42.6%	53.5%

A regression makes the same point more precisely. Regressing the monopsony indicator on a rural dummy within industry-years gives a coefficient of 0.0257 under the headcount measure and 0.0165 under the wage-bill measure. Adding the contract share does not absorb either coefficient: they change to 0.0291 and 0.0168 respectively, increases of 13 and 1.5 per cent rather than falls. Contract intensity thus acts as a mild *suppressor* of the rural gradient, not as its source: rural plants use more contract labour, while contract labour has a slightly negative association with the markdown. Controlling for it therefore makes the rural gradient marginally larger.

Two qualifications apply. First, the standard errors behind the preceding paragraph treat the estimated markdown as data, which is precisely what Section 8.4 warns against. The bootstrap-based statement about the rural gap therefore remains the one in Section 8.5, where its interval included zero under the earlier specification. What this subsection establishes concerns the *structure* of the gap rather than its magnitude: whatever

the rural gradient is, it is not made up of contract labour, and it does not disappear when contract intensity is held fixed. Second, the plant-level limitation discussed in Section 9.3 applies here too. A plant-level markdown cannot detect a wedge that falls on contract workers within a plant, so this analysis rules out a compositional explanation of the rural gradient, but not a within-plant one.

What remains, then, is a rural gradient that is not explained by industry mix (Section 7.3) or by contract composition (this section), and whose magnitude is not precisely measured (Section 8.5). That is a narrower, but more useful, object than the one with which the paper began.

10 The static welfare cost

Two Harberger-style bounds summarise what is at stake. The logic is that of the textbook welfare triangle: a wedge τ between price and marginal cost, or between the marginal revenue product and the wage, destroys surplus of approximately half the wedge times the distorted expenditure, which is the linear approximation to the trade that does not take place. We compute, year by year,

$$DWL_t^p = \sum_i \frac{1}{2} \epsilon_t \left(1 - \frac{1}{\mu_{it}}\right)^2 s_{it}, \quad DWL_t^L = \sum_i \frac{1}{2} \left(1 - \frac{1}{\psi_{it}}\right)^2 \eta_{it} \omega_{it}^w, \quad \eta_{it} = \frac{1}{\psi_{it} - 1}, \quad (19)$$

where s_{it} is the plant's share of manufacturing sales, ω_{it}^w its share of the wage bill (the markdown term is summed over plants with $\psi > 1$ only), $1 - 1/\mu_{it}$ the plant's Lerner index, ϵ_t the demand elasticity taken from the companion misallocation paper's Kimball calibration for year t , and η_{it} the labour-supply elasticity implied by the plant's own markdown. Both are Harberger triangles: quadratic in the wedge, not linear, which is what makes them convex and therefore sensitive to dispersion.

Three properties of (19) shape how it should be read. First, it is computed *plant by plant*, so dispersion matters: the loss is convex in the wedge, and a mean-preserving spread of markups raises it. This is why the product-side bound can move proportionally more than the average Lerner index. Second, it is *static and partial*: it counts the trades forgone at given technology and given assignment of activity across plants, not the misallocation across plants that the same wedges induce, and it is therefore a lower bound on the true cost. Third, it is a *first-order* approximation by design: we use it to compare the two margins and to track them over time, not as a measure of the level of welfare.

On the product side, the sales-weighted Lerner index rises from 0.083 in 2012 to a peak of 0.167 in 2020, and ends at 0.126 in 2022. The associated deadweight-loss bound rises from 3.7 to 6.0 per cent of sales over the decade, peaking at 13.6 per cent in 2019. In that year the Lerner index was 0.160, barely different from 2020, while the loss bound was a third higher: the *dispersion* of markups was wider, and the loss is convex in the

wedge. On the labour side, the markdown triangle lies between 2.6 and 3.4 per cent of the wage bill throughout. Table 14 reports both series year by year.

Two qualifications govern how these magnitudes should be read. First, the sales and wage-bill totals are sums over the *estimation sample* and do not apply the ASI survey multipliers. They are therefore sample aggregates, not population aggregates for Indian organised manufacturing, and we describe them as such. Second, the product-side calculation takes its demand elasticity from the calibration of the companion paper on misallocation rather than estimating it here, and both sides use winsorised wedges. We therefore present these series as an illustrative calculation for comparing the two margins and tracking their movement over time, not as an estimate of welfare.

Table 14: Static welfare bounds by year: sales-weighted Lerner index; product-market deadweight loss (per cent of sales); labour-market (markdown) deadweight loss (per cent of the wage bill).

Year	Lerner	DWL _p (% sales)	DWL _L (% wage bill)
2012	0.083	3.7	3.0
2013	0.085	4.5	2.8
2014	0.116	5.6	2.7
2015	0.140	6.6	2.6
2016	0.149	8.7	2.6
2017	0.148	8.5	2.8
2018	0.143	8.5	2.9
2019	0.160	13.6	2.7
2020	0.167	10.2	2.7
2021	0.140	6.5	3.2
2022	0.126	6.0	3.4

Both static bounds are small relative to the allocative cost that the same wedges induce across plants. That cost is the subject of the companion paper on misallocation, which finds that the labour-side wedge is the main driver of labour misallocation in India. Static triangles understate the cost of monopsony precisely because its harm is allocative.

11 Conclusion

11.1 What we found

This paper set out to measure how much market power Indian factories hold, on both sides of the firm, and the answer is lopsided. Product-market power is modest, spatially uniform and rising only slightly. The median markup is about 1.22, and eighteen of the twenty-three divisions lie between 1.09 and 1.41. The sales-weighted aggregate rose by 4.6 per cent over the decade, two-thirds of it because activity shifted towards plants that already had high markups rather than because the typical plant raised its own; and

concentration *fell*, with the mean employment Herfindahl index across industries declining from 0.086 to 0.066. This is not what an economy in which firms were steadily gaining pricing power would look like.

Labour-market power is a different matter. In 44.0 per cent of plant-years (48.4 per cent once the sampling multipliers are applied), the wage lies below the marginal revenue product of labour, and the figure never falls below 40.2 per cent in any year of the sample. Incidence is higher in rural than in urban plants in *every* year, at 46.4 against 41.9 per cent pooled, and the rural–urban gap approaches twenty percentage points in West Bengal and Uttar Pradesh. In the bootstrap of the earlier specification, however, the confidence interval of the pooled gap included zero, so the pattern is consistent over time without being precisely measured. The two margins do not share a geography, and they do not share an industry ranking: wearing apparel has the highest markup in Indian manufacturing but the second-lowest monopsony incidence. Whatever determines the wage-setting power that Indian factory workers face, it does not appear to be tied to the pricing power of the plants that employ them. Nor, as far as a plant-level estimator can see, is it driven by contract status. Section 9 recovers the labour categories from Block E and finds that contract labour made up 45.7 per cent of the workforce by 2022 and was paid about a quarter less per head, yet once the wage level is held fixed its association with the plant’s markdown is small—under two per cent across the whole range of contract intensity—and flat when read off the incidence profile. The mechanism that the literature most often assumes therefore finds little support at the plant level. The finding has an obvious limit: ψ averages over a plant’s whole workforce, so a flat plant-level profile cannot rule out a wedge that falls on contract workers within the plant.

Nor is labour-market power mainly a matter of the industry a plant belongs to. Only 31.5 per cent of the plant-level variance of the log markdown lies between the twenty-two divisions, and across those divisions the markup and monopsony incidence are negatively correlated (-0.53): the industries with the most product-market power are not those with the most labour-market power. This cross-industry comparison should nonetheless be read with the caution of Section 8.8, where division-level incidence proves sensitive to how labour is measured.

Nor is it driven by the industries that a state happens to host. Monopsony incidence ranges from 31.0 per cent in Tamil Nadu to 66.9 per cent in Chhattisgarh, and 73.9 per cent of the gap between the five highest- and five lowest-incidence states remains once the industry mix is held fixed: twenty-one of the twenty-two divisions common to both groups have more monopsony in the high-incidence states. What differs across state boundaries is, plausibly, the labour market rather than the technology. Incidence is also higher where organised manufacturing is smaller, with a correlation of -0.50 between incidence and the size of the sector. This is the sign that a thin-market mechanism predicts, although we cannot establish the direction of causation.

11.2 Why it matters

The question that motivated this paper was whether market power explains the long decline in labour’s share of value added in Indian manufacturing—a fall of thirteen percentage points between 1980–81 and the trough of 2007–08. This paper cannot answer that historical question directly, because the decade it measures begins after the trough. It does, however, provide something the question requires: it establishes which kind of market power is actually present, and the answer redirects the search. The superstar-firm account that dominates the literature on rich countries (Autor et al., 2020; Barkai, 2020) requires rising concentration and rising markups. Indian manufacturing over 2012–2022 shows falling concentration and an aggregate markup that rose by a few per cent, mostly through reallocation. The worker-power account (Stansbury & Summers, 2020; Manning, 2021) requires employers to hold wage-setting power, and Indian manufacturing shows that power in a large minority of plants.

The distinction is more than a matter of emphasis. It changes what should be measured next, and it changes what policy should do.

11.3 Policy implications

Four implications follow from the findings. We present them in descending order of confidence.

First, the more pressing competition question in Indian manufacturing concerns the price of labour, not the price of goods. Competition policy in India is built around product markets: combinations, abuse of dominance and cartels. Our estimates do not suggest that this is where the distortion binds. Employers’ wage-setting power in thin local labour markets is invisible to product-market antitrust, to industry-level concentration statistics, and to the price indices that competition authorities monitor. Monopsony under which 44 per cent of plant-years pay wages below the marginal revenue product of labour would leave no trace in any of them.

Second, industrial policy that targets output and investment does not reach this margin. The Make in India initiative and the Production Linked Incentive schemes are large: they cover fourteen sectors with an outlay of Rs. 1.97 lakh crore, and roughly Rs. 2.16 lakh crore of investment and about 1.44 million direct and indirect jobs have been attributed to the schemes (PIB, 2025). These are instruments for capacity, exports and competitiveness, and on their own terms they may well succeed. But they operate on the product market and on the cost of capital. If the wedge that separates workers from the value they produce reflects employers’ wage-setting power, subsidising output need not close it; and because such schemes favour larger plants, they may not be neutral with respect to it, since our size gradients show that, within industries, larger plants have *larger* markdowns. Whether the schemes widen or narrow the labour wedge is a question worth

asking directly, and our estimates provide the measure with which to ask it.

Third, sectoral priorities and the labour wedge do not line up, and this has not yet been examined. [NITI Aayog \(2026\)](#) identifies twelve priority sectors for positioning India as a global manufacturing hub, assessed on market potential, infrastructure, policy support, technological readiness and employment potential. Placing that list beside our estimates reveals a pattern that the assessment framework does not currently capture. Four of the twelve sectors correspond to divisions with some of the highest monopsony incidence we measure—steel and basic metals (0.68), chemicals (0.64), food processing (0.62) and pharmaceuticals (0.62)—while two correspond to divisions with comparatively low incidence, textiles (0.07) and leather and footwear (0.17), and automobiles (0.47) lies close to the middle. Textiles deserves particular attention: it supports more than 45 million livelihoods and is India’s second-largest employer after agriculture ([NITI Aayog, 2026](#)), and it is also a sector in which our estimates find workers comparatively unlikely to be paid below their marginal revenue product.

We do not claim that industrial policy creates these wedges, and nothing in our design could establish that. The observation is narrower, but still useful: “employment potential” as currently assessed counts jobs, not what those jobs pay relative to what they produce. Two sectors with identical employment potential can differ sharply in whether their workers capture the value they generate, and our estimates indicate that Indian manufacturing divisions do differ, by a factor of more than fifteen in monopsony incidence (from 0.05 to 0.87). These division-level figures are sensitive to how labour is measured (Section 8.8), so they should be treated as indicative rather than as a ranking to be applied mechanically. A sectoral strategy aimed at inclusive industrialisation, as called for by Sustainable Development Goal Target 9.2, nonetheless has reason to take the wedge into account. It is measurable, and it could be used as a criterion.

Fourth, the levers that would make a difference are labour-market levers, and they are geographic. If monopsony arises because workers in thin markets have few employers to choose from, the remedies are those that thicken these markets or improve workers’ outside options: physical and digital connectivity that widens commuting range, portability of benefits and employment records across employers and states, credible information about wages, and limits on contractual terms that restrict mobility. Our rural–urban gradient—present in every year and largest in West Bengal and Uttar Pradesh, although not significant in the pooled bootstrap—is suggestive about where such measures would matter most. The firmer spatial guide is the spread across states, which no resampling closed. We stop short of prescribing specific instruments, because this paper identifies where the wedge is, not which intervention would close it.

11.4 Limitations

The claims above have limits, and we prefer to state them rather than leave the reader to discover them.

Revenue, not quantities. The ASI records revenues and expenditures, not physical prices and quantities. What we estimate are therefore revenue elasticities, and the *level* of the markup is subject to the well-known critique of that practice (Bond et al., 2021; Raval, 2023). Our questions were chosen to rely on dispersion, trends, incidence and geography rather than on levels, but a reader interested in the level alone should treat it with the caution set out in Section 8.

Hicks-neutral technology. Both estimators impose Hicks neutrality. If technology is instead labour-augmenting, markups are overstated and markdowns understated (Demirer, 2025), and the two wedges are not separately identified from the first-order conditions of production alone (Rubens et al., 2026). We state the direction of the bias but do not quantify it, because doing so credibly would require a model of labour supply that these data cannot support.

The markdown is fragile where the markup is not. Section 8.2 is a limitation as much as a finding. The labour elasticity is small and loosely determined, so division-level markdowns move substantially under perturbations that leave the data untouched, and division-level incidence changes considerably when labour is measured as the wage bill rather than as a headcount (Section 8.8). Aggregates survive; the markdown levels of individual industries should not be read to three significant figures, and we have not built arguments on them.

Trends inherit the first stage. An earlier draft of this paper reported a markup that fell by about eight per cent over the decade. The decline came from a trend in the first-stage residual, which the share correction passes into every markup, and it disappeared once the first stage included year effects (Appendix L). Any trend in a production-approach markup is exposed to this channel, and we would urge readers of the wider literature to ask whether the first stage of a given study can produce one.

The geography is not a labour market. Our finest spatial unit is the state crossed with the rural–urban indicator, and neither is a labour market in the sense that a monopsony mechanism requires. Thin-market wage-setting operates within commuting zones; an Indian state is two orders of magnitude larger than a commuting zone, and “rural” within a state is a residual category rather than a place. Our rural–urban and state gradients therefore average over exactly the variation that a monopsony story is about. If employer

concentration is local, the magnitude of these gradients is a lower bound on what finer geography would reveal. This is a limitation of the source, not a modelling choice. We checked the ASI record layouts for every year of the sample: the unit-level files contain no district, tehsil, town, village or pincode field, and the only geographical information in them is the state code embedded in characters six and seven of the factory identifier, together with the rural–urban flag in item A9. Establishing that monopsony here is a *thin-market* phenomenon, rather than one that merely varies across states, would require finer geography than MOSPI releases at the unit level, or worker-flow data identifying who could plausibly have moved where. We therefore report the spatial variation as a measured fact—and Section 8.5 assesses how precisely it is measured—while treating the thin-market reading as the most plausible hypothesis rather than as a result this paper establishes.

The organised sector only. The ASI covers registered factories. Most manufacturing employment in India is in the unorganised sector, and the wage-setting power faced by those workers is invisible here—and, on any reading of the mechanism, plausibly greater. Nothing in this paper should be extrapolated to that sector.

Association, not causation. The size gradients, the geography and the industry patterns are conditional associations. We have no experiment and no instrument for employer concentration, and we do not claim that size causes wage-setting power or that thin markets cause monopsony. What we claim is measurement: these wedges exist, at this magnitude, with this distribution.

One decade. The sample begins in 2012–13, after the labour share had already fallen and begun to recover. The paper therefore measures market power in a period of partial recovery, and cannot by itself attribute the earlier decline to any cause.

11.5 Next steps

The companion papers pursue the two natural continuations: tracing what these wedges do to allocative efficiency, and asking what trade policy, exchange rates and geography do to the wedges. Two further extensions follow from the limitations above. The first is quantity-based estimation using the physical product and materials information in the ASI item blocks, which would relax the revenue-elasticity assumption at its root. The second is a model of labour supply with a credible definition of the local labour market. This is what would be needed to separate factor-augmenting technology from a genuine markdown (Rubens et al., 2026), and to move from measuring the wedge to identifying its causes.

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A Panel construction and validation

Raw structure. Each ASI round consists of ten fixed-width block files: A (identification: factory identifier, NIC-2008 activity code, state in characters 6–7 of the identifier, rural/urban status in item A9, scheme and multiplier), B (organisation), C (fixed assets, by asset class), D (working capital), E (employment and labour cost, by employee category), F (other expenses), G (other output), H (indigenous inputs, by item), I (imported inputs, by item) and J (products, by item, with gross sale and ex-factory values). Table 15 summarises the blocks and their use in this paper. Field positions differ across six format eras (from 2008–09 to 2022–23). Every block of every year is read against a column dictionary parsed from the official layout workbooks, and multi-item blocks are collapsed to the factory’s grand-total row. The rule that selects this row was validated against the additive identity of its components on the 2012–13 round.

Table 15: ASI block structure and its use in this paper.

Block	Content	Use here
A	Identification	factory id; NIC-2008 activity; state (id chars 6–7); rural/urban (A9); multiplier
B	Organisation	—
C	Fixed assets	net/gross capital, additions, depreciation (collapsed to totals)
D	Working capital	—
E	Employment & labour cost	persons engaged; wages; employee categories (collapsed)
F	Other expenses	—
G	Other output	—
H	Indigenous inputs	materials, by item (collapsed to totals)
I	Imported inputs	materials, by item (collapsed to totals)
J	Products	gross sale and ex-factory output values (collapsed)

Build chain. Within each round, the blocks are merged one-to-one on the trimmed eight-character identifier, and the rounds are then appended. This yields 935,975 factory-years for 2008–2022, with each identifier×year combination verified to be unique. The construction then proceeds in five steps.

1. Retaining operating units—those with positive output, materials, labour, capital and multiplier—leaves 634,124 factory-years.
2. Deflating at 2011-12 base prices restricts the sample to 2012–2022, leaving 495,148 plant-years. Output and materials are deflated with industry WPIs, with documented fallbacks for petroleum and repair; capital is deflated with the machinery

WPI; and all deflators are fiscal-year averages of the monthly series published by the Office of the Economic Adviser (OEA).

3. Restricting to manufacturing divisions NIC 10–33 leaves 481,322 plant-years on 142,493 factories.
4. Requiring each factory to be observed in at least two consecutive rounds, as the ACF moment conditions require, leaves the estimation sample of 423,127 plant-years on 84,298 factories.
5. The translog covers 422,820 of these plant-years (excluding NIC33), all of which lie in divisions that pass the admissibility screen. Its nine parameters are identified from the 290,023 plant-years with a valid $t - 1$ observation.

Three checks validate the build. Per-year medians evolve smoothly across the boundaries between format eras; the ratio of emoluments to wages lies between 1.10 and 1.19 throughout; and a reproduction suite re-derives the calibration and estimation outputs to machine precision.

Identifier structure (verified). Characters 6–7 of the factory identifier contain the 2011-census state code. This is stated in the official record layout and confirmed empirically: the frequency ranking reproduces India’s manufacturing geography, with Tamil Nadu accounting for 12.5 per cent, Maharashtra 11.2 per cent, Gujarat 9.2 per cent and Uttar Pradesh 8.9 per cent of estimation plant-years. Item A9 (1 = rural, 2 = urban) is present in every round and merges to 100 per cent of estimation observations.

B Year-by-year supporting series

This appendix reports two reference series that the main text summarises but does not display in full: the size and composition of the estimation sample, round by round (Table 16), and the rural–urban margin, year by year (Table 17).

C Industry-level translog estimates

Table 18 reports the translog estimates for each division, pooled over 2012–2022.

D The Cobb–Douglas benchmark

The Cobb–Douglas benchmark, estimated by LP/ACF with a single elasticity vector per industry, yields economy-wide mean markups of 1.31–1.38 across years, with no trend. Its ordering of industries is essentially the same as that of the translog, and it produces

Table 16: Sample coverage by round. Plant-years are the *Cobb–Douglas/ACF* estimation sample, summing to 423,127; the translog is estimated and aggregated on 422,820 of these, in 23 divisions, reported by year in Table 5. The rural share refers to the raw factory panel, a wider frame than either estimation sample.

Year	Plant-years	Rural share (%)
2012	33,199	41.8
2013	36,396	42.6
2014	38,436	43.0
2015	37,358	43.0
2016	39,099	43.8
2017	39,227	44.3
2018	39,412	44.0
2019	39,750	44.8
2020	40,267	45.2
2021	40,905	45.1
2022	39,078	45.6

Table 17: The rural–urban margin, year by year. The markup columns are means weighted by the ASI survey multiplier and so refer to the organised-manufacturing population; the incidence columns are *unweighted* plant-year shares. The two halves of the table are therefore not on a common weighting basis, and the rural–urban markup and incidence gaps should not be compared as though they were. The rural excess in incidence appears in every round on either basis.

Year	Weighted mean μ		$\Pr(\psi > 1)$	
	Rural	Urban	Rural	Urban
2012	1.251	1.208	0.514	0.481
2013	1.242	1.206	0.500	0.470
2014	1.243	1.213	0.482	0.452
2015	1.262	1.236	0.451	0.418
2016	1.270	1.248	0.441	0.402
2017	1.269	1.248	0.444	0.400
2018	1.264	1.245	0.454	0.402
2019	1.272	1.267	0.434	0.373
2020	1.273	1.281	0.439	0.376
2021	1.244	1.237	0.476	0.418
2022	1.234	1.226	0.485	0.429

Table 18: Translog DLW estimates by NIC-2008 division, pooled 2012–2022. N = plant-years over which elasticities are evaluated (summing to 422,820); the stage-2 GMM that identifies the parameters uses the subset with a valid $t - 1$ observation, 290,023 in total and reported per division in the replication files. $\bar{\theta}$ = mean elasticities; RTS = returns to scale; $\Pr(\psi > 1)$ = monopsony incidence; medians of winsorised μ and ψ . † NIC18 is excluded from markdown aggregates (Section 4.6); its markdown columns are shown for completeness. Every division passes the admissibility screen. All rows are the estimator applied to the original sample within the bootstrap (replication 0), so that each point estimate and its standard error come from one computation. Convergence flags are reported in Appendix F.

NIC2	N	$\bar{\theta}^L$	$\bar{\theta}^K$	$\bar{\theta}^M$	RTS	$\Pr(\psi > 1)$	med μ	med ψ
10	62,535	0.080	0.013	0.927	1.019	0.624	1.103	1.358
11	9,375	0.125	0.052	0.874	1.051	0.412	1.339	0.672
12	5,542	0.186	0.000	0.855	1.041	0.390	1.701	0.662
13	29,560	0.034	-0.028	0.992	0.999	0.065	1.217	0.191
14	18,947	0.019	0.026	0.964	1.009	0.049	1.724	0.051
15	8,448	0.056	0.092	0.881	1.028	0.171	1.246	0.225
16	9,049	0.279	0.053	0.788	1.120	0.869	1.063	2.879
17	12,113	0.080	0.005	0.923	1.008	0.384	1.185	0.780
18†	6,690	0.040	0.039	0.859	0.938	0.004	0.995	0.135
19	5,182	0.088	0.000	0.936	1.024	0.675	1.147	2.113
20	28,962	0.127	0.030	0.881	1.038	0.640	1.217	1.343
21	15,237	0.270	-0.020	0.845	1.094	0.621	1.402	1.437
22	22,780	0.081	0.013	0.937	1.031	0.380	1.245	0.788
23	39,980	0.168	0.036	0.823	1.027	0.249	1.367	0.586
24	24,838	0.088	0.002	0.928	1.018	0.679	1.100	1.774
25	22,969	0.122	0.014	0.898	1.034	0.418	1.225	0.845
26	8,497	0.191	-0.049	0.925	1.067	0.422	1.431	0.786
27	22,357	0.150	0.021	0.875	1.046	0.635	1.147	1.348
28	27,019	0.129	0.036	0.869	1.034	0.358	1.302	0.675
29	18,629	0.140	-0.008	0.944	1.076	0.470	1.325	0.935
30	7,775	0.073	0.020	0.912	1.005	0.258	1.253	0.334
31	5,593	0.174	0.007	0.867	1.048	0.519	1.260	1.050
32	10,743	0.120	0.060	0.840	1.020	0.318	1.291	0.579

the cross-estimator agreement statistics reported in Section 8. Table 19 reports the series by year, and Table 20 the estimates by industry. The benchmark was estimated once and has not been re-estimated for this draft. Its markups divide by the *uncorrected* materials share, θ^M/α^M , so they do not pass through the first-stage residual at all; that they show no trend is consistent with the diagnosis in Appendix L, where the decline in the earlier translog series is traced to that residual.

A note on the 2019 labour composite. The composite column of Table 19 shows 3.640 in 2019, against 2.08–2.53 in every other year. This is an artefact of the untrimmed mean, not an economic event. In 2019 the median composite is 1.002, the 90th percentile 4.155 and the 99th percentile 16.773, all in line with the other years; the median is in fact the lowest of the eleven years. The entire excess lies beyond the 99th percentile: the standard deviation is 315.4, against 5.1–14.2 in other years, and the maximum is 62,831. Eight plant-years out of 39,738 (0.02 per cent) exceed 100, two exceed 1,000, and one exceeds 10,000.

Removing the top 0.1 per cent brings the 2019 mean down to 1.921, and a symmetric 1/99 within-year trim brings it to 1.732, compared with 1.743 in 2017 and 1.728 in 2020. The 2020 value shows a milder version of the same problem (maximum 15,797). We report the untrimmed series for transparency and flag the issue here rather than trimming it silently. The markup and markdown columns, which are winsorised, show no such behaviour, and no result in the paper is computed from the composite column.

Table 19: Cobb–Douglas benchmark, by year: mean markup, labour-based composite $\theta^L/\alpha^L = \mu\psi$, and markdown. The composite is an *untrimmed* mean and its 2019 value is not comparable to the others: see the note below the table. N is the Cobb–Douglas/ACF estimation panel (423,127 plant-years in total, as in Table 16); the composite column is computed on the 422,820 of those for which it is defined, 307 fewer, so the per-year counts behind that column run a few dozen below N — 39,738 against 39,750 in 2019.

Year	$\bar{\mu}^{CD}$	$\overline{\mu\psi}^{CD}$	$\bar{\psi}^{CD}$	N
2012	1.321	2.532	1.911	33,199
2013	1.310	2.264	1.859	36,396
2014	1.318	2.211	1.787	38,436
2015	1.346	2.132	1.679	37,358
2016	1.356	2.080	1.619	39,099
2017	1.353	2.128	1.623	39,227
2018	1.345	2.097	1.673	39,412
2019	1.367	3.640	1.618	39,750
2020	1.376	2.415	1.605	40,267
2021	1.327	2.217	1.805	40,905
2022	1.323	2.260	1.865	39,078

Table 20: Cobb–Douglas estimates by division: elasticities, returns to scale, markup and markdown.

NIC2	θ^M	θ^L	θ^K	RTS	μ^{CD}	ψ^{CD}
10	0.953	0.098	0.028	1.079	1.188	3.379
11	0.862	0.138	0.022	1.022	1.433	1.812
12	0.859	0.179	0.010	1.048	2.043	1.806
13	0.951	0.041	-0.013	0.978	1.301	0.507
14	0.881	0.100	0.009	0.990	1.658	0.555
15	0.886	0.098	0.012	0.996	1.303	0.876
16	0.915	0.142	0.025	1.082	1.290	1.970
17	0.919	0.127	0.026	1.073	1.225	1.862
18	0.850	0.044	0.029	0.923	1.202	0.346
19	0.948	0.073	0.012	1.033	1.231	3.235
20	0.876	0.124	0.019	1.019	1.305	2.237
21	0.895	0.220	-0.035	1.079	1.664	1.677
22	0.913	0.119	0.022	1.054	1.253	1.832
23	0.828	0.173	0.017	1.019	1.544	1.388
24	0.912	0.057	-0.024	0.945	1.119	2.296
25	0.894	0.114	0.012	1.020	1.298	1.358
26	0.882	0.127	-0.000	1.009	1.494	1.109
27	0.902	0.121	0.022	1.046	1.243	1.884
28	0.874	0.097	0.002	0.972	1.369	0.806
29	0.908	0.087	0.001	0.997	1.309	0.826
30	0.933	0.099	0.021	1.053	1.367	1.025
31	0.854	0.168	0.010	1.032	1.316	1.761
32	0.853	0.126	0.040	1.019	1.418	1.634

E Estimation implementation

First stage. Φ_t in (11) is a third-order polynomial in (k, l, m) with year effects, estimated by OLS within each industry. Because $\hat{\Phi}$ removes only ε , all information about β is deferred to the second stage, following [Akerberg et al. \(2015\)](#).

Second stage. For each candidate β (the nine translog coefficients), productivity is $\omega_{it}(\beta) = \hat{\Phi}_{it} - f(\cdot; \beta)$; the function g in (12) is a cubic in ω_{it-1} ; and the innovation is the residual of that projection. The GMM objective, $Q(\beta) = \left[\frac{1}{N} \sum \xi_{it}(\beta) \mathbf{z}_{it} \right]' W[\cdot]$, with the fixed weight matrix $W = (\mathbf{Z}'\mathbf{Z}/N)^{-1}$, is minimised in two passes: Nelder–Mead from an OLS translog starting point, followed by a BFGS refinement started at the simplex solution. The BFGS solution is retained only if the routine exits cleanly *and* does not worsen the objective.

Two features of this procedure should be stated explicitly rather than left in the code. First, when the BFGS pass obtains its gradient by central differences, it fails in 18 of the 23 divisions with a “could not calculate numerical derivatives” error, and the refinement is discarded. On an objective of order 10^{-6} whose surface is nearly flat in several directions, differencing returns rounding noise rather than a derivative, so under a finite-difference gradient the estimates in those 18 divisions are simply the Nelder–Mead solutions. We therefore supply the analytic gradient of Q , derived in closed form and verified in two ways: against central differences on synthetic data, where the finite-difference benchmark is valid (relative error 5.5×10^{-11} , with clean second-order convergence in the step size), and element by element against a reference implementation (to within 10^{-14}). With the

analytic gradient, the refinement is accepted in 17 of the 23 divisions and lowers the attained objective by a median factor of about 4.9×10^3 , reaching 10^{-9} – 10^{-11} where the finite-difference version stalled at 10^{-5} – 10^{-6} .

Second, the acceptance rule is deliberately conservative, and we have not relaxed it. In the six divisions where BFGS still exits uncleanly, the Nelder–Mead point is retained even where the rejected refinement reached a lower objective—in NIC24, by five orders of magnitude. This is defensible precisely because, as explained below, the attained objective is not a measure of solution quality in this estimator, so preferring the solution from the pass that ran cleanly is the safer rule. It is also the rule that the bootstrap applies, replication by replication.

We emphasise that this is a *just-identified* estimator: nine translog parameters are matched to nine moments, so the objective can in principle be driven to zero for any weight matrix. A small attained Q is therefore a statement about the numerical solver, not a test of instrument validity or of overidentifying restrictions, and we do not use it as one. The point is not merely formal, and the switch to the analytic gradient illustrates it twice. Under the finite-difference optimiser, the two industries that did *not* report convergence, NIC14 and NIC28, attained the two *smallest* objectives in the entire set (1.15×10^{-8} and 2.73×10^{-8} , against a typical value of about 10^{-5}), exactly as one would expect if Q carries no information about solution quality. And under the analytic gradient the attained objective falls by up to five orders of magnitude in seventeen divisions, while the economic estimates barely move: median markups shift by +0.19 per cent on average, and in the median division not at all. A quantity that can be reduced by several orders of magnitude without materially changing the parameters is not measuring fit, and we do not report it as though it were.

Convergence and sensitivity to starting values are reported and diagnosed in Appendix F. Industries with fewer than 500 usable plant-years are not estimated under the translog; only NIC 33 (repair and installation of machinery) is excluded on this ground. A trial run on a single industry (NIC 31) preceded the full run, and is reproduced by the verification suite to machine precision (a deviation of 4.5×10^{-17}).

Standard errors. For the Cobb–Douglas benchmark, the reported standard errors are the analytic (Wooldridge-form) ones, which lie between 0.0015 and 0.019. A 50-replication plant-block bootstrap was also run, and we report what it actually produced rather than treating it as corroboration: its dispersion is not credible. Four of the twenty-three industries return a standard error of *exactly* zero for the capital elasticity, and nine return standard errors for the materials and labour elasticities below 10^{-5} , three orders of magnitude below the analytic values; the bootstrap standard error of the materials elasticity ranges from 1.4×10^{-6} to 0.0037 across industries. Standard errors of exactly zero from fifty resamples indicate that the replications are not producing genuine variation in the

estimate, not that the estimate is precise. We therefore do *not* claim that the two procedures agree, and we do not use this bootstrap to validate the analytic standard errors. Fifty replications are, in any case, too few for final inference on a non-linear estimator with convergence failures. Inference for the translog statistics instead rests on the plant-clustered bootstrap over the full estimation pipeline reported in Section 8.4, which propagates first-stage as well as second-stage uncertainty and counts failed replications rather than discarding them.

Order of operations. Estimation $\rightarrow$ plant-level μ and $\psi \rightarrow$ winsorisation within NIC-2 $\times$ year cells (1/99 for μ , 2/98 for ψ) $\rightarrow$ aggregation (with multipliers where statements about the population are made). Winsorising *after* estimation and *before* aggregation ensures that the tails do not drive the aggregates, while the estimation sample remains the one the model defines.

F Convergence and starting-value sensitivity

Every diagnostic in this appendix was computed under the earlier first stage, without year effects (Appendix L). We retain it because it concerns the shape of the GMM objective, which the year effects do not alter in kind; it has not been re-run.

Under the finite-difference optimiser, two of the twenty-three estimated divisions—NIC14 (wearing apparel) and NIC28 (machinery n.e.c.)—carried a convergence flag of zero, and the BFGS refinement was rejected in favour of the Nelder–Mead solution in 18 of the 23. We report this rather than omit it, because the diagnosis is informative about the estimator.

The non-convergence turns out to be an artefact of *how the gradient was obtained*, not of the data or of identification. With the analytic gradient described in Appendix E, all twenty-three divisions converge, including NIC14 and NIC28, whose attained objectives fall to 9.4×10^{-11} and 6.0×10^{-10} . The multi-start evidence reported in this appendix was gathered under the finite-difference optimiser, and we retain it unchanged for two reasons. It remains the direct evidence on the question that a convergence flag cannot answer—whether different starting values recover the same parameter vector—and it was the basis on which these estimates were originally judged sound. The analytic gradient now corroborates that judgement by an independent route rather than superseding it.

The convergence flag should not be read as evidence that the estimates are unreliable, nor should its absence be read as evidence that they are sound; the reason is visible in the results themselves. The attained GMM objectives for the two non-converged divisions are 1.15×10^{-8} (NIC14) and 2.73×10^{-8} (NIC28), the two *smallest* in the entire set, against values of order 10^{-5} for divisions whose optimiser did report convergence. In a just-identified system the moment conditions can be satisfied almost exactly at the solution,

so the objective is uninformative about solution quality in either direction. What the flag records is whether the simplex met an internal tolerance before reaching its iteration limit, which is a statement about the solver’s path rather than about identification.

Distinguishing between these possibilities requires perturbing the starting values, which the baseline estimation does not do: it uses a single OLS translog start for every division. We therefore re-estimate both divisions from six documented sets of starting values—the default OLS translog, a Cobb–Douglas start with the second-order terms set to zero, the OLS vector scaled by one half, the OLS first-order terms alone, a start built from mean cost shares with a capital coefficient of 0.10, and a seeded random perturbation of the OLS vector—under both optimisers. For every start, whether it succeeds or fails, we record convergence status and economic admissibility (positive mean θ^L and θ^M , and the share of plants with non-positive elasticities). If different starts recover the same parameter vector, the flag is an artefact of the solver’s stopping rule and the estimates stand; if they scatter, the divisions have an identification problem and must be excluded from the wedge aggregates on the same footing as NIC18. Table 21 reports every start attempted, successes and failures alike.

Table 21: Starting-value sensitivity of the translog GMM. Every start attempted is shown. “Adm.” marks solutions that are economically admissible, requiring mean $\theta^L > 0$ and mean $\theta^M > 0$. Elasticity means are over the stage-2 estimation sample, so they differ slightly from Table 18, which averages over all plant-years in the division.

NIC2	Starting values	Q	conv	$\bar{\theta}^L$	$\bar{\theta}^M$	med μ	Adm.
14	OLS translog (default)	1.15e-08	0	0.116	0.868	1.566	yes
14	Cobb–Douglas OLS	1.07e-09	1	0.118	0.865	1.563	yes
14	OLS $\times 0.5$	2.74e-05	1	−0.207	1.250	2.250	no
14	OLS first-order only	1.53e-05	1	−0.052	1.047	1.886	no
14	mean cost shares	4.60e-02	1	1.783	−1.031	−1.913	no
14	OLS + random perturbation	2.53e-05	1	0.109	0.867	1.570	yes
28	OLS translog (default)	2.73e-08	0	0.107	0.894	1.345	yes

Three results follow, and the first two are cautionary.

First, the objective surface has multiple local optima, and three of the six starts reach inadmissible ones. The starts from the halved OLS vector and from the first-order terms alone return *negative* mean labour elasticities, and the cost-share start returns a negative materials elasticity and a negative median markup. These are not marginal cases. A single-start procedure with no admissibility screen could have reported any of them.

Second, neither the convergence flag nor the value of the objective identifies the good solutions on its own. All three inadmissible solutions report `conv` = 1, whereas the default start, which is one of the two best, reports `conv` = 0: a rule of “keep the converged solution” would have selected a nonsensical one. Nor is the objective alone sufficient: the first-order-only start attains $Q = 1.53 \times 10^{-5}$ with $\bar{\theta}^L < 0$, while the random-perturbation

start attains a *worse* value, $Q = 2.53 \times 10^{-5}$, and is admissible. Selection requires both criteria together.

Third, under a stated rule the estimates are nevertheless well determined. We adopt the rule: *among economically admissible solutions, retain the one with the lowest objective*. For NIC14 this selects the Cobb–Douglas start, at $Q = 1.07 \times 10^{-9}$, between one and seven orders of magnitude below every other solution. It agrees closely with the default start used in the reported estimates ($\bar{\theta}^L$ of 0.118 against 0.116, and a median μ of 1.563 against 1.566, a difference of 0.2 per cent), and the random-perturbation start independently recovers the same point. The reported estimate for NIC14 is therefore sound; only the optimiser’s convergence status was in question.

For NIC28, the default start reproduces the reported objective exactly (2.7327×10^{-8}) and is admissible. The remaining starts did not complete within a practical run time, and we do not report results that we did not obtain: the NIC28 solution is certified as admissible and reproducible, but not as the best available on its surface. As a partial check on the wider question, all 23 divisional solutions were screened against the admissibility criterion. Twenty-two satisfied it under the earlier first stage. The single exception was NIC30, whose mean labour elasticity was -0.032 under the finite-difference optimiser and -0.020 under the analytic gradient; with year effects in the first stage it is $+0.073$ (Table 18), and all 23 divisions now pass (Section 4.6). We report as a limitation that the other divisions were not subjected to the full multi-start diagnostic, which at the observed run times is not feasible for all 23.

G State and state $\times$ industry detail

This appendix provides the supporting tables for Section 7. Markups are computed over all 23 divisions; markdowns and incidence exclude NIC18, as throughout the paper. “Labour share” is emoluments over gross output, the paper’s α^L ; it is the ratio as reported, not a KLEMS-style construction (see the caution in Section 1).

The complete cell-level results—493 state $\times$ division cells and 799 state $\times$ three-digit cells, each with at least 100 plant-years—are not reproduced here, as they would run to many pages. They are provided as `state_nic2.csv` and `state_nic3.csv` in the replication package, together with `state_summary.csv` and, for the decomposition in (18), `state_decomposition.csv`. Table 22 summarises every state with at least 1,000 plant-years, and Table 23 gives division-level detail for the five highest- and five lowest-incidence states.

H Data-quality diagnostics

Three diagnostics support the design choices described in the main text.

Table 22: Market power by state, pooled 2012–2022. Every state with at least 1,000 plant-years, ordered by size of the organised manufacturing sector. Median markup and markdown; monopsony incidence $\Pr(\psi > 1)$; median labour share α^L ; median plant size in persons engaged.

State	Plant-years	med μ	med ψ	$\Pr(\psi > 1)$	α^L	size
Tamil Nadu	52,347	1.251	0.545	0.310	0.116	118
Maharashtra	47,501	1.217	0.753	0.389	0.106	137
Gujarat	38,814	1.213	0.936	0.474	0.086	114
Uttar Pradesh	37,799	1.201	0.944	0.481	0.094	79
Karnataka	24,163	1.222	0.701	0.375	0.120	97
Haryana	23,363	1.228	0.760	0.411	0.102	132
Rajasthan	19,730	1.208	0.919	0.471	0.088	64
West Bengal	17,954	1.198	1.008	0.503	0.086	77
Punjab	17,935	1.204	0.910	0.471	0.093	50
Andhra Pradesh	15,039	1.203	1.063	0.524	0.078	50
Madhya Pradesh	14,560	1.187	1.161	0.554	0.077	39
Kerala	12,848	1.195	0.678	0.360	0.130	44
Telangana	12,496	1.222	0.936	0.479	0.097	90
Uttarakhand	10,509	1.232	1.133	0.551	0.077	160
Assam	10,347	1.222	0.880	0.450	0.094	98
Bihar	9,004	1.477	0.487	0.351	0.274	82
Himachal Pradesh	7,468	1.204	1.145	0.544	0.078	102
Delhi	7,143	1.193	0.742	0.398	0.123	34
Odisha	6,841	1.177	1.177	0.570	0.078	29
Chhattisgarh	6,673	1.149	1.616	0.669	0.056	30
Jharkhand	5,230	1.183	1.230	0.569	0.075	31
Jammu & Kashmir	3,858	1.175	1.397	0.622	0.073	35
Tripura	3,626	1.504	0.640	0.229	0.211	81
Daman & Diu	3,356	1.239	0.861	0.436	0.085	154
Goa	2,738	1.221	0.734	0.391	0.124	105
Puducherry	2,363	1.214	0.901	0.469	0.103	74
Dadra & N. Haveli	2,119	1.193	0.970	0.487	0.063	173
code 14	1,602	1.506	0.638	0.229	0.235	27
Meghalaya	1,212	1.140	1.492	0.627	0.063	24
code 13	1,183	1.408	0.733	0.379	0.209	21

Table 23: State $\times$ division detail for the five highest- and five lowest-incidence states: the six largest divisions in each state by plant-years. Cells require at least 100 plant-years. Markdown levels at this resolution are noisier than division-level ones and should be read for ordering rather than for their second decimal (Section 8.2).

State	Division	Plant-years	med μ	med ψ	Pr($\psi>1$)	
<i>Five highest-incidence states</i>						
Chhattisgarh	Basic metals	1,490	1.065	2.951	0.883	
	Food	1,161	1.092	1.834	0.680	
	Non-metallic	759	1.287	0.860	0.419	
	Fabricated metal	559	1.220	1.228	0.596	
	Chemicals	488	1.179	1.691	0.742	
	Machinery	322	1.263	1.263	0.590	
Odisha	Food	1,278	1.096	1.418	0.670	
	Basic metals	1,191	1.133	1.536	0.713	
	Non-metallic	740	1.316	0.780	0.342	
	Chemicals	497	1.186	1.495	0.676	
	Fabricated metal	444	1.179	1.299	0.622	
	Electrical eq.	316	1.116	1.856	0.718	
Jharkhand	Non-metallic	910	1.381	0.517	0.263	
	Basic metals	687	1.086	2.070	0.693	
	Food	674	1.080	1.782	0.702	
	Chemicals	444	1.173	2.062	0.779	
	Motor vehicles	380	1.256	1.709	0.768	
	Coke/petro.	333	1.150	1.881	0.667	
Madhya Pradesh	Food	2,666	1.065	2.513	0.790	
	Chemicals	1,287	1.192	1.522	0.713	
	Non-metallic	1,268	1.273	0.808	0.356	
	Rubber/plastic	1,155	1.220	1.026	0.513	
	Basic metals	878	1.104	1.779	0.691	
	Electrical eq.	859	1.129	1.699	0.703	
Uttarakhand	Pharma	1,145	1.457	1.688	0.683	
	Rubber/plastic	1,030	1.276	0.873	0.406	
	Electrical eq.	1,016	1.163	1.801	0.785	
	Motor vehicles	980	1.321	1.233	0.618	
	Food	978	1.114	1.033	0.507	
	Chemicals	808	1.246	1.829	0.750	
<i>Five lowest-incidence states</i>						
Maharashtra	Food	6,422	1.117	1.225	0.593	
	Machinery	4,463	1.314	0.494	0.239	
	Motor vehicles	3,715	1.313	0.943	0.470	
	Fabricated metal	3,627	1.241	0.650	0.280	
	Chemicals	3,556	1.224	1.053	0.526	
	Electrical eq.	2,827	1.159	1.106	0.552	
Karnataka	Food	4,168	1.108	1.196	0.563	
	Machinery	2,178	1.328	0.481	0.194	
	Apparel	1,603	1.740	0.008	0.015	
	Electrical eq.	1,592	1.158	1.026	0.518	
	Chemicals	1,556	1.223	1.032	0.514	
	Non-metallic	1,458	1.281	0.674	0.302	
Kerala	Food	96	3,620	1.132	0.859	0.427
	Non-metallic		1,002	1.344	0.424	0.154
	Textiles		967	1.229	0.109	0.019
	Rubber/plastic		874	1.206	0.710	0.235

(i) Negative implied value added. In the cleaned manufacturing sample, 32,989 observations report $\alpha^M > 1$ and 69,031 report $\alpha^L + \alpha^M > 1$. Their distribution across industries is uneven, but they are not concentrated in the two flagged industries. NIC 18 (printing) stands out at 36.4 per cent, by a wide margin the highest rate of any estimated division, consistent with the reporting pattern of job work and contract manufacturing, in which the principal’s materials do not pass through the contractor’s accounts. Elsewhere the rates are not in the low single digits: the all-division rate is 14.1 per cent, NIC 30 (other transport equipment) is at 14.9 per cent, and four divisions that we *retain* have higher rates still—NIC 26 (19.7), NIC 13 (19.5), NIC 24 (17.0) and NIC 10 (16.7). This diagnostic therefore justifies excluding NIC 18 from the markdown aggregates, and gives no ground for treating NIC 30 differently from any other division (Section 4.6). Markup results are in any case insensitive to the anomaly, because anomalies in the materials share are winsorised.

(ii) Coherence of the wage bill. The ratio of total emoluments to basic wages lies between 1.10 and 1.19 in every round, which is the expected loading of bonus, provident fund and welfare contributions over wages, and it shows no discontinuity across format eras. This validates the construction of the wage bill, which takes account of the format era.

(iii) Continuity across format eras. The medians of output, materials, employment and capital evolve smoothly across the six format eras of the raw data, with no breaks at the era boundaries, where parsing errors would appear.

I Division bias in the size gradient, and its resolution

A regression of $\ln \psi_{it}$ on contemporaneous $\ln L_{it}$ is mechanically biased. From (14), $\ln \psi = \ln \theta^L - \ln \alpha^L - \ln \mu$, and $\alpha^L = wL/pY$, so current employment enters the dependent variable (with a negative sign): this is classical division bias. An exact log-decomposition, computed on the common sample of the finite-difference estimates ($N = 408,438$, with the identities holding to within 8×10^{-20}), shows how the bias operates. Under all three size measures, the elasticity component $\ln \theta^M$ contributes *positively* to the size gradient; the negative raw *markdown*–employment gradient is carried entirely by the labour-share channel. The markup gradient behaves differently and the distinction matters: it is *positive* in the raw data (+0.0166 within NIC-4 $\times$ year, +0.0464 within plants), and the elasticity and materials-share channels push it the same way rather than offsetting.

The solution is to measure size before the markdown is determined. On *lagged* size, the within-plant markdown gradient is +0.096 (lagged output) and +0.052 (lagged market share), and the corresponding industry $\times$ year gradients are +0.036 and +0.041. All four

are positive, as the granular-employer model predicts, and free of the mechanical channel; their conventional t -statistics are large, but, as Section 8.4 shows, conventional standard errors overstate the precision of such regressions, so we rely on these estimates for their sign. That the within-plant and cross-sectional gradients agree in sign is itself informative, although on lagged output the within-plant gradient is between two and three times the cross-sectional one. Under the finite-difference optimiser, by contrast, the industry $\times$ year gradients were indistinguishable from zero; Section 5.5 explains what changed and why. We report contemporaneous gradients only together with this caveat.

The sample size quoted above for the log-decomposition refers to the common sample of the finite-difference estimates. The lagged regressions reported in the main text use $N = 282,652$ plant-years, because lagging removes each plant’s first observed year; the contemporaneous regressions use 422,820 plant-years for the markup and 403,196 for the markdown.

J Aggregation concepts for the markdown

Let plant weights be s_i (counts, employment or sales). The *typical-plant* aggregate is $\bar{\psi} = N^{-1} \sum_i \psi_i$; the weighted mean is $\bar{\psi}_s = \sum_i s_i \psi_i$; and the harmonic aggregate of Yeh et al. (2022), derived by aggregating the plants’ first-order conditions, is

$$\bar{\psi}^H = \left[\sum_i \chi_i \psi_i^{-1} \right]^{-1}, \quad \chi_i \text{ output-elasticity-adjusted sales weights.} \quad (20)$$

Because $x \mapsto 1/x$ is convex, harmonic aggregation gives most weight to the plants with *low* ψ . Under the bimodal Indian distribution—a mode below one and a heavy tail of monopsonistic plants—the harmonic aggregate falls below one, to 0.408 on our common sample, even though 40–50 per cent of plants mark wages down. The Cobb–Douglas harmonic aggregate (1.366; Table 7) does not fall below one, because Cobb–Douglas removes the heterogeneity in elasticities that interacts with the weights. Across the concept–estimator pairs in Table 7 whose ψ lies in the stable region of the mapping (15), the implied elasticity then ranges over $\varepsilon_L \in [0.35, 2.73]$ on identical data. The choice of aggregation concept, rather than of sample or estimator, is therefore the first-order choice, and studies should state it.

K Sensitivity to the optimiser: finite-difference versus analytic-gradient estimates

This appendix compares the translog estimates obtained with an optimiser that computes its gradient by central differences with those obtained using the analytic gradient, which this paper retains. Both runs use the earlier first stage without year effects, so

neither column is the set of estimates reported in the main text; Appendix L compares the analytic-gradient run with those. As Appendix E explains, the finite-difference gradient cannot be computed reliably on this objective: under it, the BFGS refinement was discarded in 18 of the 23 divisions, and two divisions did not report convergence. The analytic gradient resolves all three symptoms. We report its effect on the estimates in full, division by division, because a change of numerical method that moves the estimates should be shown rather than merely described. In Tables 24 and 25, the columns labelled “pub” report the finite-difference estimates, and those labelled “rep” the analytic-gradient estimates.

Three points frame the comparison. First, this is a robustness result, not a correction of an error in the data or the model: the finite-difference estimates solve the same moment conditions, reached by a solver that stopped early, and nothing about the data, the sample or the identification differs. Second, the comparison is like for like: the moment conditions and starting values of the two runs are provably identical (Appendix E), and only the numerical path to the optimum differs. Third, the analytic gradient improves every diagnostic: 23 of 23 divisions converge, against 21 under the finite-difference gradient; the refinement is accepted in 17 divisions, against 5; and the attained objective falls by a median factor of about 4.9×10^3 .

Table 24: Finite-difference (“pub”) versus analytic-gradient (“rep”) estimates, by division, both under the earlier first stage without year effects. NIC30 failed the admissibility screen under both; with year effects it passes (Appendix L).

NIC2	$\bar{\theta}^L$		$\bar{\theta}^M$		med μ		$\Pr(\psi > 1)$	
	pub	rep	pub	rep	pub	rep	pub	rep
10	0.083	0.083	0.924	0.923	1.102	1.101	0.628	0.644
11	0.132	0.094	0.879	0.899	1.345	1.342	0.424	0.338
12	0.179	0.179	0.868	0.868	1.739	1.739	0.376	0.376
13	0.061	0.054	0.966	0.972	1.189	1.197	0.179	0.140
14	0.121	0.116	0.868	0.863	1.556	1.553	0.203	0.238
15	0.118	0.122	0.867	0.865	1.231	1.229	0.231	0.272
16	0.133	0.134	0.889	0.888	1.194	1.194	0.580	0.474
17	0.114	0.083	0.897	0.913	1.152	1.173	0.599	0.420
18	0.059	0.037	0.855	0.864	0.996	1.002	0.002	0.004
19	0.090	0.095	0.986	0.998	1.189	1.203	0.662	0.684
20	0.129	0.133	0.882	0.871	1.218	1.203	0.622	0.676
21	0.231	0.272	0.848	0.843	1.422	1.400	0.570	0.642
22	0.116	0.113	0.887	0.890	1.179	1.183	0.598	0.585
23	0.159	0.170	0.824	0.833	1.385	1.384	0.279	0.145
24	0.087	0.090	0.923	0.929	1.090	1.100	0.673	0.692
25	0.053	0.053	0.998	0.998	1.363	1.363	0.068	0.068
26	0.127	0.167	0.878	0.847	1.362	1.313	0.322	0.386
27	0.068	0.068	0.950	0.950	1.249	1.249	0.136	0.136
28	0.115	0.129	0.892	0.885	1.337	1.326	0.311	0.350
29	0.103	0.016	0.908	0.987	1.275	1.385	0.346	0.104
30	-0.032	-0.020	1.011	0.996	1.386	1.364	0.053	0.068
31	0.182	0.182	0.861	0.861	1.253	1.253	0.540	0.540
32	0.079	0.105	0.875	0.858	1.338	1.316	0.203	0.218

The pattern is the one anticipated in Section 8.2. The materials elasticity is essentially unchanged: its median change across the 22 admissible divisions is zero to four decimal places. The markup follows it, with a median change of zero and a largest change of

8.6 per cent. The labour elasticity is a different matter, and the markdown inherits its instability. Four divisions move by more than ten percentage points in monopsony incidence, and one by more than twenty: NIC29 (motor vehicles), where $\bar{\theta}^L$ falls from 0.103 to 0.016 and incidence from 0.35 to 0.10, a move of 24 points, followed by NIC17 (−17.9), NIC23 (−13.4) and NIC16 (−10.6). Four divisions—NIC12, 25, 27 and 31—are unchanged to three decimal places, as expected: in these divisions the refinement was rejected under both optimisers, so both runs report the same Nelder–Mead solution.

Table 25: Finite-difference (“pub”) versus analytic-gradient (“rep”) estimates, year by year, both under the earlier first stage. Incidence excludes NIC18 and NIC30 in both columns, the rule of that draft.

Year	med μ		med ψ		Pr($\psi > 1$)	
	pub	rep	pub	rep	pub	rep
2012	1.248	1.252	0.870	0.848	0.466	0.443
2013	1.227	1.232	0.844	0.817	0.457	0.432
2014	1.241	1.247	0.807	0.780	0.441	0.412
2015	1.272	1.273	0.722	0.708	0.405	0.384
2016	1.260	1.260	0.693	0.679	0.391	0.367
2017	1.241	1.243	0.690	0.678	0.390	0.371
2018	1.225	1.228	0.708	0.690	0.398	0.379
2019	1.215	1.216	0.658	0.643	0.376	0.357
2020	1.231	1.231	0.670	0.661	0.379	0.363
2021	1.197	1.196	0.745	0.733	0.415	0.398
2022	1.156	1.154	0.758	0.753	0.419	0.408

At the level at which the paper’s arguments are made, the change of optimiser matters little (Table 25). Every markup series moves by less than 0.01 in every year. The median markdown series shifts down by between 0.005 and 0.027 without changing shape, still falling to a trough in 2019 and then recovering. Monopsony incidence falls by between 1.1 and 2.9 percentage points in each year, leaving the U-shape intact, and its annual range moves from 37.6–46.6 to 35.7–44.3 per cent. With one exception—the cross-sectional size gradient discussed in Section 5.5—no qualitative claim in this paper depends on which of the two optimisers is used. This is the useful content of the appendix: it establishes that the markup findings are insensitive to the optimiser and, at the same time, measures precisely how sensitive the division-level markdown findings are, which is why the latter are reported with bootstrap intervals, while the former can be read as point estimates.

L The first stage with and without year effects

The first stage of the estimator regresses log output on a third-order polynomial in the inputs and keeps the fitted value $\hat{\phi}_{it}$ and the residual $\hat{\varepsilon}_{it}$. The residual enters the markup through the share correction, $\mu_{it} = \theta_{it}^M / (\alpha_{it}^M e^{\hat{\varepsilon}_{it}})$, which removes measurement error in

output from the denominator (De Loecker & Warzynski, 2012). Earlier drafts of this paper estimated the polynomial without year effects. Any common movement in output that the polynomial in inputs could not absorb was then left in $\hat{\varepsilon}_{it}$, and from there it passed into every markup.

That is what happened. Table 26 shows that under the earlier first stage the mean residual rose from -0.030 in 2012 to $+0.054$ in 2022. A rise of 0.084 in $\hat{\varepsilon}$ lowers every markup by a factor of $e^{-0.084} \approx 0.92$, so it accounts by itself for an eight per cent fall in the measured markup, with no change in any plant’s pricing. The median markup under that specification fell from 1.253 to 1.155 . The uncorrected ratio θ^M/α^M , which does not involve the residual, rose slightly over the same years, from 1.195 to 1.201 , and the Cobb–Douglas benchmark of Appendix D, which also divides by the uncorrected share, shows no trend either. Adding year effects to the polynomial makes the residual mean-zero in every year by construction. The median markup then runs from 1.199 to 1.213 , and the uncorrected ratio still moves with it.

Table 26: First stage without (“old”) and with (“new”) year effects, by year. All 23 estimated divisions in both columns. $\bar{\hat{\varepsilon}}$: mean first-stage residual under the old specification (zero in every year under the new one). θ^M/α^M : median of the elasticity over the uncorrected materials share. Medians of winsorised μ and ψ . Source: `firststage_compare/by_year.csv`.

Year	$\bar{\hat{\varepsilon}}$ (old)	med μ		med θ^M/α^M		med ψ	
		old	new	old	new	old	new
2012	-0.030	1.253	1.199	1.195	1.180	0.832	0.968
2013	-0.018	1.232	1.191	1.193	1.180	0.804	0.932
2014	-0.023	1.248	1.199	1.199	1.184	0.767	0.891
2015	-0.028	1.274	1.218	1.224	1.211	0.694	0.803
2016	-0.017	1.261	1.221	1.230	1.216	0.665	0.768
2017	-0.012	1.244	1.221	1.227	1.214	0.662	0.770
2018	$+0.002$	1.228	1.221	1.221	1.209	0.677	0.792
2019	$+0.022$	1.217	1.239	1.240	1.227	0.631	0.730
2020	$+0.016$	1.233	1.258	1.245	1.234	0.645	0.741
2021	$+0.025$	1.198	1.221	1.206	1.197	0.718	0.837
2022	$+0.054$	1.155	1.213	1.201	1.191	0.735	0.858

Year effects enter only the first stage, but the second stage uses $\hat{\phi}_{it}$ to construct productivity, so the output elasticities are re-estimated too, and that is how the markdown changes. The markdown does not contain $\hat{\varepsilon}$, because the share correction appears in both of its ratios and cancels; it moves only through θ^L and θ^M . At the plant level the two specifications agree closely on the markup (correlation 0.904) and somewhat less on the markdown (0.852). The pooled median markdown rises from 0.708 to 0.821 , and the median markdown is higher in every year.

At the division level the change is larger, which is the pattern Section 8.2 leads one to expect. Under the old first stage the median markup had a negative linear trend in 17 of

23 divisions; under the new one, in 4. Monopsony incidence moves by tens of percentage points in several divisions: fabricated metal products from 6.8 to 41.8 per cent, electrical equipment from 13.6 to 63.5, motor vehicles from 10.3 to 47.0 and wood products from 48.0 to 86.9, while wearing apparel falls from 23.8 to 4.9 and rubber and plastics from 58.5 to 38.0. The mean labour elasticity of other transport equipment (NIC30) changes sign, from -0.020 to $+0.073$, so the division now passes the admissibility screen and enters every aggregate.

Two checks support the new specification as a correction rather than a choice among equals. First, the code that estimates it reproduces the earlier estimates exactly when the year effects are switched off (checked on three divisions, NIC11, NIC12 and NIC31), so the year effects are the only difference between the two. Second, the earlier decline did not appear in any quantity that bypasses the residual: not in θ^M/α^M and not in the Cobb–Douglas markups. A fall in markups that is visible only through the first-stage residual is a fall in that residual, and we report it as such.